COMPUTER NETWORKS

CS610

Lecture-1

Hammad Khalid Khan

Course Answers the Basic Question:

How do COMPUTER NETWORKS and INTERNET Operate?

Textbook

- Title

  Computer Networks and Internets, with Internet Applications, 4th edition

- Author Douglas E. Comer
- Publisher Pearson Education
- Book website

  http://www.netbook.cs.purdue.edu

Reference Books

- “Computer Networks” by Andrew S. Tanenbaum, Prentice Hall Publishers
- “Networking: A Top-Down Approach Featuring the Internet” by James F. Kurose and Keith W. Ross

Course Composition

- Four Main Parts
  - Using and Building Internet Applications (Chap. 1 - 3)
  - Data Transmission (Chap. 4 - 6)
  - Packet Transmission (Chap. 7-16)
  - Interntworking (Chap. 17-27)
  - Network Applications (Chap. 28-41)

Part 1: Data Transmission

- Transmission Media
- Local Asynchronous Communication (RS 232)
- Long Distance Communication (Modems and Carriers)

Part 2: Packet Transmission

- Packets and Frames
- Local Area Networks (LANs)
  - Hardware Addressing
  - Ethernet, WLAN and Bluetooth
  - Extending LANs: Fiber Modems, Repeaters, Bridges, Switches

Part 2: Packet Transmission (Cont'd)

- WAN Technologies and Routing
- Network Ownership and Service Paradigm
- Protocols and Layering

Part 3: Interntworking

- Concepts, Architecture and Protocols
- IP Addresses
- Binding Protocol Addresses

Part 3: Interntworking (Cont'd)

- IP Datagrams and Datagram Forwarding
- IP Encapsulation, Fragmentation and Reassembly
- TCP/IP

Part 4: Network Applications

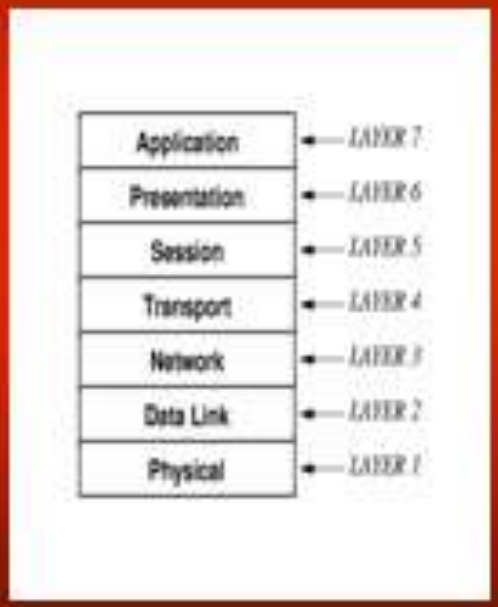
- Client-Server Interaction
- The Socket Interface
- Example Applications

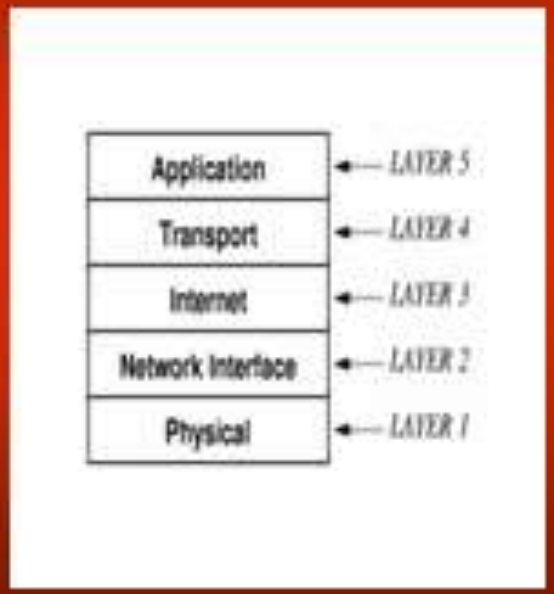
Grade Distribution

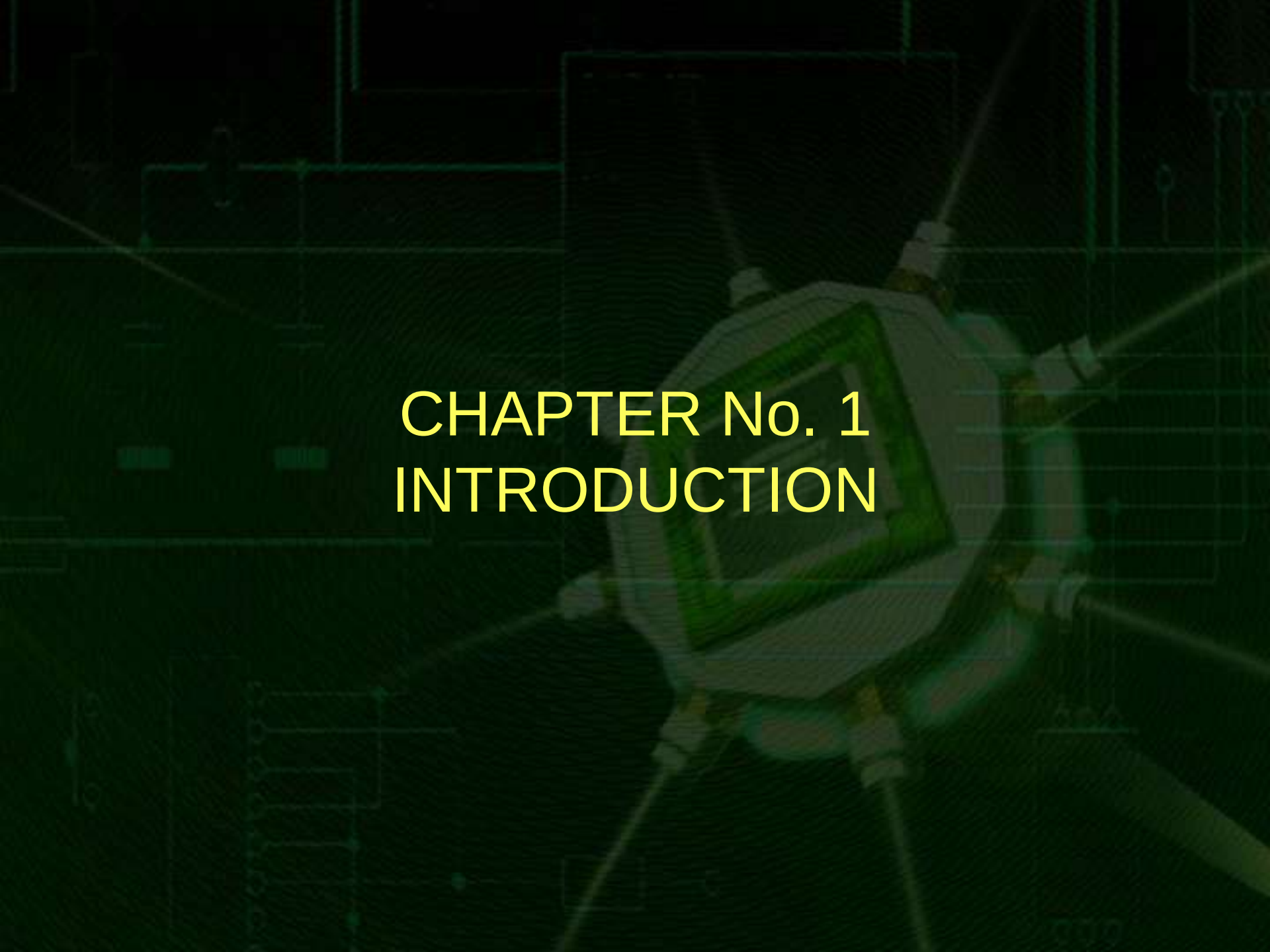
- Assignments 15%
- Midterm Examination 35%
- GMDB 5%
- Final Examination 45%

COURSE INTRODUCTION

ISO Model and TCP/IP Protocol







CHAPTER No. 1
INTRODUCTION

Network and Internet

- **Network**

  System for connecting computers using a single transmission technology

- **Internet**

  Set of networks connected by routers that are configured to pass traffic among any computers attached to any network in the set

Classification of Networks

- By SIZE
  - Local Area Network (LAN) versus Wide Area Network (WAN)
- By Connectivity
  - Point-to-Point versus Broadcast
- By Communication Medium
- By Mobility
  - Fixed versus Mobile

Networks in Daily Life

- Now, networks are an important part of everyday activities
  - Business
  - Home
  - Government
  - Education

Mastering The Complexity

- Concentrate on understanding the concepts
- Learning the Networking Terminology
  (summarized in a Glossary in Appendix 1)

Complexity in Network Systems

- Many different technologies exist
- No single underlying theory or model
- Models are either so simplistic or so complex
- No simple or uniform terminology

Review Lecture 1

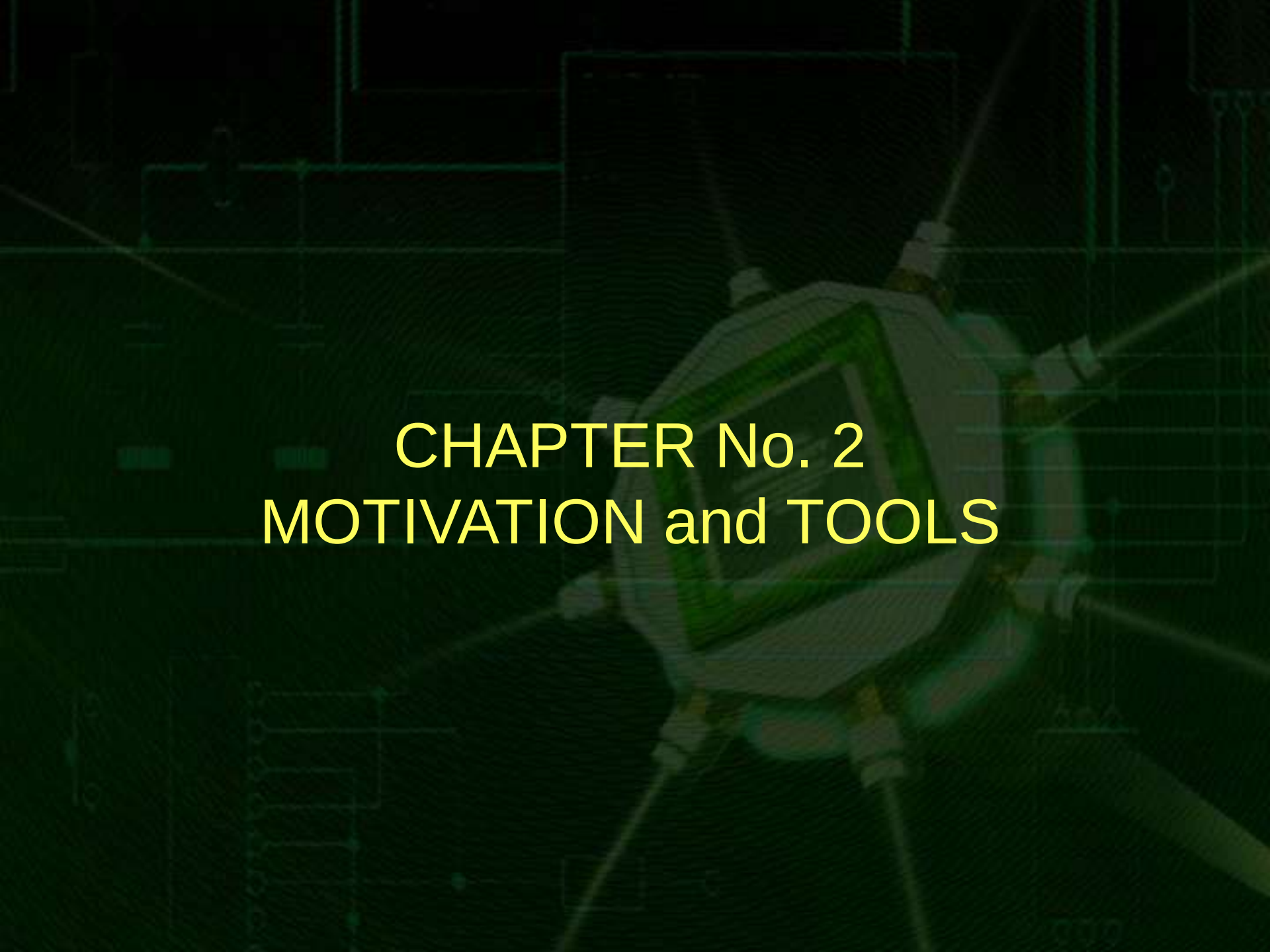
- Course Introduction
- Network and Internet
- Classification of Networks
- Networks in Daily Life
- Complexity in Computer Networking
- Mastering the Complexity

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Lecture-2

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CHAPTER No. 2
MOTIVATION and TOOLS

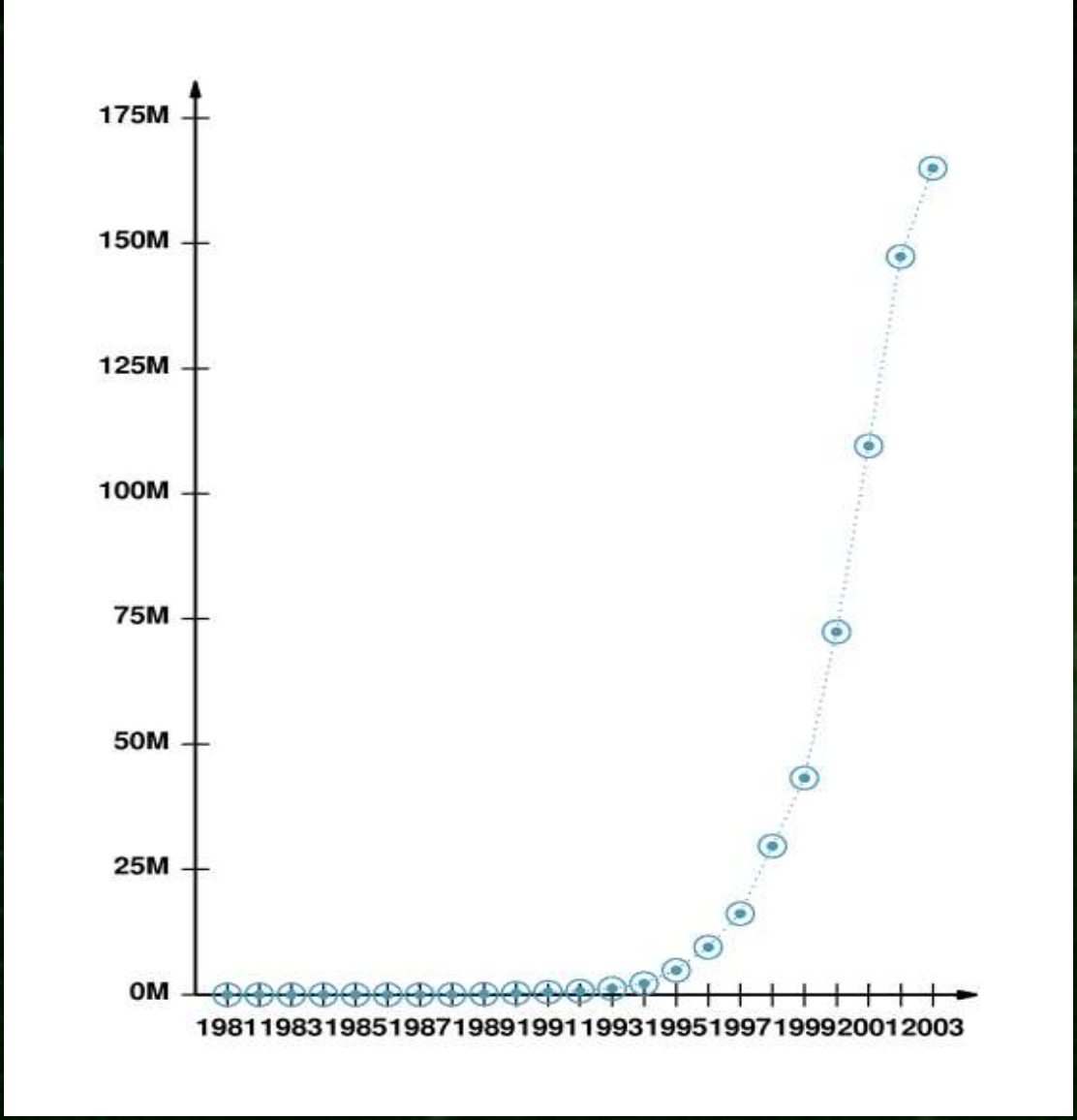
Resource Sharing

- Sharing a peripheral device such as a printer or a disk
- Sharing a large-scale computational power, because
  - Computers were extremely expensive
  - The government budgets were not sufficient to provide computers for all scientists and engineers

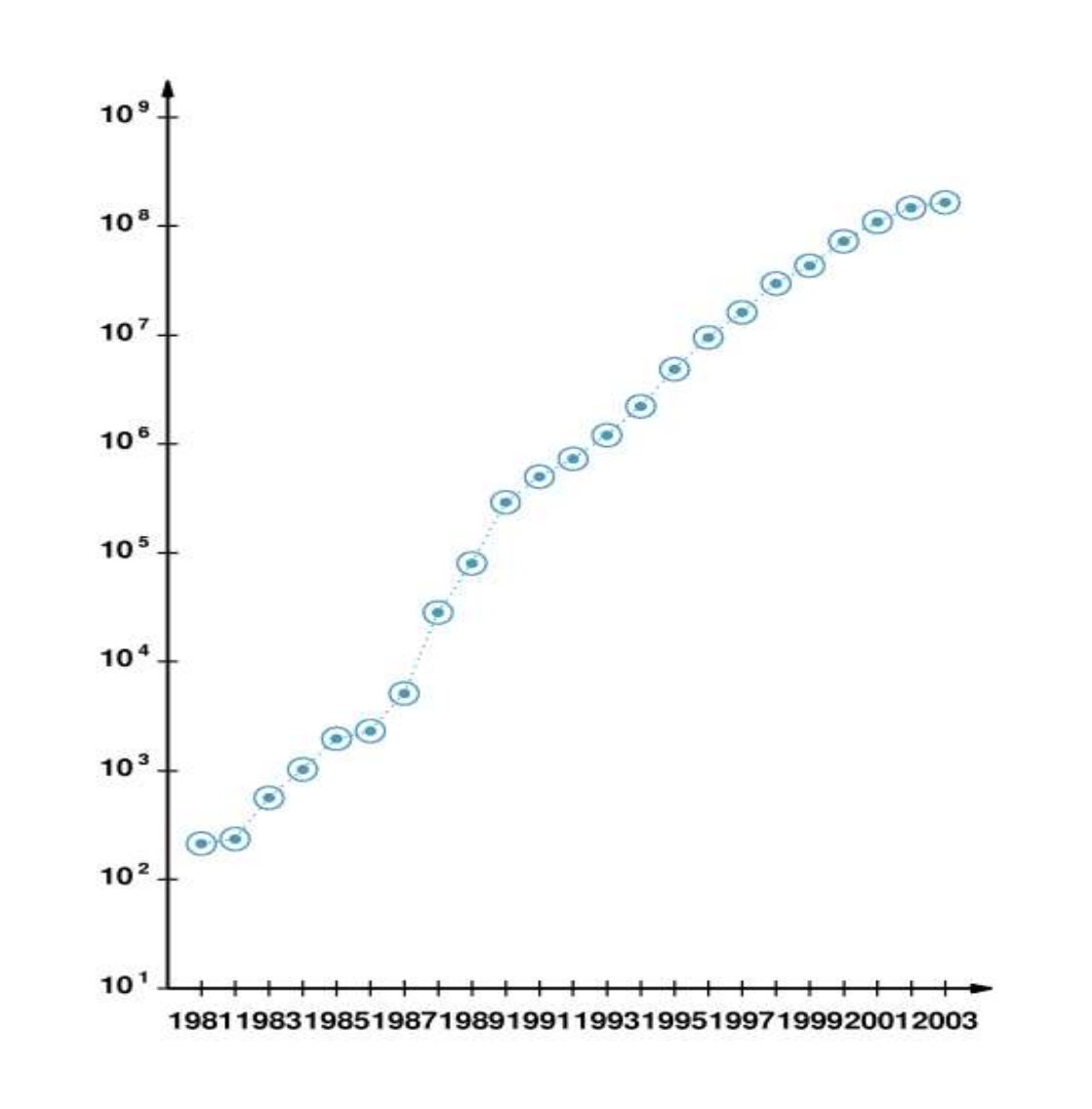
Growth of the Internet

- U.S Department of Defence Advanced Research Projects Agency (ARPA) started ARPANET project
- The early internet emerged in 1970s
- Has grown drastically since then

Growth of the Internet

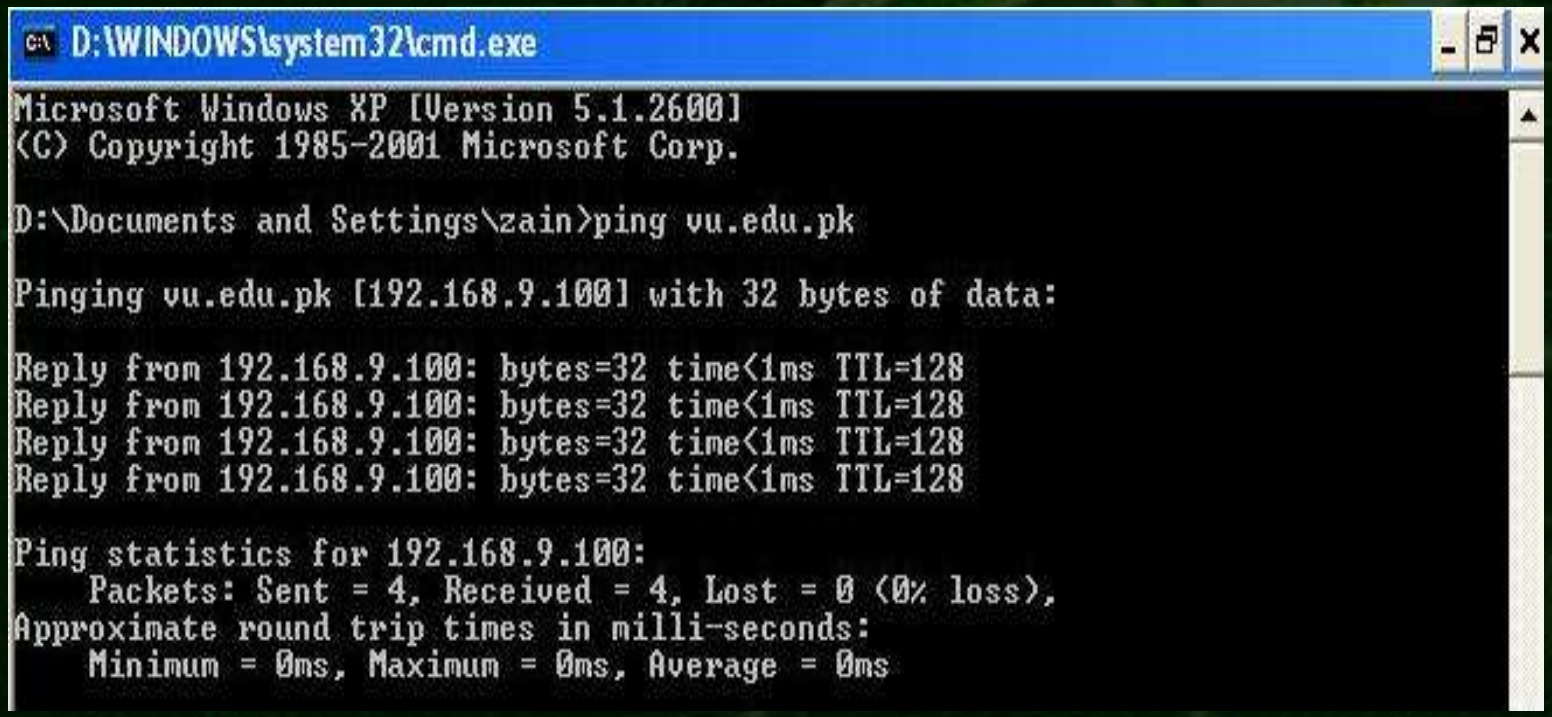


Growth of the Internet



Probing The Internet

- “PING” program



Probing The Internet

- “PING” program

```
PING sears.com: 56 data bytes
64 bytes from 32.97.168.129: icmp_seq=0. time=49. ms
64 bytes from 32.97.168.129: icmp_seq=1. time=50. ms
64 bytes from 32.97.168.129: icmp_seq=2. time=48. ms
64 bytes from 32.97.168.129: icmp_seq=3. time=50. ms
64 bytes from 32.97.168.129: icmp_seq=4. time=48. ms
----sears.com PING Statistics----
5 packets transmitted, 5 packets received, 0% packet loss
round-trip (ms)  min/avg/max = 48/49/50
```

Probing The Internet

- Problems with the ping
  - “Ping” cannot tell the reason of the problem
  - Some networks or computers reject the ping packets
  - “Traceroute”

Probing The Internet

- Tracing A Route

```
traceroute to DANDELION-PATCH.MIT.EDU (18.181.0.31), 40 byte packets
 1  cisco1 (128.10.2.250)  2 ms  1 ms  2 ms
 2  cisco-tel-252.tcom.purdue.edu (128.210.252.22)  2 ms  1 ms  1 ms
 3  abilene.tcom.purdue.edu (192.5.40.10)  6 ms  8 ms  7 ms
 4  clev-ipls.abilene.ucaid.edu (198.32.8.26)  14 ms  14 ms  12 ms
 5  nycm-clev.abilene.ucaid.edu (198.32.8.30)  24 ms  27 ms  24 ms
 6  192.5.89.45 (192.5.89.45)  31 ms  34 ms  35 ms
 7  192.5.89.10 (192.5.89.10)  33 ms  33 ms  33 ms
 8  NW12-RTR-FDDI.MIT.EDU (18.168.0.16)  59 ms  34 ms  33 ms
 9  DANDELION-PATCH.MIT.EDU (18.181.0.31)  62 ms *  79 ms
```

Summary

- Resource Sharing
- Growth of the Internet
  - Linear Scale
  - Log Scale
- Tools for Probing the Internet
  - PING
  - TRACE ROUTE

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Lecture-3

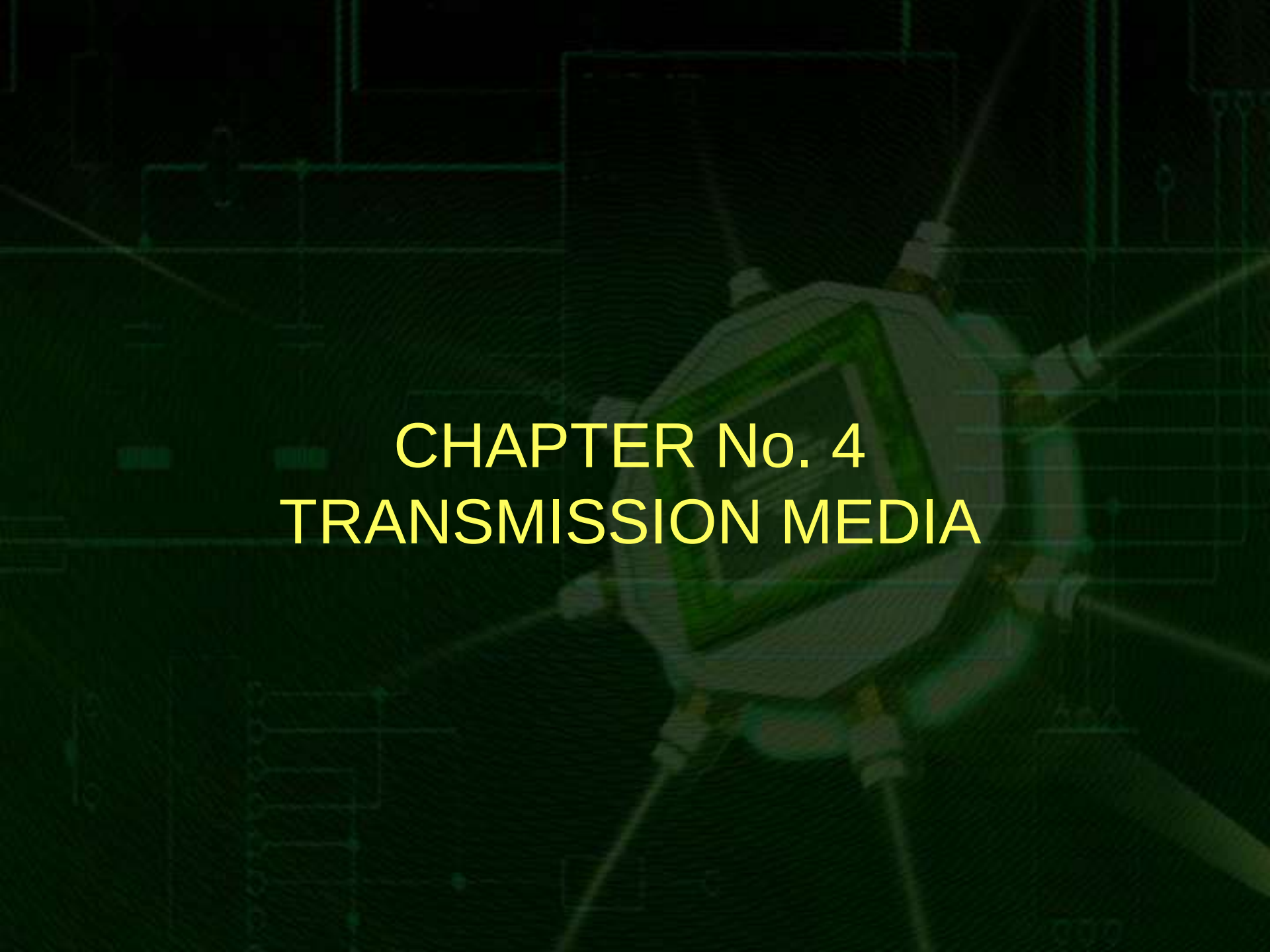
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Types of Multiplexing

- Frequency Division Multiplexing (FDM)
- Wave Division Multiplexing (WDM0)
- Time Division Multiplexing (TDM)

Review Lecture 2

- Resource Sharing
- Growth of the Internet
  - Linear Scale
  - Log Scale
- Tools for Probing the Internet
  - PING
  - TRACE ROUTE



CHAPTER No. 4
TRANSMISSION MEDIA

Basic Idea

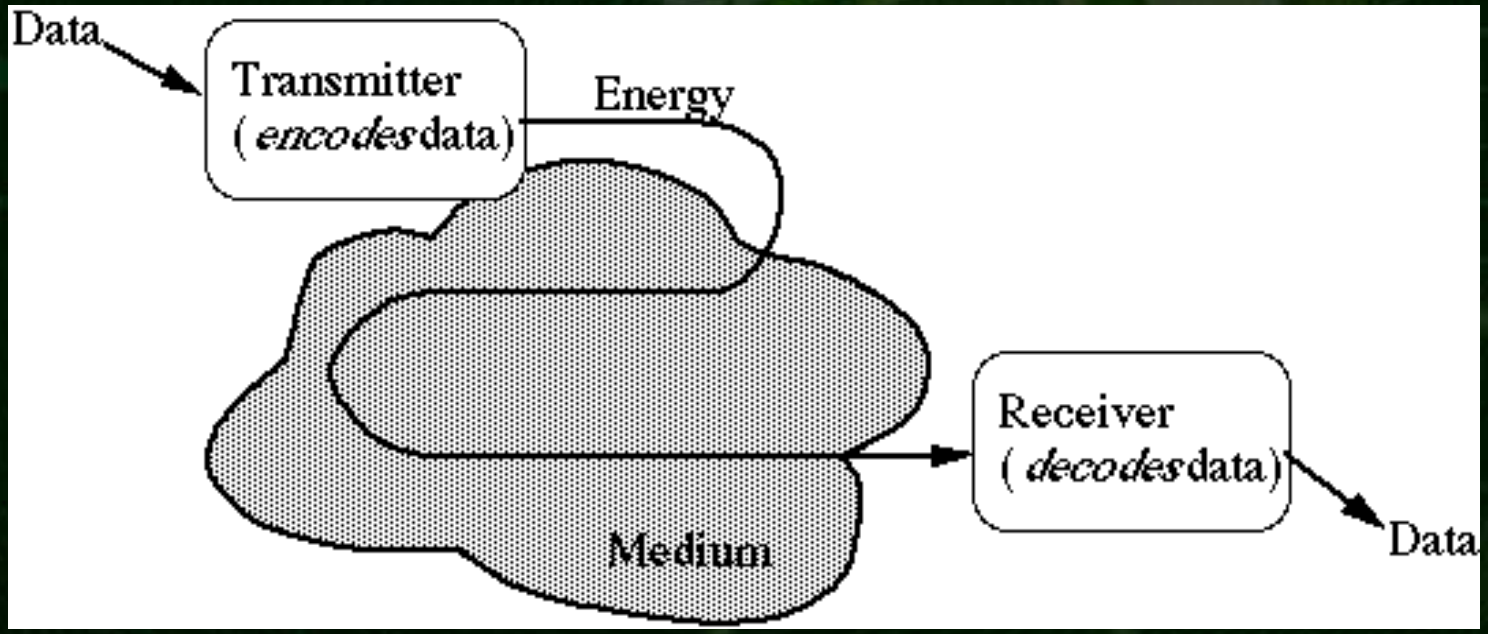
- Encode data as energy and transmit energy
- Decode energy at destination back into data
- Energy can be electrical, light, radio, sound, ...
- Each form of energy has different properties and requirements for transmission

Transmission media

- Transmitted energy is carried through some sort of medium
- Transmitter encodes data as energy and transmits energy through medium
  - Requires special hardware for data encoding
  - Requires hardware connection to transmission medium

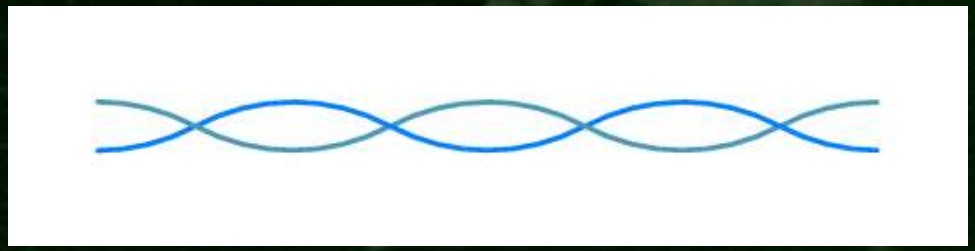
Transmission media

- Media can be copper, glass, air, ...



Copper Wires

- Widely used because of low resistance to electric current, inexpensive and easy to install
- Twisted pair wires
  - Two insulated copper wires twisted together to reduce Interference
  - Examples: Telephone wiring (one pair).LAN wiring UTP cable (4 pairs of copper wire)

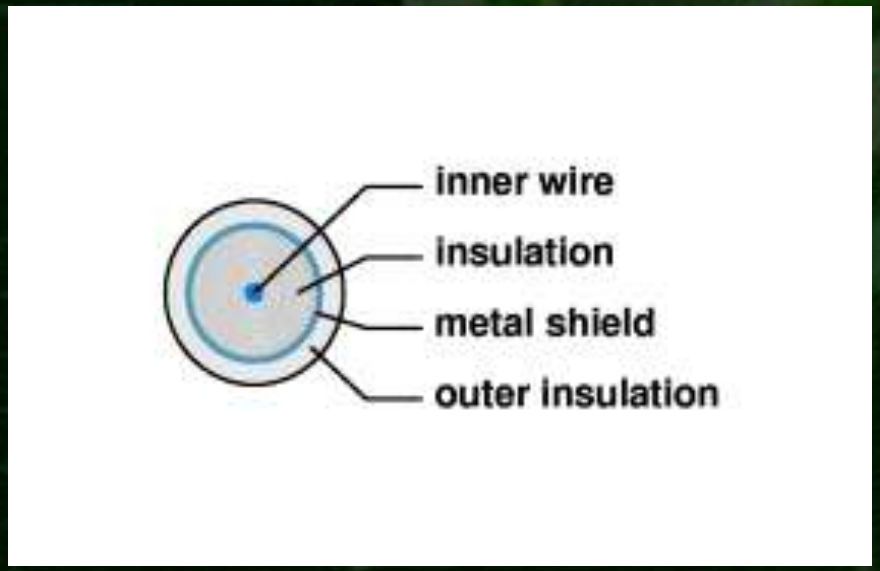


Types of Media

- GUIDED Media:
- Uses physical path
  - Examples: copper wires, optical fiber, coaxial cables etc.
- UNGUIDED Media:
- Uses no physical path
  - Example: Radio waves

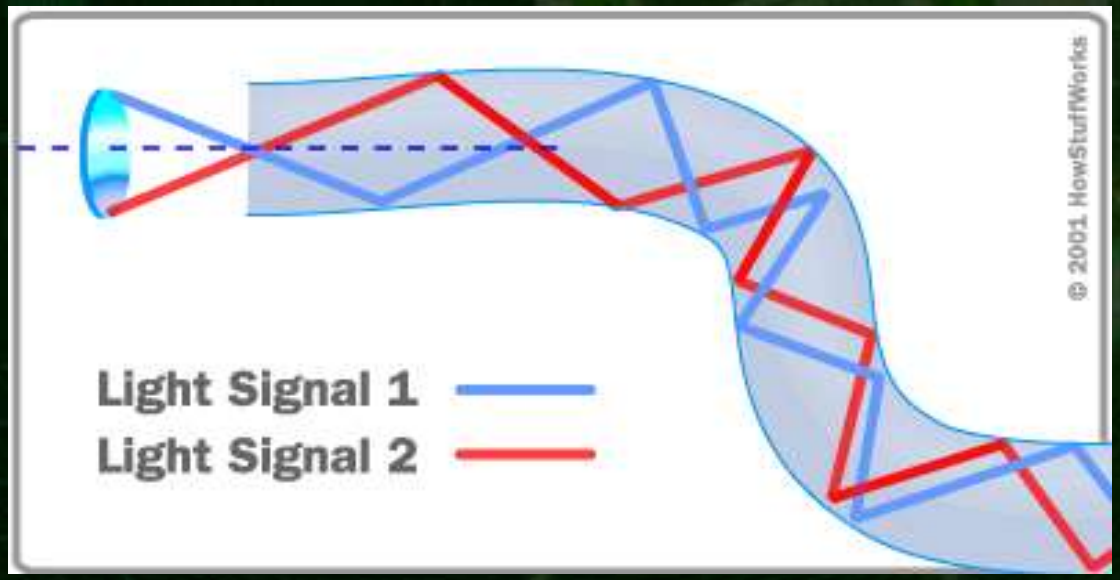
Coaxial Cable

- Single wire surrounded by an insulation and a metal shield against interference
- Example use: Cable TV wiring and early LAN wirings



Optical Fibers

- Uses light to transport data



Advantages of Optical Fibers

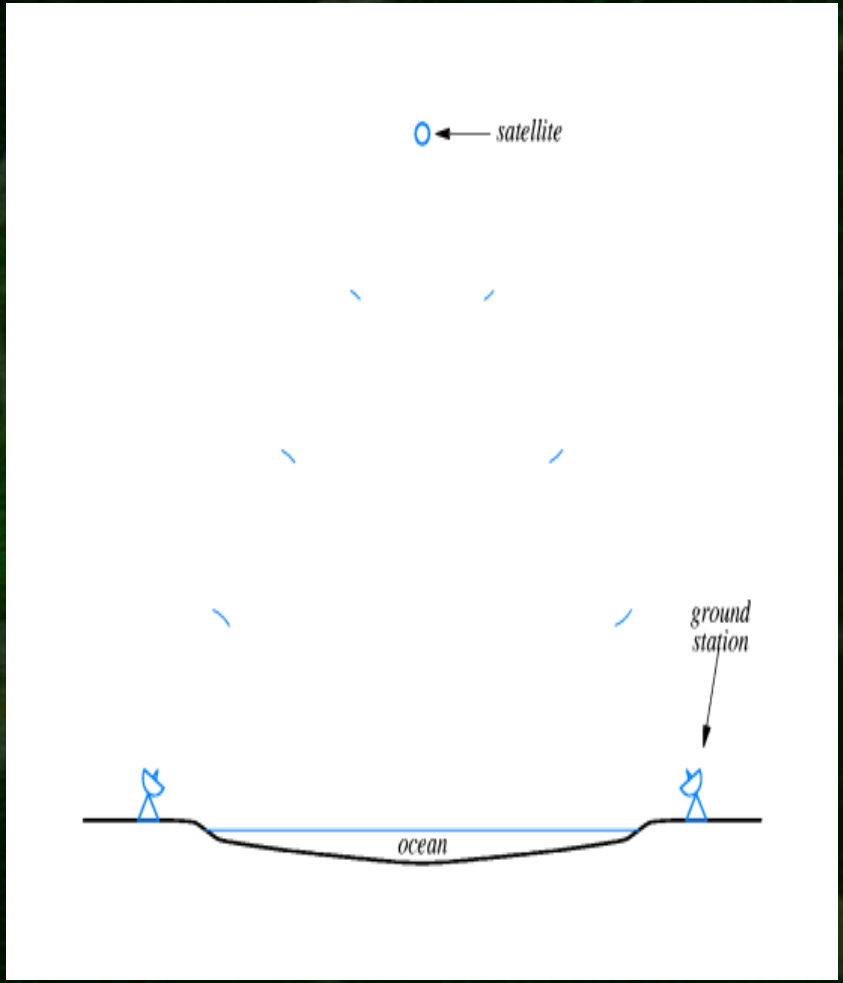
- Neither causes nor susceptible to electrical interference
- Can carry pulse much farther than copper
- Light can encode more information than electric
- One single fiber is sufficient to carry data

Radio

- Used for public radio and TV broadcast
- Data transmission is said to operate at Radio Frequency (RF)
- Transmission range depends on antenna size

Satellites

- Satellites for long distance transmission
- Satellite transponders receive, amplify, and transmit the RF signal back to the ground



Geosynchronous Satellites

- Placed in an orbit (at 35,785 km or 22,223 miles above the earth) that is exactly syncronized with the rotation of the earth
- From the ground, it appears to stay at the same position at all times

Low Earth Orbit Satellites

- Rotates faster than the rotation of the earth at 200 to 400 miles above the earth
- They do not appear to remain stationary
- Low Earth Orbit Satellite Arrays is a set of satellites communicating with each other and with the earth surface

Microwave

- Electromagnetic radiation beyond the frequency range used for radio and TV
- It can be aimed in a single direction
- Can carry more information than low frequency RF transmissions

Infrared

- Remote controls used with TV etc uses IR
- Inexpensive wireless communication
- Limited to a small area
- Convenient for portable computers

CHAPTER No. 5
LOCAL ASYNCHRONOUS COMMUNICATION (RS-232)

Introduction

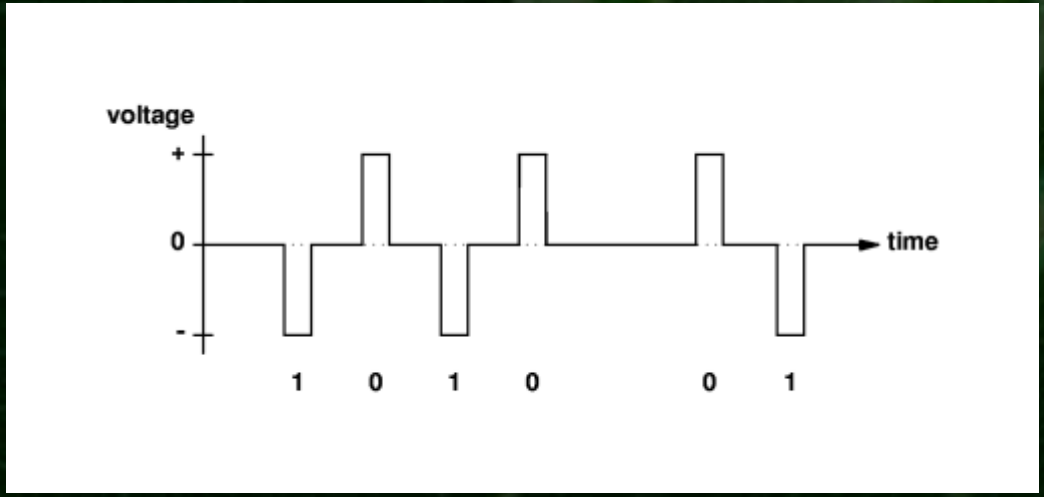
- Computers use binary digits (0s and 1s) (bits) to represent data
- Computers communicate by sending bits through a transmission medium (i.e. cable, air).
- This chapter describes how electric current is used to send digital information across **Short Distances**

The Need For Async Communication

- In **asynchronous** communication the sender and receiver do not need to coordinate (synchronize) before the transmission begins
- In an asynchronous system, the receiving hardware must be ready to accept and interpret the signal whenever it arrives

Using Electric Current to Send Bits

- A small electric current is used to encode data
- Example: Negative voltage to represent a **0** bit, positive voltage to represent a **1** bit.



Standards For Communication

- Standard specifies both the timing of the signals and the electrical details of voltage and current
- **RS232-C** (by EIA) is one of the most widely used standard to transfer characters between a computer and a device such as modem, keyboard etc.
- RS-232 defines a **serial** (bits are sent one after another) **asynchronous** communication

RS232-C Specifications

- Cable length < 50 feet (20mt)
- -15v (bit 1) , +15v (bit 0)
- One character => 7 or 8 data bits
- No delay between character bits
- When idle, puts -15v (bit 1) on the wire
- Each character start with a start bit(0), and end with a stop bit(1)

Baud Rate, Framing, And Errors

- Baud rate:
  - The number of changes in the signal per second
- For simple RS-232, Baud rate = Number of bits/sec
- If the voltage measurements do not all agree or if the stop bit does not occur exactly at the same time expected, framing errors occur

Full-Duplex Asynchronous Comm

- Electrical circuits require at least two wires, one of them to carry the data signal, the other is used as the return path, called ground
- Simultaneous transfers in two directions are known as full-duplex transmission

Full-Duplex Asynchronous Comm

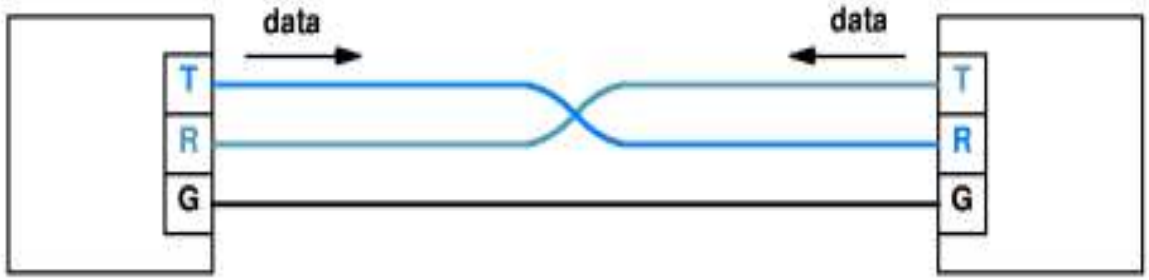


Figure 5.3 The minimal wiring required for full-duplex RS-232 communication. Although the two circuits carry data independently, it is possible for them to share a single ground wire.

CHAPTER No. 6
LONG DISTANCE COMMUNICATION
(Carriers, Modulation and Modems)

Long Distance Communication

- Electric current becomes weaker as it travels over copper
- Signal loss occurs when RS232 is attempted to use to communicate to a remote site
- A continuous, oscillating signal, called a Carrier, propagates farther than other signals

Modulation

- To send data, a transmitter slightly modifies the original information by using a carrier
- Such modifications are called “Modulation”
- The receiver
  - monitors the incoming carrier,
  - detects modulation,
  - reconstructs the original data, and
  - discards the carrier

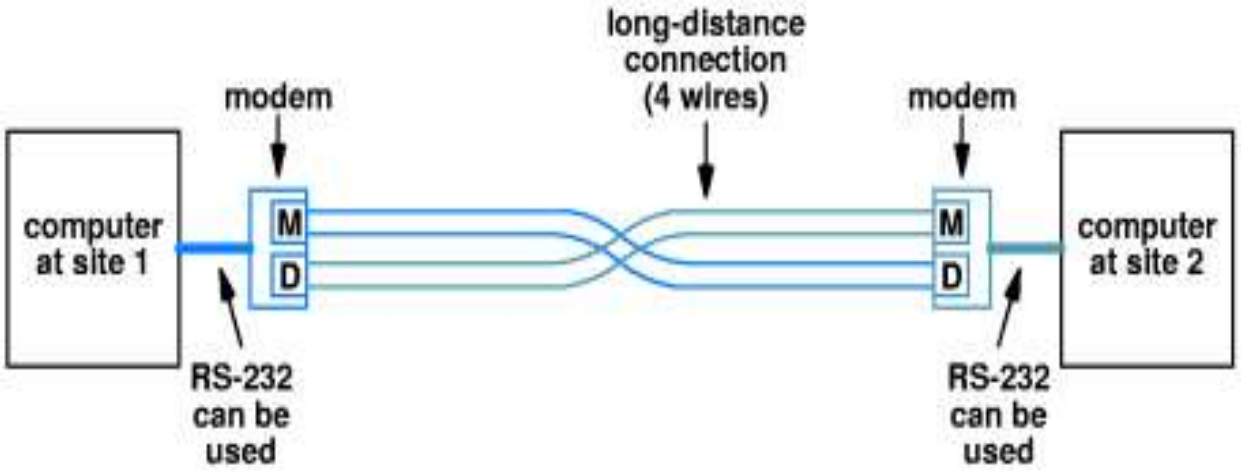
Types of Modulation

- Three of the modulation schemes are:
  - Amplitude Modulation (AM)
  - Frequency Modulation (FM)
  - Phase Modulation (PM)

Modem

- A Modulator is a hardware circuit that
  - accepts a sequence of data bits
  - applies modulation to a carrier wave according to the bits
- A Demodulator is a hardware circuit that
  - accepts a modulated carrier wave and
  - recreates the sequence of data bits
- A Modem is a hardware that combines both modulator and demodulator in a single device

Modem


long-distance
connection
(4 wires)
modem
modem
computer
at site 1
M
D
M
D
computer
at site 2
RS-232
can be
used
RS-232
can be
used

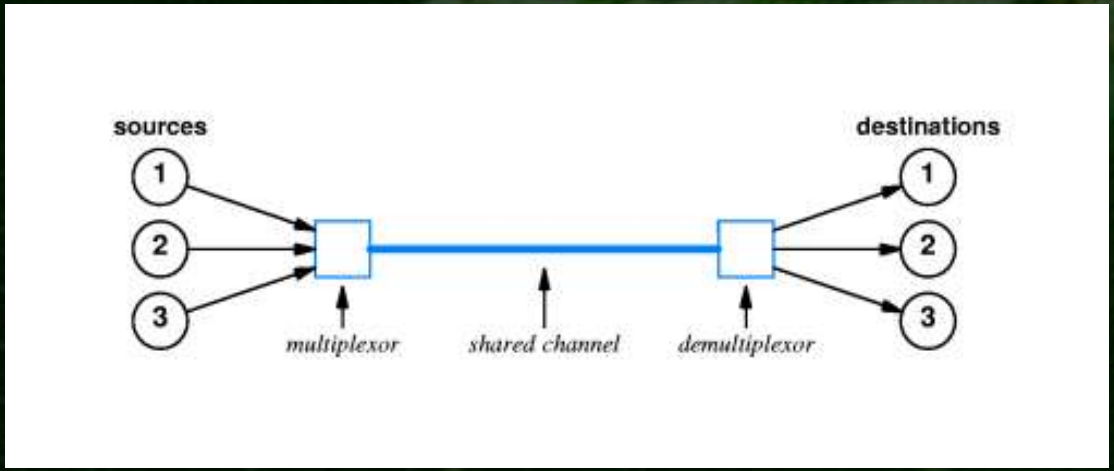
Multiplexing

- Q: How can two or more signals can be transmitted simultaneously over a single wire ?

- A: By using different carrier frequencies

Career Frequencies And Multiplexing

- Frequency division multiplexing (FDM)
  - Using multiple carrier frequencies to allow independent signals, to travel through a medium

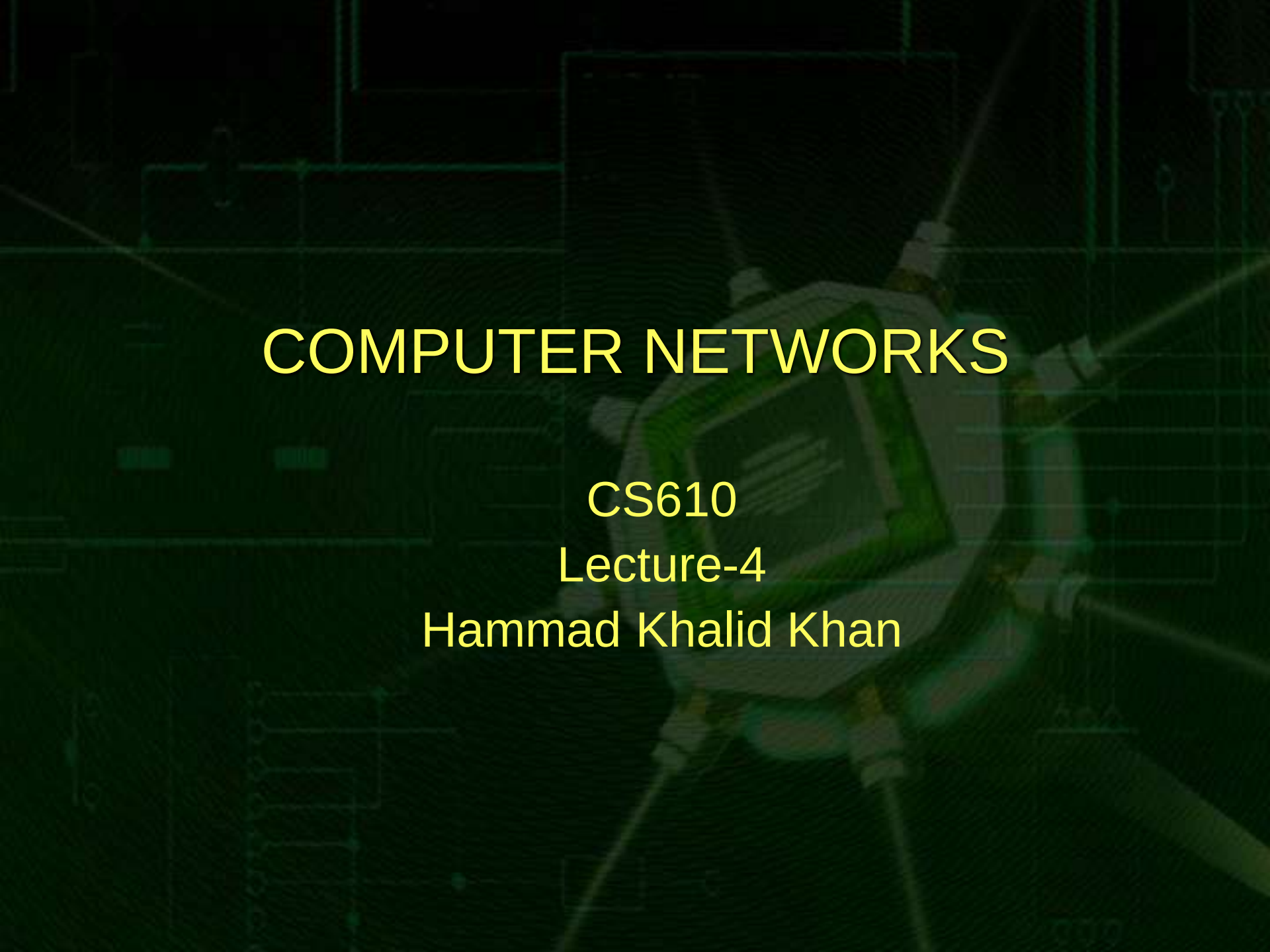


Wavelength Division Multiplexing (WDM)

- FDM concept applied to the optical transmission systems (optical fibers) is known as Wave Division Multiplexing

Time Division Multiplexing (TDM)

- Sources sharing a medium “take turns” to send data
- Synchronous TDM gives each source an opportunity to send data
- Statistical TDM does not give chance to a source that does not have data to send

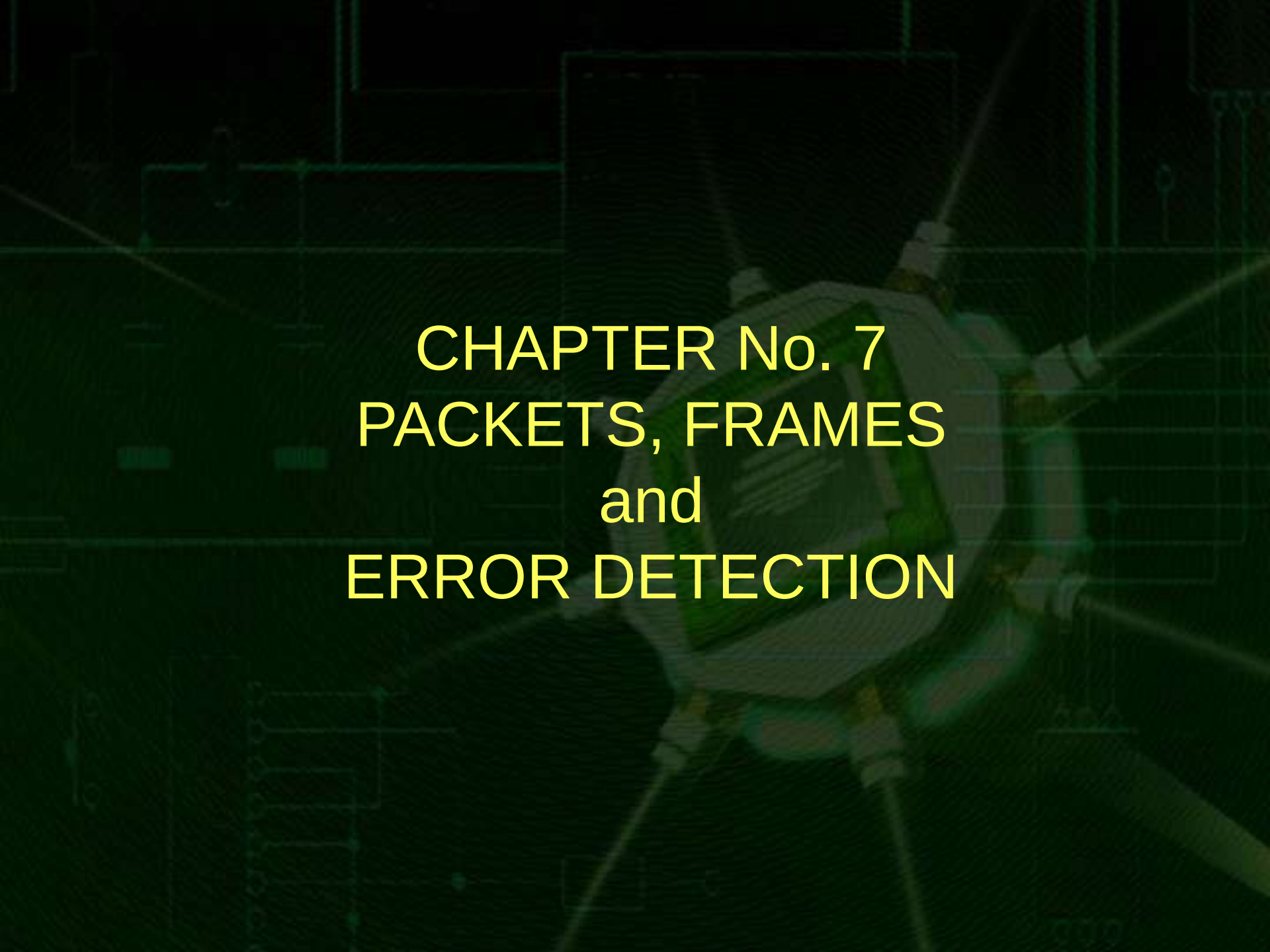


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Lecture-4

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CHAPTER No. 7
PACKETS, FRAMES
and
ERROR DETECTION

Introduction

- Previous chapters describe how bits are transmitted across a physical network
- This chapter introduces the concept of packets

The Concept Of Packets

- The problems:
  - A sender and a receiver need to coordinate to detect transmission errors
  - Individual connections between each pair of computers is not possible

The Concept Of Packets

- The solution:
  - Shared network connections among many workstations.

Reasons For Using Packets

- Resource Sharing
- To Detect The Errors

Use of Packets



Figure 7.1 An illustration of one reason computer networks use packets. While one pair of computers communicate, others must wait.

Problems with the Sharing

- Demand high – Many computers need to use the shared networks
- Some applications have large data transfers – Holds the network for a long time
- Some applications cannot wait
- Need mechanism for fairness

Solution for Fairness

- Divide the data into small blocks/chunks called “PACKETS”
- Computers take turns to send one packet at a time over the shared connection.
- Because each packet is small, no computer experiences a long delay

Example



Figure 7.1 An illustration of one reason computer networks use packets. While one pair of computers communicate, others must wait.

Example

- WITHOUT PACKETS
  - 5MB file transferred across network with 56Kbps capacity will require 12 minutes:

$$\frac{5\text{x}10^{6}\ \text{bytes} * 8\ \text{bits/byte}}{60\ \text{secs/minute} * 56\text{x}10^{3}\ \text{bits/second}} = 11.9\ \text{minutes}$$

  - All other computers will be forced to wait 12 minutes before initiating other transfers

Example

- WITH PACKETS
  - If file is broken into packets, other computers must only wait until packet (not entire file) has been sent
  - Suppose file is broken into 1000 byte packets

$$\frac{1000 \text{ bytes} * 8 \text{ bits/byte}}{56x10^3 \text{ bits/second}} = .143 \text{ seconds}$$

  - Now each packet takes less than .2 seconds to transmit

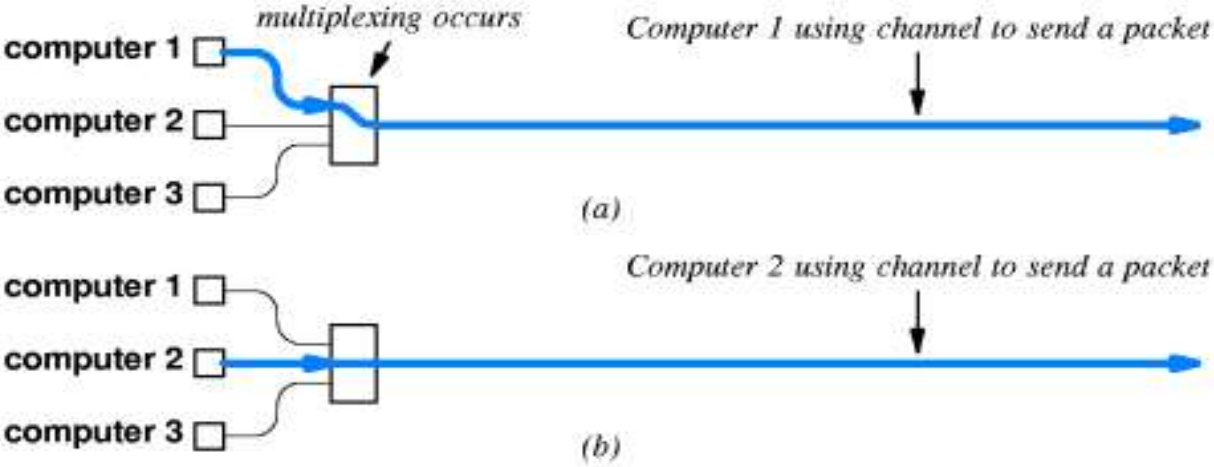
Advantage

- Other computer must only wait .143 seconds before beginning to transmit
- Note:
  - If both files are both 5MB long, each now takes 24 minutes to transmit
  - BUT if second file is only 10KB long, it will be transmitted in only 2.8 seconds, while 5MB file still takes roughly 12 minutes

Packets and TDM

- Dividing data into small packets allows Time-Division Multiplexing
- Each packet leaves the source and is switched onto the shared communication channel through a Multiplexer
- At the destination, the packet is switched through a Demultiplexer to the destination

Packets and TDM



Packets and Frames

- *Packet* is ``generic'' term that refers to a *small block of data*
- Each hardware technology uses different packet format
- *Frame* or *hardware frame* denotes a packet of a specific format on a specific hardware technology

Frame Format

- Need to define a standard format for data to indicate the beginning and end of the frame
- Header and trailer used to “frame” the data

Frame Format



Disadvantages of Framing

- Incurs extra overhead - soh and eot take time to transmit, but carry no data

Advantages

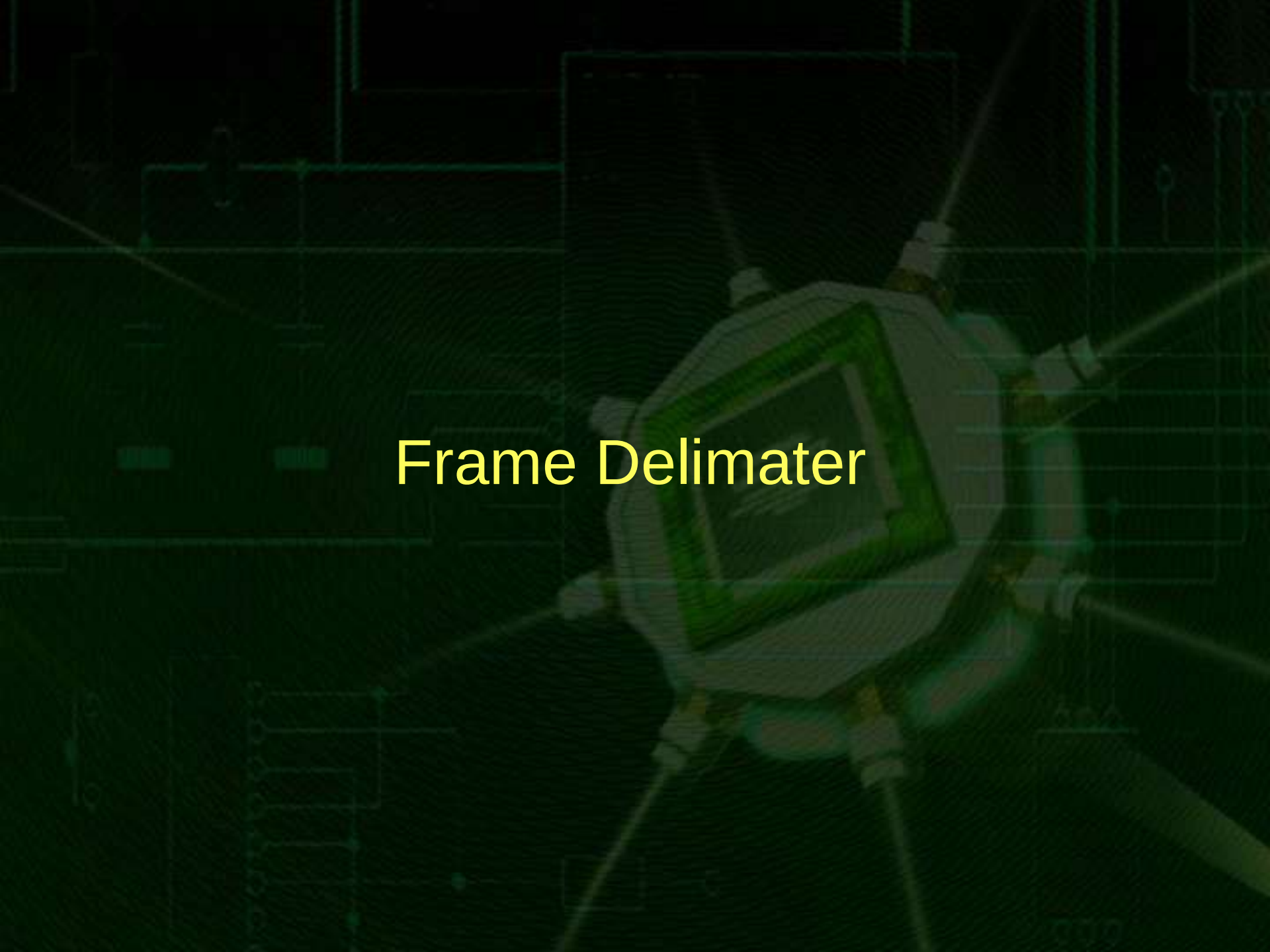
- Accommodates transmission problems:
  - Missing eot indicates sending computer crashed
  - Missing soh indicates receiving computer missed beginning of message
  - Bad frame is discarded

Data Stuffing

- Sometimes, the special characters (i.e. *soh* or *eot*) may appear in the data
- Problem: *soh* and *eot* as part of data will be misinterpreted as framing data
- Solution: *Data stuffing*

Summary

- The Concept of Packets
- Why use Packets?
- Packets and Frames
- Data Stuffing


Frame Delimater

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Review Lecture 4

- The Concept of Packets
- Why use Packets?
- Packets and Frames
- Byte Stuffing

Byte Stuffing

- Sometimes, the special characters (i.e. soh or eot) may appear in the data
- Problem: soh and eot as part of data will be misinterpreted as framing data
- Solution: Byte stuffing

Byte Stuffing (cont'd)

- Byte stuffing translates each reserved byte into two unreserved bytes
- For example, can use esc as prefix, followed by x for soh, y for eot and z for esc

Byte Stuffing (cont'd)

- The sender replaces each occurance of special characters with a combination of two characters (i.e.esc x, esc y, esc z)
- The receiver then replaces each occurence of esc x, esc y, esc z, by the corresponding single character

Byte Stuffing (cont'd)

Character In Data	Characters Sent
soh	esc x
eot	esc y
esc	esc z

Byte Stuffing (cont'd)

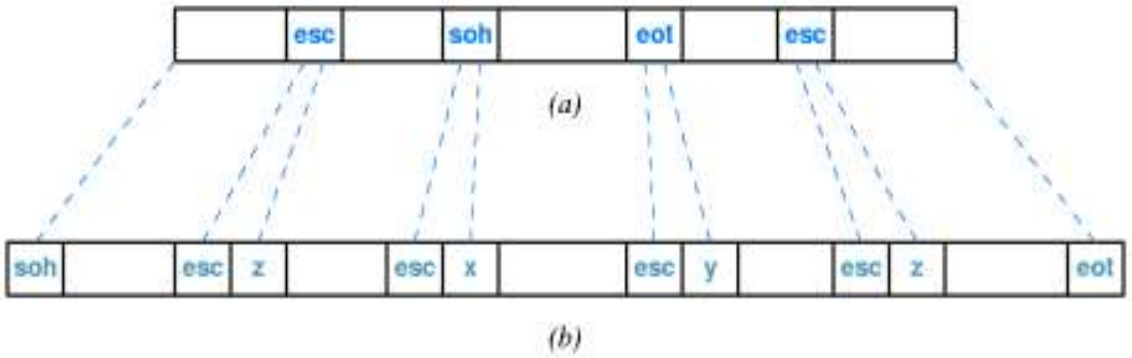


Figure 7.5 Illustration of byte stuffing, where (a) is an example of data that includes characters such as *soh*, and (b) is the frame after byte stuffing. The dashed lines show the locations in the original data where characters have been replaced or new characters added.

Transmission Errors

- Interference or power surges may destroy data during transmission
  - Bits may be lost
  - Bit values may be changed

- Frames include additional information to detect / correct errors
  - Inserted by the sender
  - Checked by the receiver

Error Detection & Correction

- Error detection - send additional information so that incorrect data can be detected and rejected
- Error correction - send additional information so that incorrect data can be corrected and accepted

Parity Checking

- Parity refers to the number of bits set to 1 in the data item

  - Even parity - An even number of bits are 1
  - Odd parity - An odd number of bits are 1

Parity Checking

- A Parity Bit is an extra bit transmitted with a data item, chose to give the resulting bits even or odd parity
  - Even parity - data: 10010001, parity bit 1
  - Odd parity - data: 10010111,
  - parity bit 0

Parity and Error Detection

- If noise or other interference introduces an error, one of the bits in the data will be changed from a 1 to a 0 or from a 0 to a 1

- Parity of resulting bits will be wrong
  - Original data and parity: 10010001+1 (even parity)
  - Incorrect data: 10110001+1 (odd parity)

Limitations of Parity Checking

- Parity can only detect errors that change an odd number of bits
  - Original data and parity: 10010001+1 (even parity)
  - Incorrect data: 10110011+1 (even parity!)
- Parity usually used to catch one-bit errors

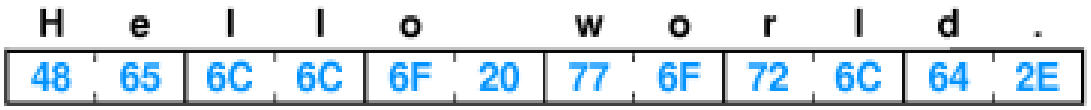
Alternative Error Detection Schemes

- Alternative error detection mechanisms have been introduced
- Differences among mechanisms:
  - The size of the additional information (transmission overhead)
  - Computational complexity of the algorithm (computational overhead)
  - The number of bit error that can be detected (how well errors are detected)

Checksum

- Checksum is calculated before transmission
  - Treats data as a sequence of integers
  - Computes their arithmetic sum
  - Carry bits are added into the final sum
- Calculated checksum is sent along with data
- Receiver then performs the same calculation

Checksum



4865 + 6C6C + 6F20 + 776F + 726C + 642E + carry = 71FC

Checksum

- Can be 8-, 16- or 32-bit integers
- Easy to do - uses only addition
- May not detect all errors

Data Item In Binary	Checksum Value	Data Item In Binary	Checksum Value
0001	1	0011	3
0010	2	0000	0
0011	3	0001	1
0001	1	0011	3
totals	7		7

Cyclic Redundancy Check (CRC)

- Problem : How can a network system detect more errors without increasing the amount of information in each packet?
- Answer: Cyclic Redundancy Check (CRC)

Cyclic Redundancy Check (CRC)

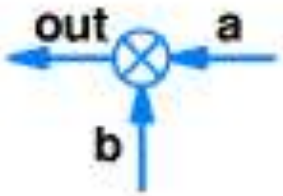
- Consider data in message as coefficients of a polynomial
- Divide that coefficient set by a known polynomial
- Transmit remainder as CRC
  - Good error detection properties
  - Easy to implement in hardware

Hardware Components

- Uses two hardware components:
  - Shift Register
  - Exclusive or (xor) unit

Hardware Components

- Exclusive OR (XOR) unit



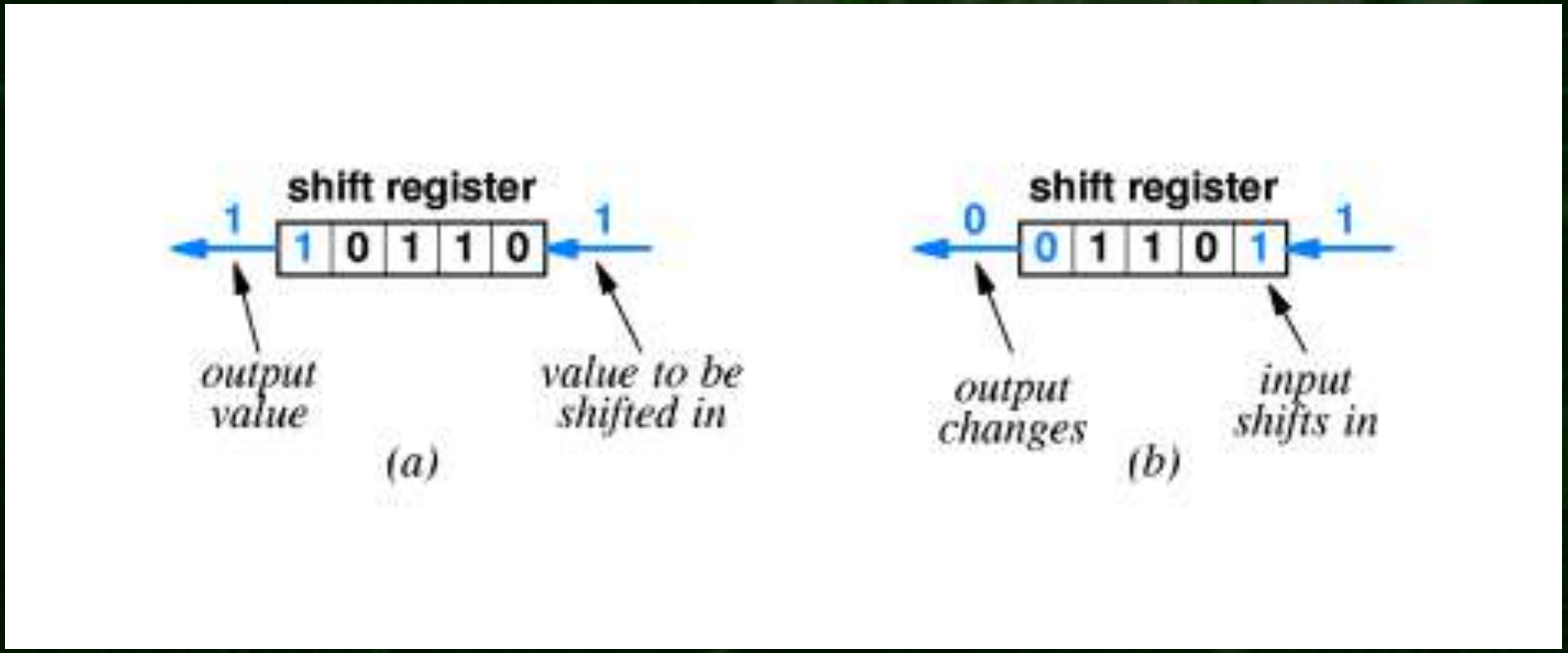
(a)

a	b	out
0	0	0
0	1	1
1	0	1
1	1	0

(b)

Hardware Components

- Shift Register: Performs two operations:
  - Initialize : Sets all bits to zero
  - Shift: Moves all bits to the left one position



SUMMARY

- Byte Stuffing
- Transmission Errors
- Error Detection
  - Parity Check
  - Checksum
  - CRC

COMPUTER NETWORKS

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Lecture-6

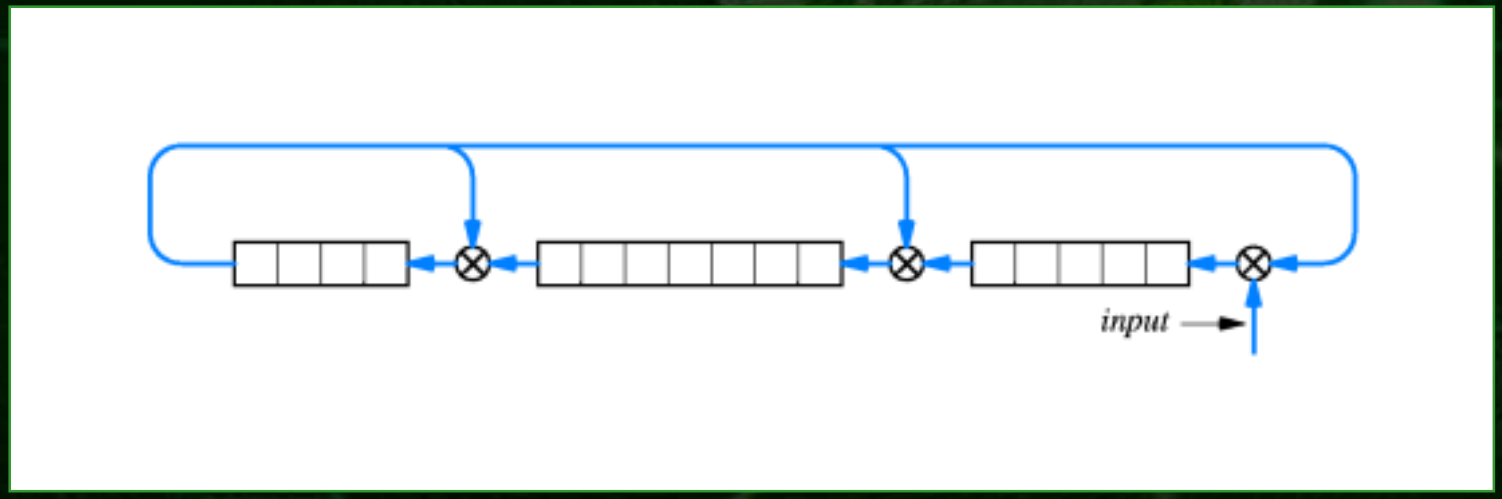
Hammad Khalid Khan

Review of lecture 5

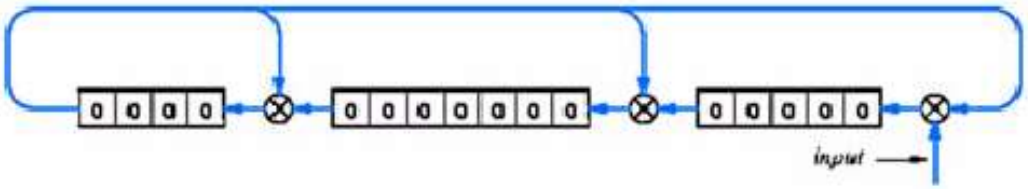
- Byte Stuffing
- Transmission Errors
- Error Detection
  - Parity Check
  - Checksum
  - CRC

Chapter 7 (cont'd) Combining Building Blocks

- Example 16-bit CRC hardware that uses three shift registers and three exclusive or (xor) units



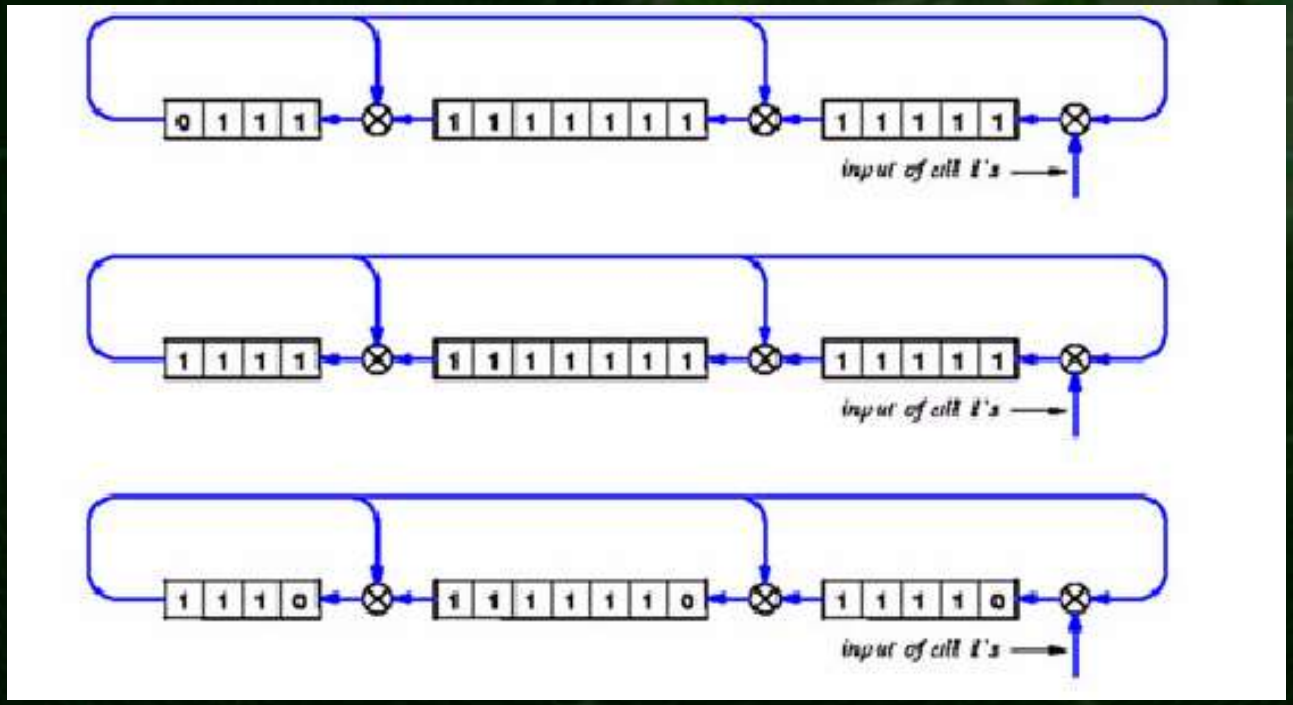
Combining Building Blocks (cont'd)



- Computes 16-bit CRC
- Registers initialized to zero
- Bits of message shifted in
- CRC found in registers

Combining Building Blocks (cont'd)

-Input data is all 1s
-CRC shown after 15,16,17 bits shifted
-Feedback introduces 0s in CRC




DIRECT POINT-TO-POINT
COMMUNICATION

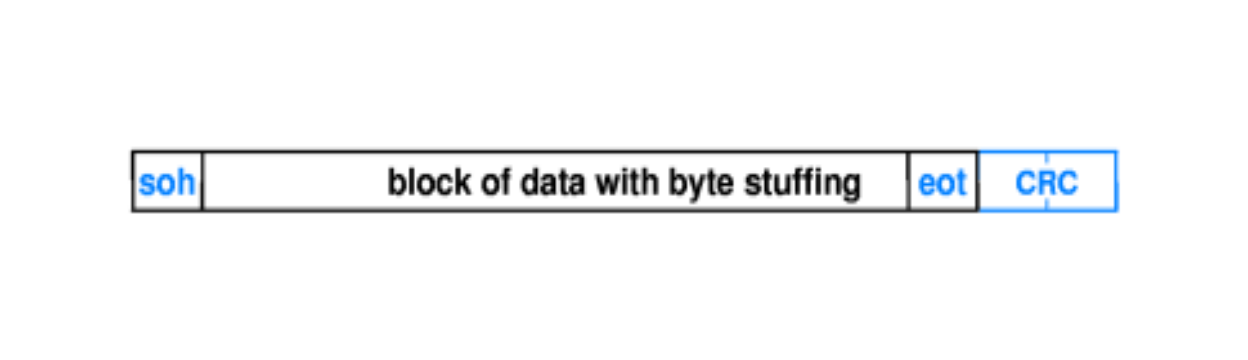
Types of Errors

- CRCs can detect the following errors better than checksums
  - Vertical errors: Occur due to a hardware failure e.g. Damaged second bit of every character
  - Burst errors: Changing a small set of bits near a specific location, caused by lightening, electric motor start, etc.

Frame Format And Error Detection

- Modified frame format including CRC
- Error in frame typically causes receiver to discard frame
- Is Byte stuffing also required for CRC itself?

Frame Format And Error Detection



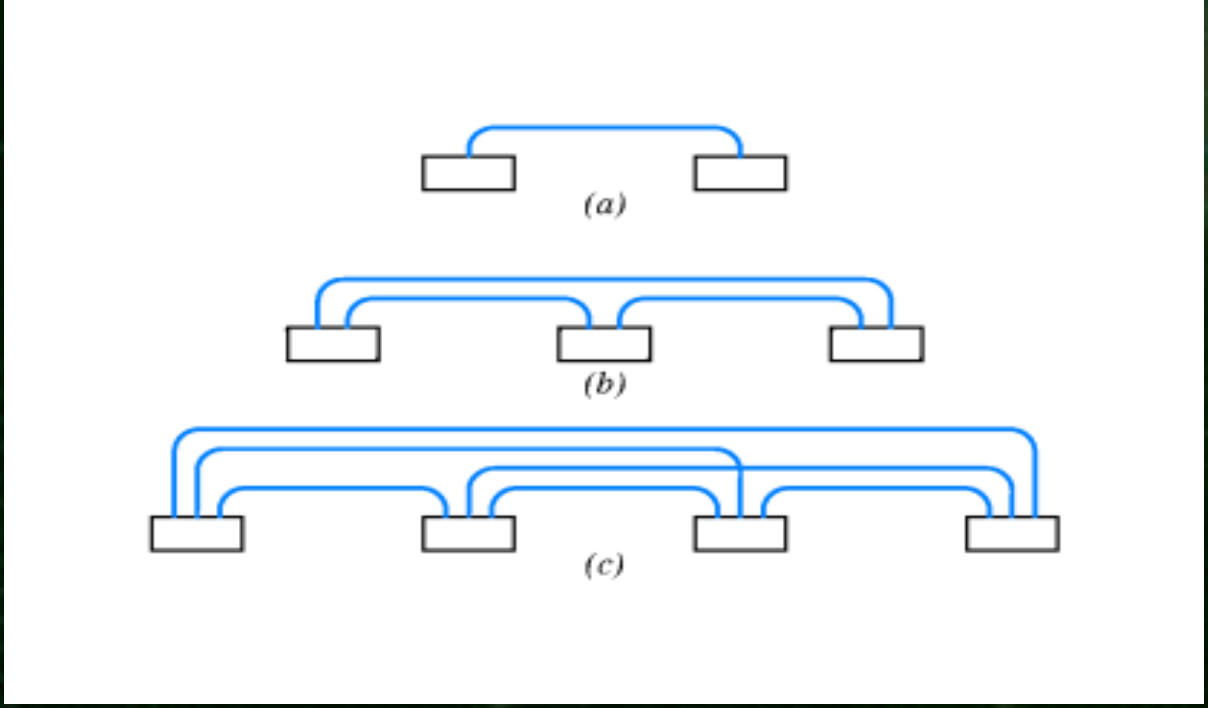
CHAPTER No. 8
LAN TECHNOLOGIES
and
NETWORK TOPOLOGIES

Introduction

- Most networks are Local and are designed to share resources among multiple computers
- Hardware technologies used for Local networks allow multiple devices to connect with a shared network
- The computers must take turns using the shared medium

Direct Point-to-Point Communication

- Each communication channel connects exactly two computers
- Form a MESH or a POINT-to-POINT network



Direct Point-to-Point Communication

- Advantages:
  - The connection type of individual connections can be different
  - Individual connection can choose a different frame format, error detection mechanisms, etc
  - Easy to enforce Security and Privacy

Direct Point-to-Point Communication

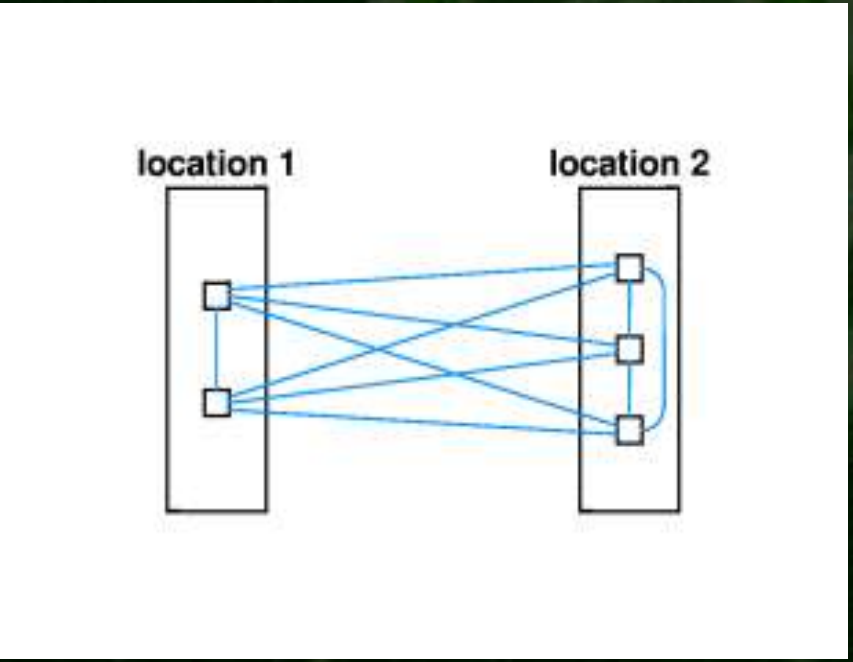
- Disadvantages:
  - The number of connections grow more rapidly than the number of computers
    - For ‘N’ computers:

$$\text{Connections} = \frac{N^2 - N}{2}$$

  - Most connections use the same physical path
  - Expensive

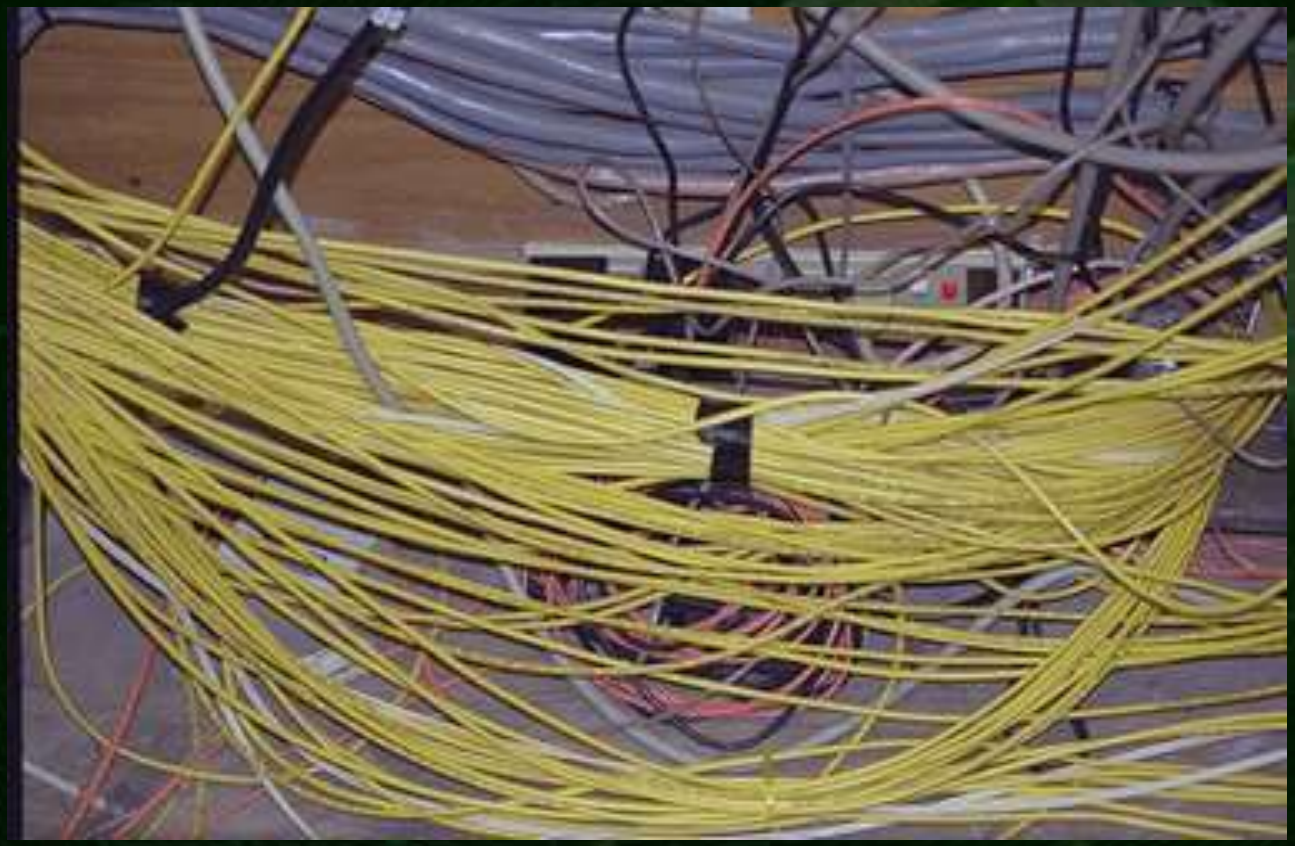
Direct Point-to-Point Communication

- Disadvantages:
  - Adding a new computer to the network requires ‘N-1’ new connections



Star Topology in Practice

- The Result is:



Summary

- CRC Hardware Components
- Frame Format and CRC
- LAN technology and Network topology (Chapter 8)
- Direct Point-to-Point Communication

COMPUTER NETWORKS

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Lecture-7

Hammad Khalid Khan

Review Lecture 6

- CRC Hardware Components
- Frame Format and Error Detection
- LAN technology and Network Topology (Chapter 8)
- Direct Point-to-Point Communication

Shared Communication Channels

- LANs developed in late 1960s and early 1970s
- Key idea
  - Reduce number of connections by sharing connections among many computers

Shared Communication Channels

- Each LAN consists of a single shared medium
- The computers take turns using the medium
- Sharing a single medium over long distances is inefficient, due to the long delays

Growth of LAN Technologies

- LAN technologies reduce cost by reducing number of connections
- But ... attached computers compete for use of shared connection
- Local communication almost exclusively LAN
- Long distance almost exclusively point-to-point

Significance Of LANs And Locality Of Reference

- LANs are most popular form of computer networks
- LAN technologies are inexpensive
- The demand for LANs is related to the “Locality of Reference” principle

Locality Of Reference Principle

- Principle of Locality of Reference helps predict computer communication patterns:
  - Spatial (or physical) locality of reference
    - Computers are likely to communicate with other computers that are located nearby
  - Temporal locality of reference
    - Computers are likely to communicate with the same computers repeatedly

Locality Of Reference Principle

- Thus
  - LANs are effective because of spatial locality of reference,

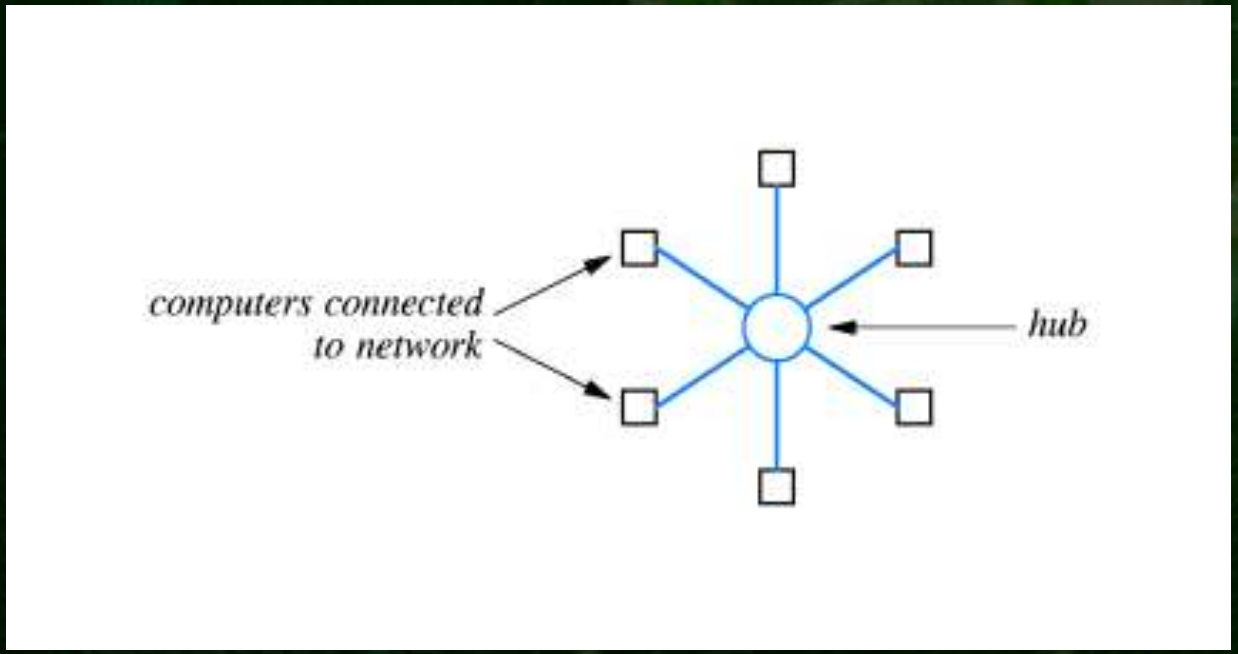
    and
  - Temporal locality of reference may give insight into which computers should be on a LAN

LAN Topologies

- Networks may be classified by shape
- Three most popular:
  - Star
  - Ring
  - Bus

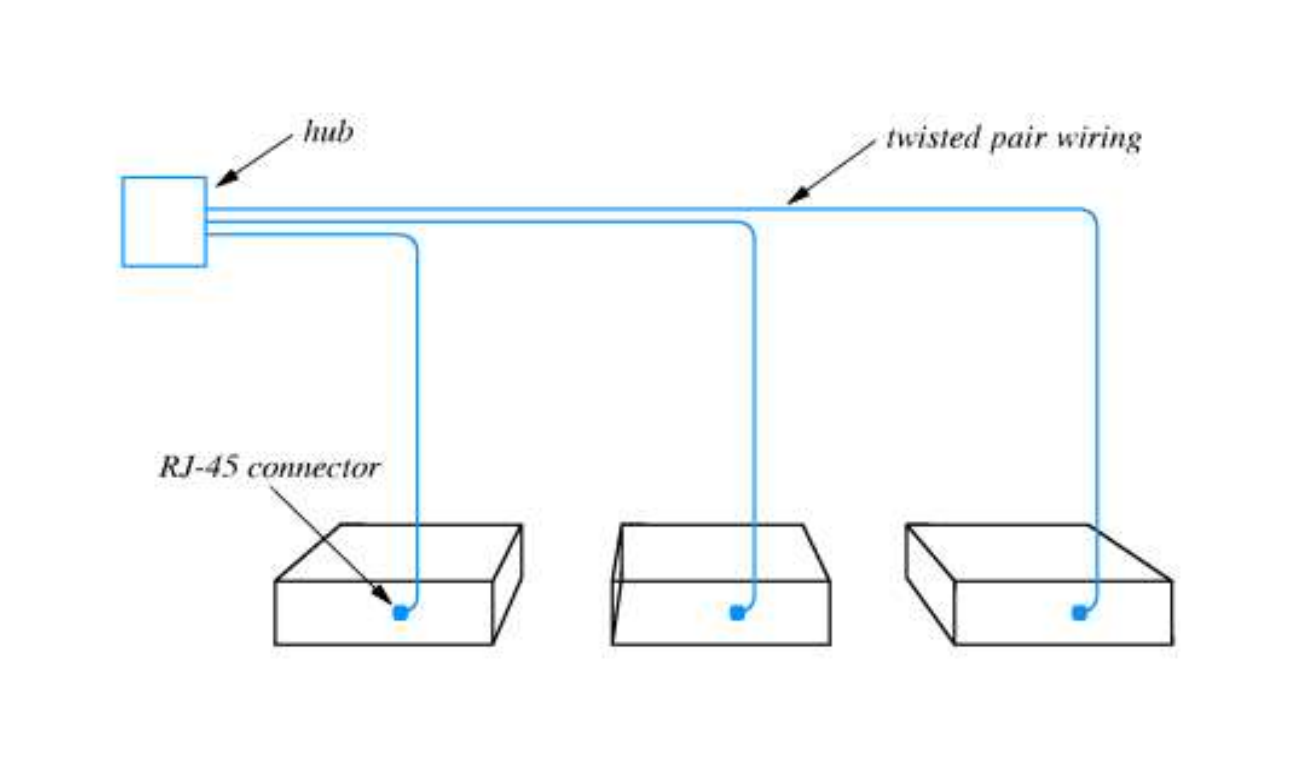
Star Topology

- All computers are attached to a central point
- Center of star is sometimes called a “Hub”
- Logical versus Physical Star



Star Topology in Practice

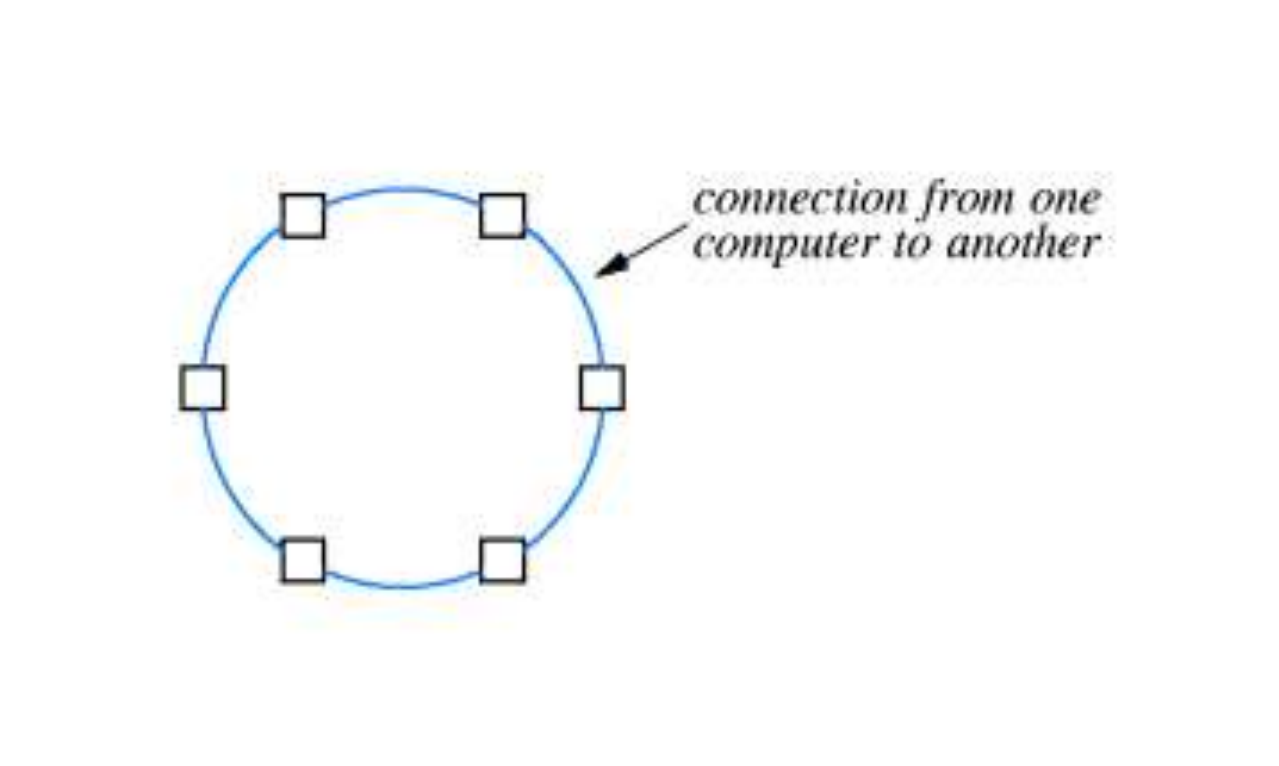
- Previous diagram is idealized



Ring Topology

- Computers connected in a closed loop
- First passes data to second, second passes data to third, and so on
- Logical versus Physical Ring

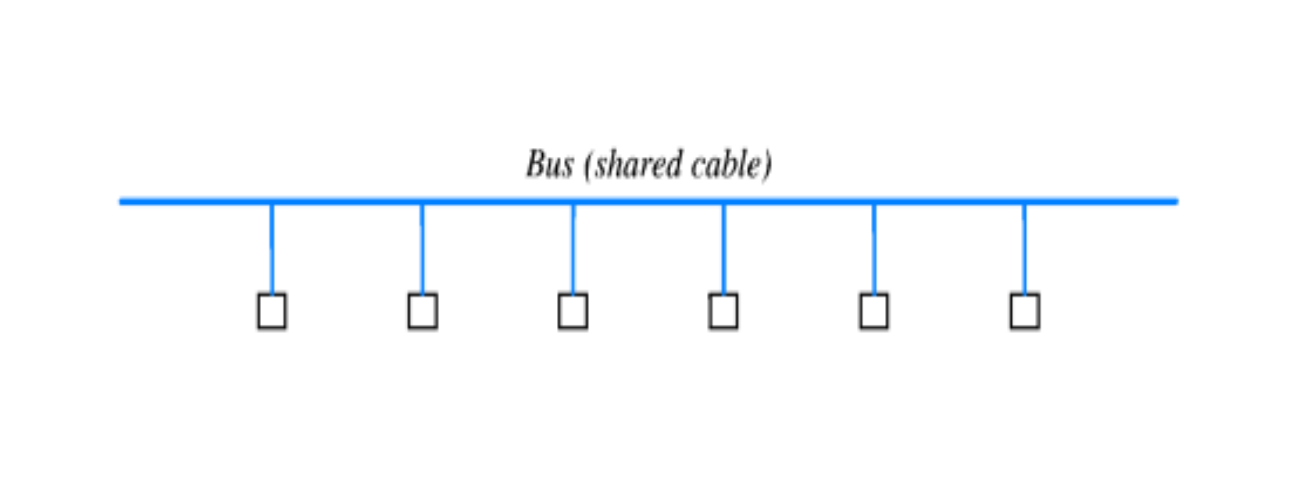
Ring Topology



Bus Topology

- All computers are attached to a single long cable
- Any computer can send data to any other computer
- Coordination required to decide which computer uses the line at what time

Bus Topology



Reason for Multiple Topologies

- Each topology has advantages and disadvantages

  - In a Ring, it is easy to coordinate access, however entire network is disabled if a cable cut occurs
  - In a Star, only one computer is affected, when a cable cut occurs
  - In a Bus, needs fewer wires than a star, however entire network is disabled when a cable cut occurs

Example Bus Network: Ethernet

- Widely used LAN technology
  - Invented at Xerox PARC (Palo Alto Research Center) in 1970s
  - Defined in a standard by Xerox, Intel and Digital Equipment Corporation - *DIX* standard
  - Standard now managed by IEEE (802.3) defines formats, voltages, cable lengths, ...

Example Bus Network: Ethernet

- Uses Bus Topology
  - Single coax cable - the Ether
  - Multiple computers connect

- One Ethernet cable is sometimes called a Segment
  - Limited to 500 meters in length
  - Minimum separation between connections is 3 meters

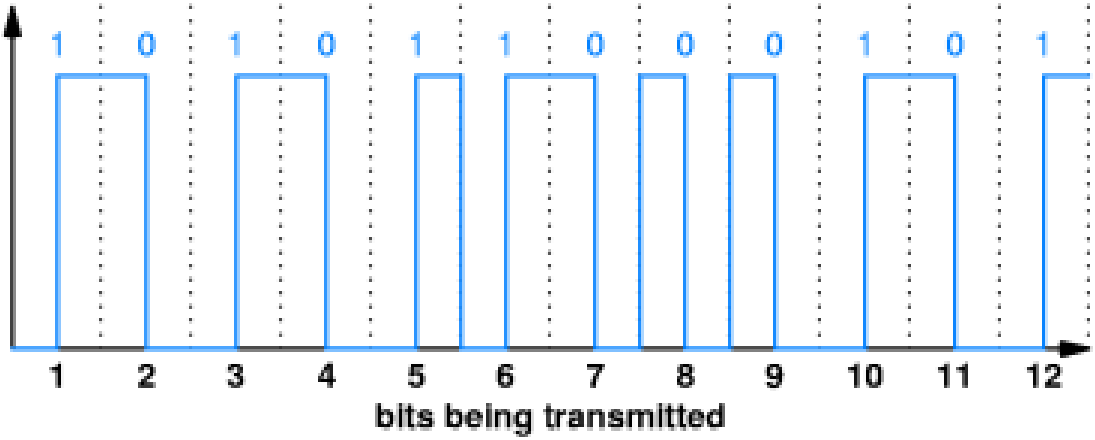
Ethernet Speeds

- Originally 3Mbps
- Current standard is 10Mbps
- Fast Ethernet operates at 100Mbps
- Giga-Bit Ethernet

Encoding used in Ethernet

- Manchester Encoding: Uses signal changes to encode data
  - A change from positive voltage to 0 encodes a 0 bit
  - A change from 0 voltage to positive voltage encodes a 1 bit

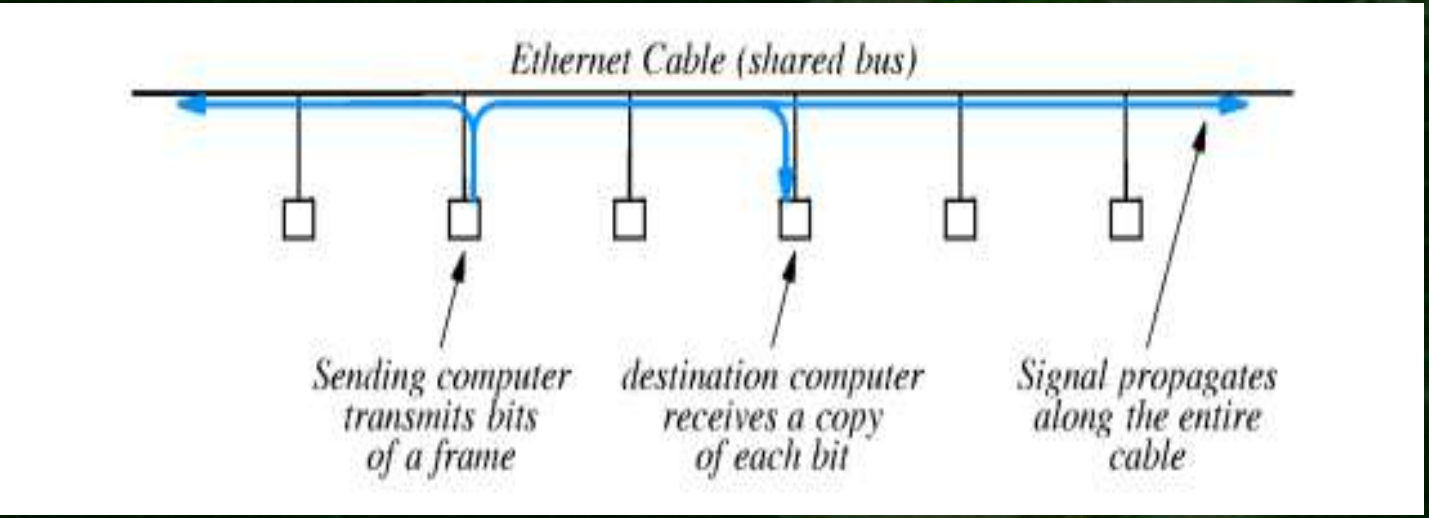
Encoding used in Ethernet



Ethernet Operation

- One computer transmits at a time
- Signal propagates from transmitter in both directions along length of segment

Ethernet Operation



Carrier Sense Multiple Access Networks (CSMA)

- No central control management when computers transmit on Ether
- Ethernet employs CSMA to coordinate transmission among multiple attached computers

Summary

- Shared Communication Channel
- Locality of Reference Principle
- LAN Topologies
  - Star
  - Ring
  - Bus
- Ethernet

COMPUTER NETWORKS

CS610

Lecture-8

Hammad Khalid Khan

Review Lecture 7

- Shared Communication Channel
- Locality of Reference Principle
- LAN Topologies
  - Star
  - Ring
  - Bus
- Ethernet

Carrier Sense Multiple Access

- A coordination scheme that defines how to take turns using a shared cable

- A computer listens to the cable (carrier sense)
  - If the cable is idle then starts transmitting
  - If the cable is in use, then waits

- If Simultaneous transmissions occur, the frames interfere with each other called a Collision

Collision Detection - CD

- Signals from two computers will interfere with each other
- Overlapping frames is called a Collision
  - No harm to hardware
  - Data from both frames is Garbled

Ethernet CD

- Ethernet interfaces include hardware to detect transmission
  - Monitor outgoing signal
  - Garbled signal is interpreted as a collision
- After collision is detected, computer stops transmitting
- So, Ethernet uses CSMA/CD to coordinate transmissions

Recovery from Collision

- Computer that detects a collision sends special signal to force all other interfaces to detect collision
- Computer then waits for ether to be idle before transmitting
  - If both computers wait same length of time, frames will collide again
  - Standard specifies maximum delay, and both computers choose random delay less than maximum

Recovery from Collision

- After waiting, computers use carrier sense to avoid subsequent collision
  - Computer with shorter delay will go first
  - Other computers may transmit later

Exponential Backoff

- Even with random delays, collisions may occur
- Especially likely with busy segments
- Computers double delay with each subsequent collision
- Reduces likelihood of sequence of collisions

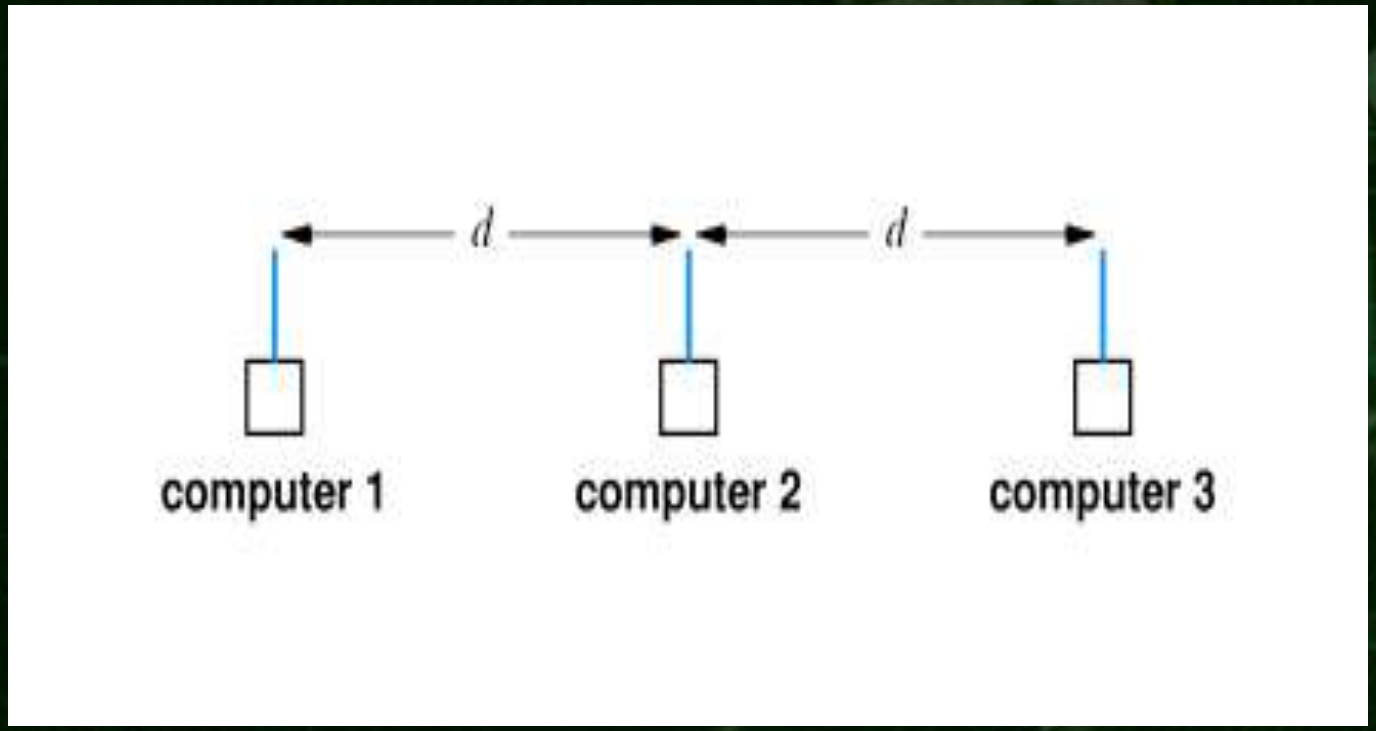
802.11 Wireless LANs And CSMA/CA

- Use radio signals at 2.4 GHz (IEEE 802.11)
- Data rate at 11Mbps
- Older devices use radio signals at 900MHz
- Data rate of 2Mbps
- Bluetooth specifies a wireless LAN for short distances
- Shared medium - radio instead of coax

Limited Connectivity with Wireless

- In contrast with wired LAN, not all participants may be able to reach each other
  - Low signal strength
  - Propagation blocked by walls, etc.
- Can't depend on CD; not all participants may hear

Limited Connectivity with Wireless



CSMA/CA

- Wireless uses Collision Avoidance rather than Collision Detection
  - Transmitting computer sends very short message to receiver
  - Receiver responds with short message reserving slot for transmitter
- Response from receiver is broadcast, so all potential transmitters receive reservation

Collisions

- Receiver may receive simultaneous requests
  - Results in collision at receiver
  - Both requests are lost
  - Neither transmitter receives reservation; both use backoff and retry
- Receiver may receive closely spaced requests
  - Selects one
  - Selected transmitter sends message
  - Transmitter not selected uses backoff and retries

Limited Connectivity with Wireless



Local Talk

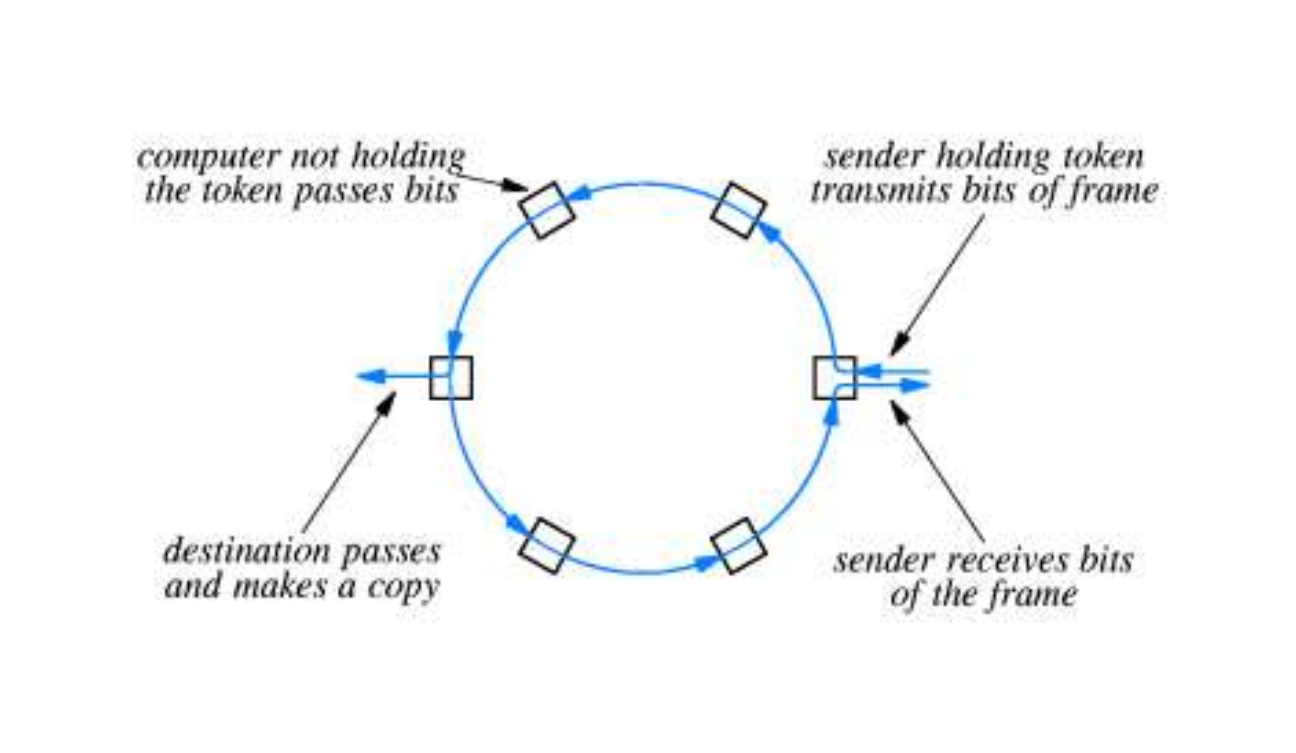
- LAN technology invented by Apple that uses bus topology
- Interface included with all Macintosh computers
- Relatively low speed - 230.4Kbps
- Low cost (``free'' with a Macintosh); easy to install and connect
- Uses CSMA/CA

Token Ring

- Many LAN technologies that use ring topology use token passing for synchronized access to the ring
- Ring itself is treated as a single, shared communication medium
- Bits pass from transmitter, past other computers and are copied by destination

Token Ring

- Hardware must be designed to pass token even if attached computer is powered down



Using the token

- When a computer wants to transmit, it waits for the token
- After transmission, computer transmits token on ring
-
- Next computer ready to transmit receives token and then transmits

Token and Synchronization

- Because there is only one token, only one computer will transmit at a time
  - Token is short, reserved frame that cannot appear in data
  - Hardware must regenerate token if lost

Token and Synchronization

- Token gives computer permission to send one frame
  - If all ready to transmit, enforces ``Round-Robin'' access
  - If none ready to transmit, token circulates around ring

IBM Token Ring

- Very widely used
- Originally 4mbps, now 16Mbps
- Uses special connector cable between computer and ring interface

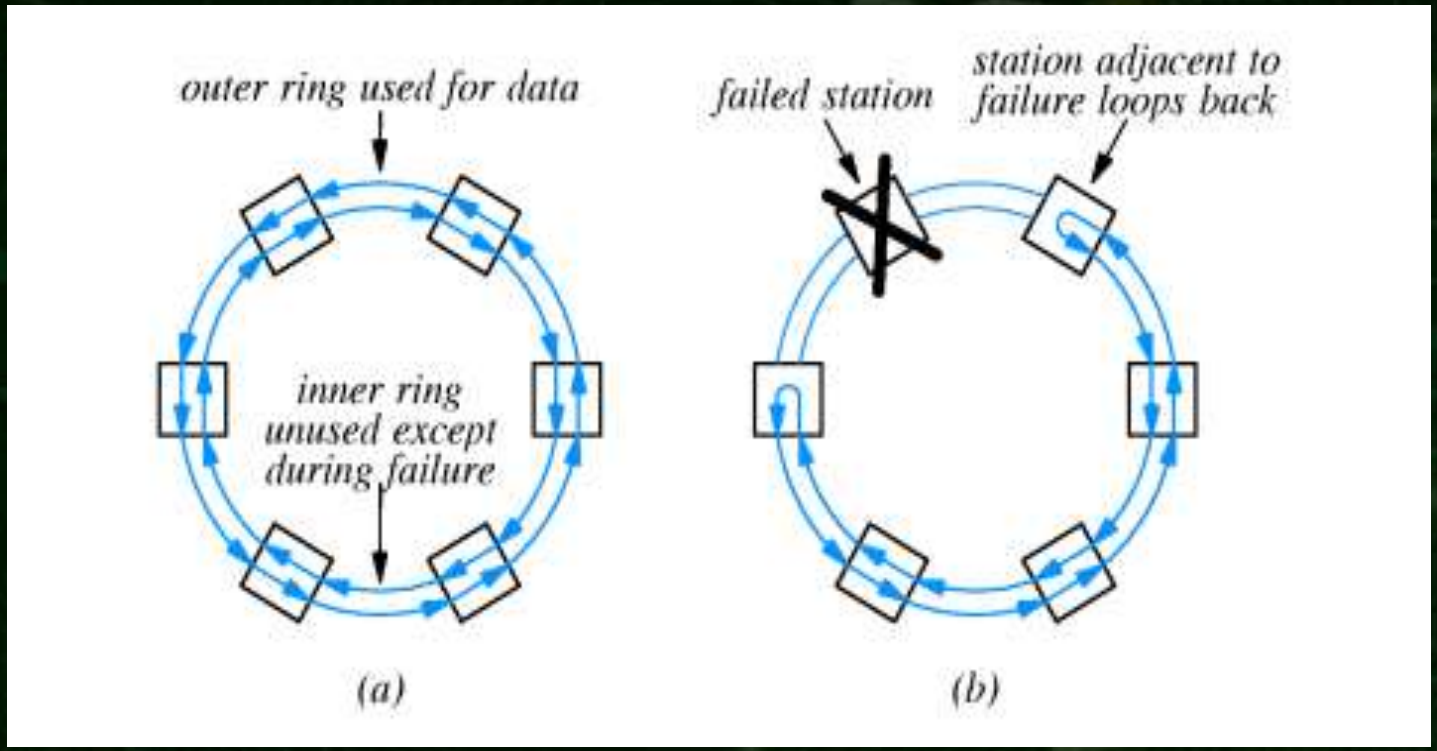
FDDI

- Fiber Distributed Data Interconnect (FDDI) is another ring technology
  - Uses fiber optics between stations
  - Transmits data at 100Mbps
- Uses pairs of fibers to form two concentric rings

FDDI and Reliability

- FDDI uses counter-rotating rings in which data flows in opposite directions
-
- In case of fiber or station failure, remaining stations loop back and reroute data through spare ring
- All stations automatically configure loop back by monitoring data ring

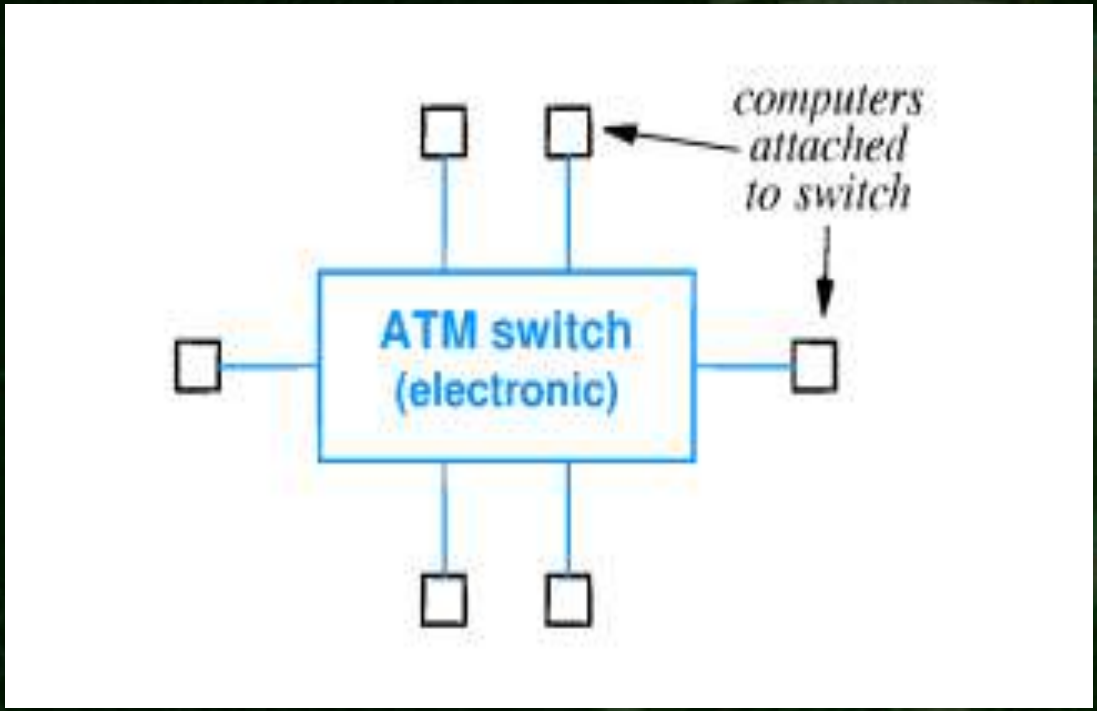
FDDI and Reliability



ATM – Star Network

- Asynchronous Transfer Mode technology consists of electronic packet switches to which computers can connect
- ATM switches form hub into which computers connect in a star topology
- Computers get point-to-point connections - data from transmitter is routed directly through hub switches to destination
  - If none ready to transmit, token circulates around ring
  - If all ready to transmit, enforces ``Round-Robin'' access

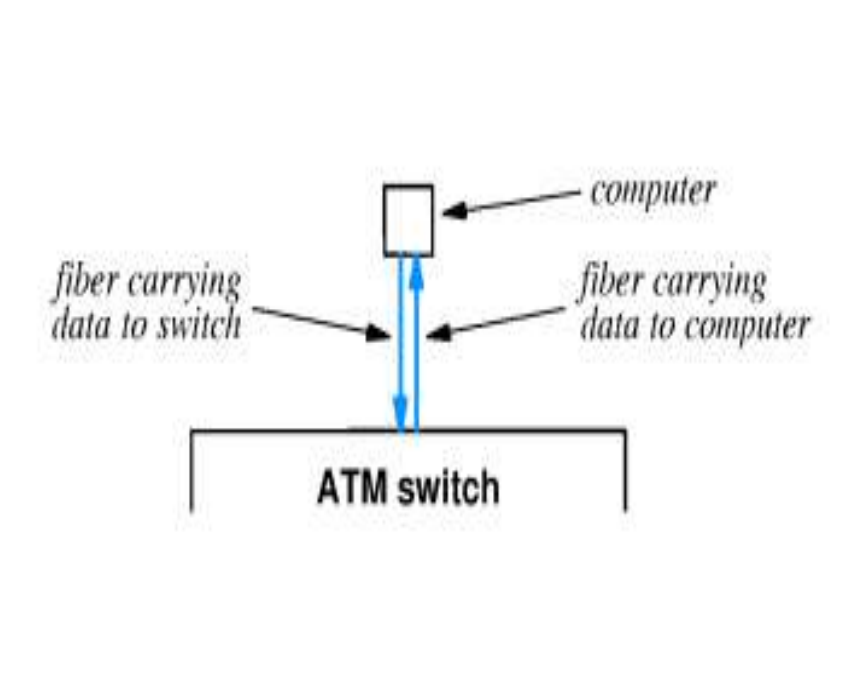
ATM – Star Network



ATM Details

- Transmits data at over 100Mbps
- Uses fiber optics to connect computer to switch
- Each connection includes two fibers

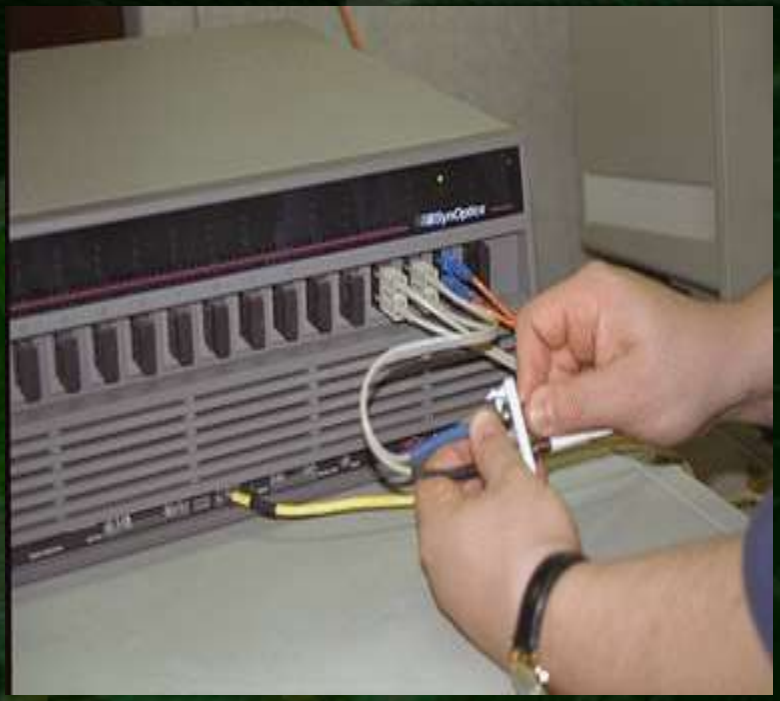
ATM Details


computer
fiber carrying
data to switch
fiber carrying
data to computer
ATM switch



ATM Switches





Summary

- CSMA/CD
- Wireless LANs and CSMA/CA
- Token Ring
- FDDI
- ATM

COMPUTER NETWORKS

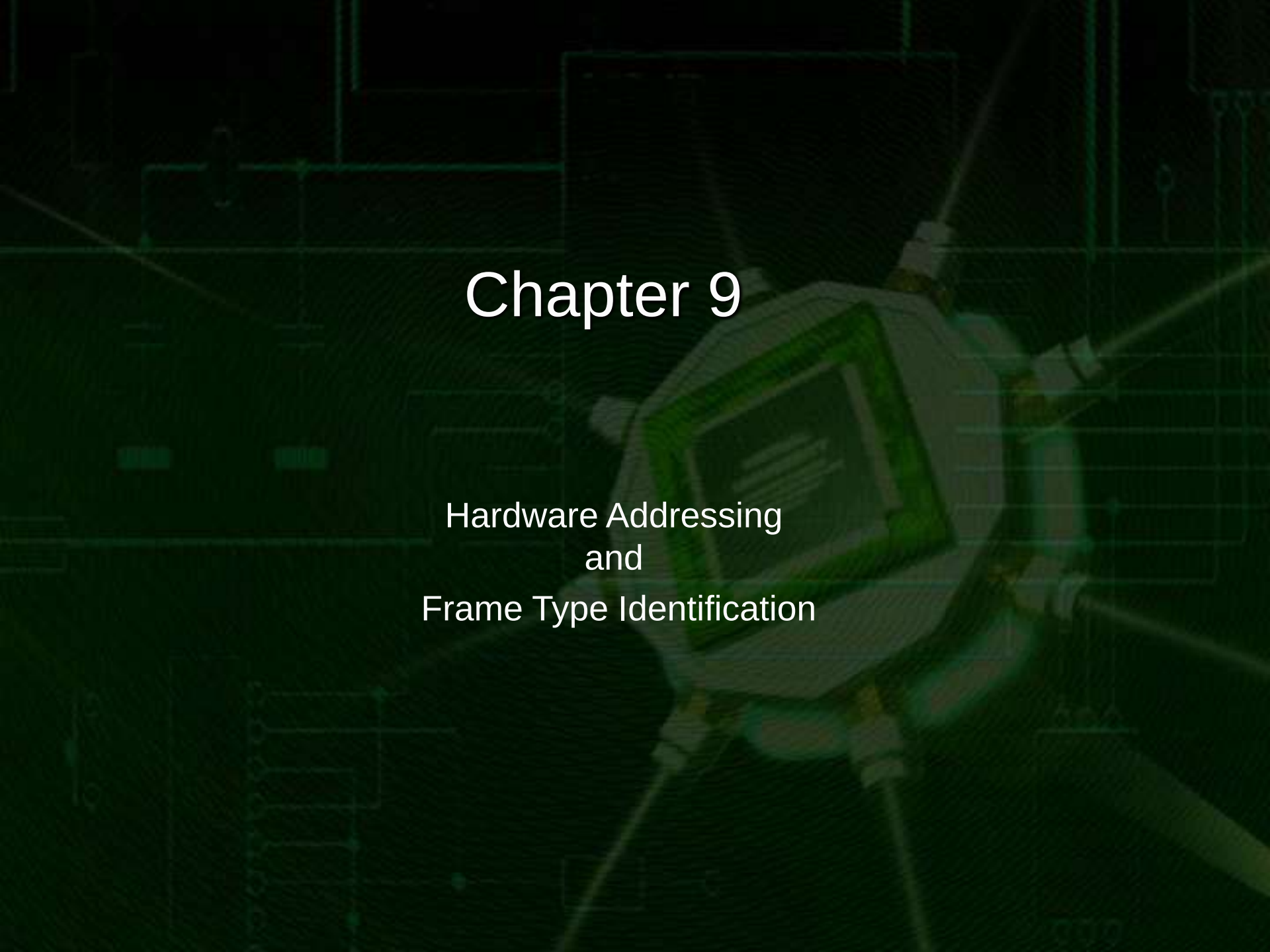
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Lecture-9

Hammad Khalid Khan

Review Lecture 8

- CSMA/CD
- Wireless LANs and CSMA/CA
- Token Ring
- FDDI
- ATM



Chapter 9

Hardware Addressing
and
Frame Type Identification

Introduction

- Need to devise technique for delivering message through LAN medium to single, specific destination computer
-
- Sending computer uses a Hardware Address to identify the intended destination of a frame
-
- Sending computer also identifies Type of data carried in the frame

Specifying a Destination

- Data sent across a shared network reaches all attached stations - for all LAN topologies
- Interface hardware detects delivery of frame and extracts frame from medium
- But ... most applications want data to be delivered to One specific application on another computer - not All computers

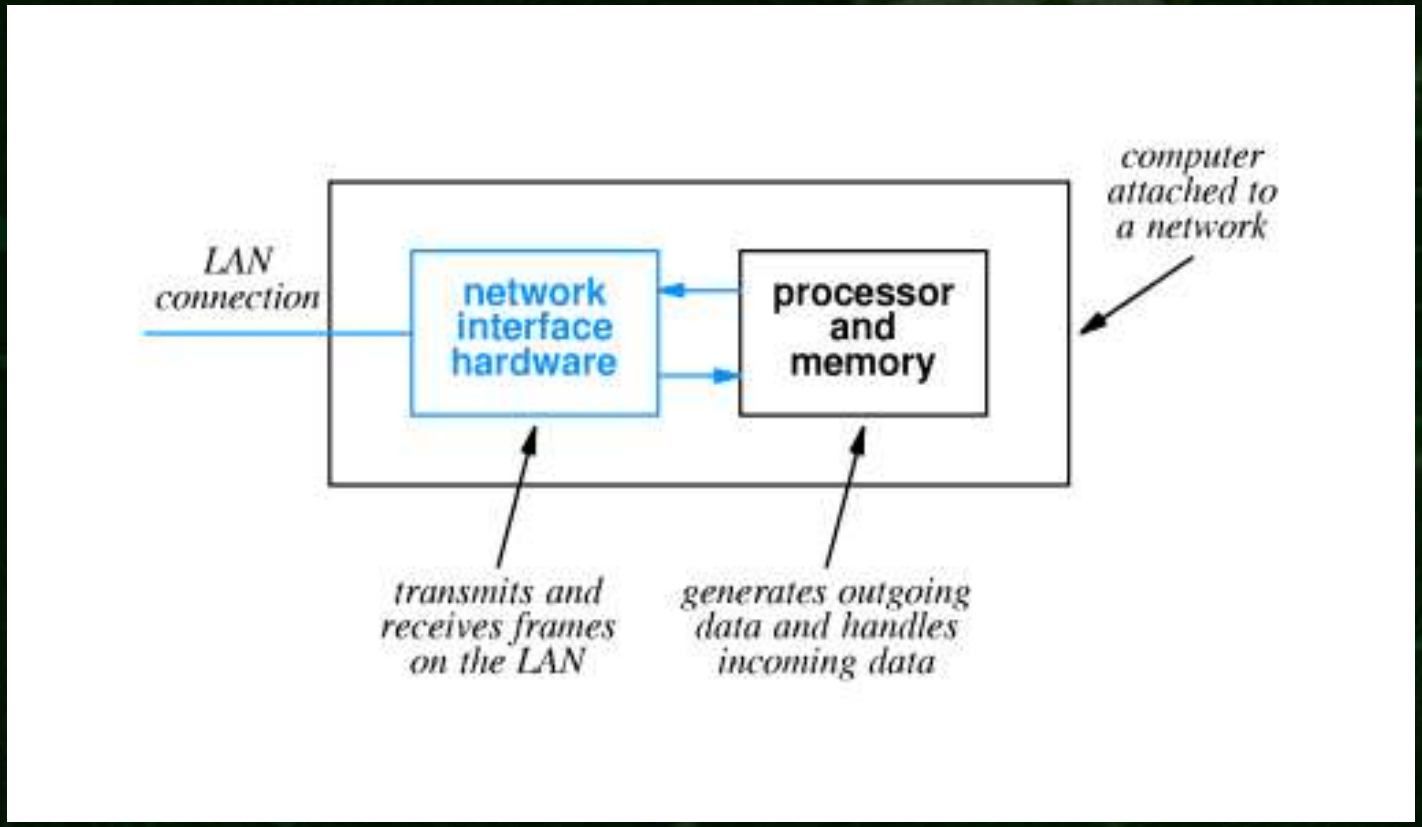
Hardware Addressing

- Most network technologies have a hardware addressing scheme that identifies stations on the network
- Each station is assigned a numeric Hardware Address or Physical Address

Hardware Addressing

- Sender includes hardware address in each transmitted frame
- Only station identified in frame receives copy of frame
- Most LAN technologies include sender's hardware address in frame, too

LAN Hardware and Packet Filtering



LAN Interface

- LAN interface handles all details of frame transmission and reception
  - Adds hardware addresses, error detection codes, etc. to outgoing frames
  -
  - May use DMA to copy frame data directly from main memory
  -
  - Obeys access rules (e.g., CSMA/CD) when transmitting

LAN Interface

- Checks error detection codes on incoming frames
- May use DMA to copy data directly into main memory
- Checks destination address on incoming frames
  - Station's address, a copy of the frame is passed to the attached computer
- Frames not addressed to the local computer are ignored and don't affect the local computer in any way

Format of Hardware Addresses

- Numeric value
- Size selected for specific network technology
- Length is one to six bytes

Assigning Hardware Addresses

- Hardware addresses must be unique on a LAN
- How can those address be assigned and who is responsible for uniqueness?

Assigning Hardware Addresses

Static	Hardwar e manufacturer assigns permanent physical address to each network interface .	Manufacturer must ensure every interface has a unique address
Configurable	Address can be set by end user, either manually (e.g., switches or jumpers on the inte rface) or electronically (e.g., through software	System administrators must coordinate to avoid conflict
Dynamic	Interfac e automatically assigns physical address each time it is powered up	Automatic scheme must be reliable to prevent conflicts


Broadcasting

Problems with Broadcasting

- For every broadcast frame on the network, each computer uses computational resources:
  - Places contents into memory
  - Interrupts the CPU
  - Allows system software to make a decision whether to discard or use the frame

Problems with Broadcasting

- If a pair of computers, use broadcasting, instead of sending them directly, all other computers waste CPU time while discarding the frames.

Multicasting

- Solution: The restricted form of broadcasting: Multicasting:
  - It works like broadcasting
  - However it does not forward frames automatically to the CPU
  - The interface hardware is programmed in advance to accept certain frames that have multicast address as the destination address

Summary

- Hardware Addressing
- Specifying a Destination
- Packet Filtering
- Assigning Hardware Addresses
- Broadcasting
- Multicasting

COMPUTER NETWORKS

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Lecture-10

Hammad Khalid Khan

Review Lecture 9

- Hardware Addressing
- Specifying a Destination
- Packet Filtering
- Assigning Hardware Addresses
- Broadcasting
- Multicasting

Problems with Broadcasting

- For every broadcast frame on the network, each computer uses computational resources:
  - Places contents into memory.
  - Interrupts the CPU.
  - Allows system software to make a decision whether to discard or use the frame.

Problems with Broadcasting

- If a pair of computers, use broadcasting, instead of sending them directly, all other computers waste CPU time while discarding the frames.


Multicasting

Multicast Addressing

- If an application program wishes to receive certain frames, then
  - it programs the interface hardware to accept an additional set of addresses.
- The interface hardware then begins accepting:
  - those frames (multicast frames),
  - broadcast frames and
  - frames that are destined to the station itself.

Multicast Addressing

- For example: Computers running an audio application
  - can receive audio frames, if the interfaces are programmed to receive them.
  - The other computers that are not running that audio application will not waste resources.

Identifying Packet Contents

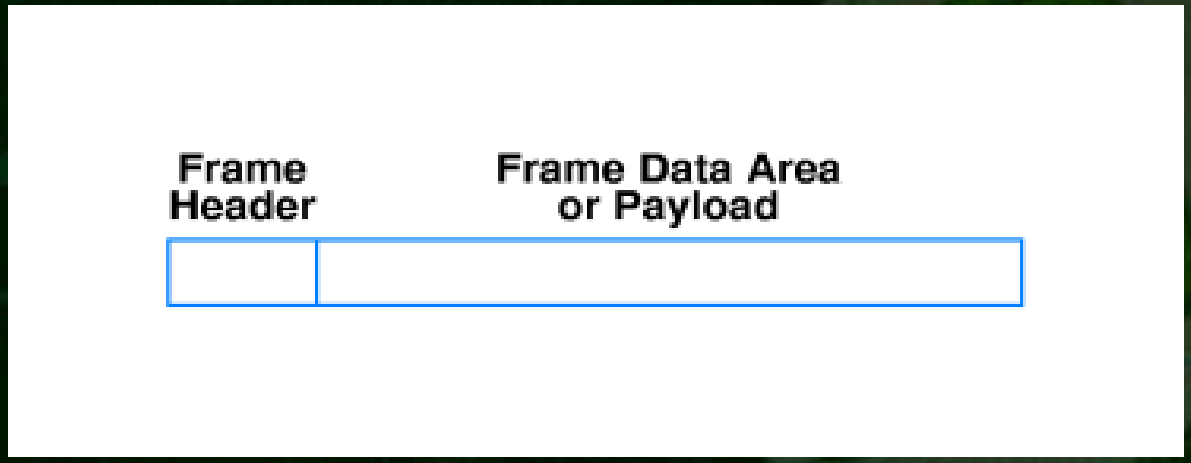
- Destination must get some clue about how to interpret frame data.
- Can use:
  - Explicit Frame Type - Identifying value included with frame describes type of included data.
  - Implicit Frame Type - Receiver must infer type from frame data.

Headers and Frame Formats

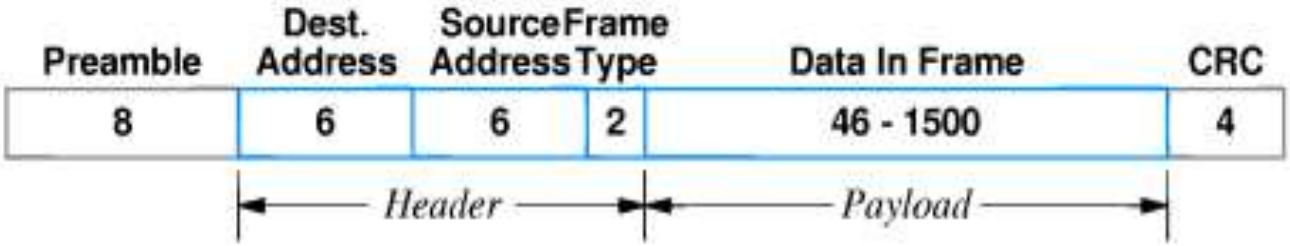
- LAN technology standards define frame format for each technology.

- All contemporary standards use the following general format:
  - Frame header has address and other identifying information.
  - Information typically in fields with fixed size and location.
  - Data area may vary in size.

Headers and Frame Formats



Ethernet Frame Format



Ethernet Frame Format

Field	Purpose
Preamble	Receiver synchronization
Dest. addr.	Identifies intended receiver
Source addr.	Hardware address of sender
Frame type	Type of data carried in frame
Data	Frame payload
CRC	32-bit CRC code

Ethernet Fields

- Preamble and CRC often not shown
- Destination address of all 1s is the broadcast address.
- Special values are reserved for frame type field.

Ethernet Fields

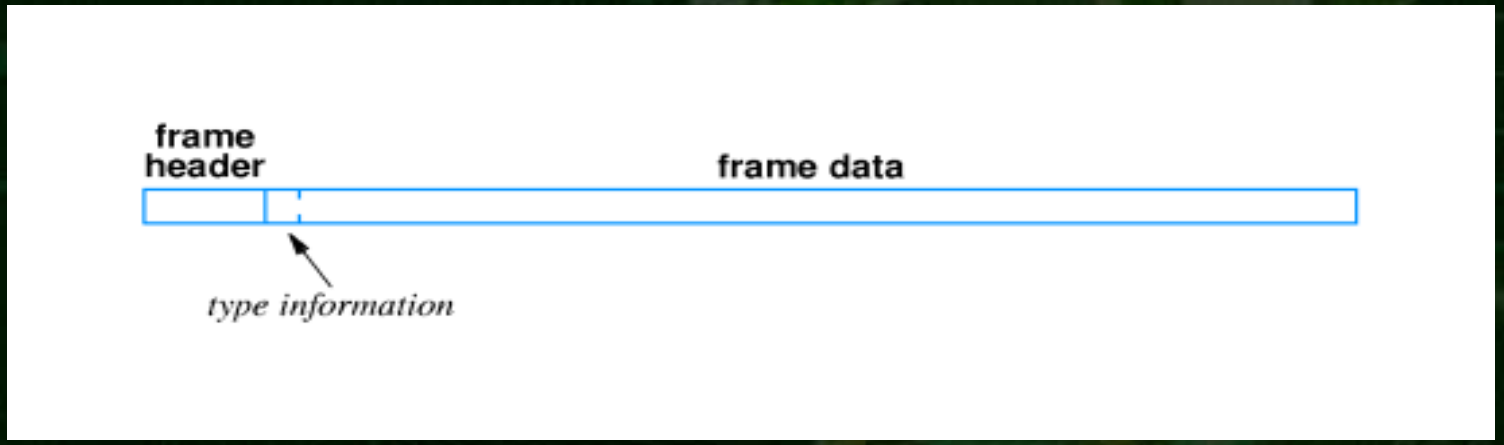
Value	Meaning
0000-05DC	Reserved for use with IEEE 802.3
0800	Internet IP Version 4
0805	CCITT X.25
0900	Ungermann-Bass Corporation network debugger
0BAD	Banyan Systems Corporation VINES
1000-100F	Berkeley UNIX Trailer encapsulation
6004	Digital Equipment Corporation LAT
6559	Frame Relay
8005	Hewlett Packard Corporation network probe
8008	AT&T Corporation
8014	Silicon Graphics Corporation network games
8035	Internet Reverse ARP
8038	Digital Equipment Corporation LANBridge
805C	Stanford University V Kernel
809B	Apple Computer Corporation AppleTalk
80C4-80C5	Banyan Systems Corporation
80D5	IBM Corporation SNA
80FF-8103	Wellfleet Communications
8137-8138	Novell Corporation IPX
818D	Motorola Corporation
FFFF	Reserved

Frames without Type Fields

- Some LAN technologies do not include a type field.
- Sender and receiver can agree on interpretation:
  - Agree on a single data format and use only that format.
  - Limits LAN to one type of data.
  - All computers on LAN must use one format.
  - Agree to encode the data format in the first few bytes of the data field.

Encoding the Data Type

- Illustration of using data area to encode data type:



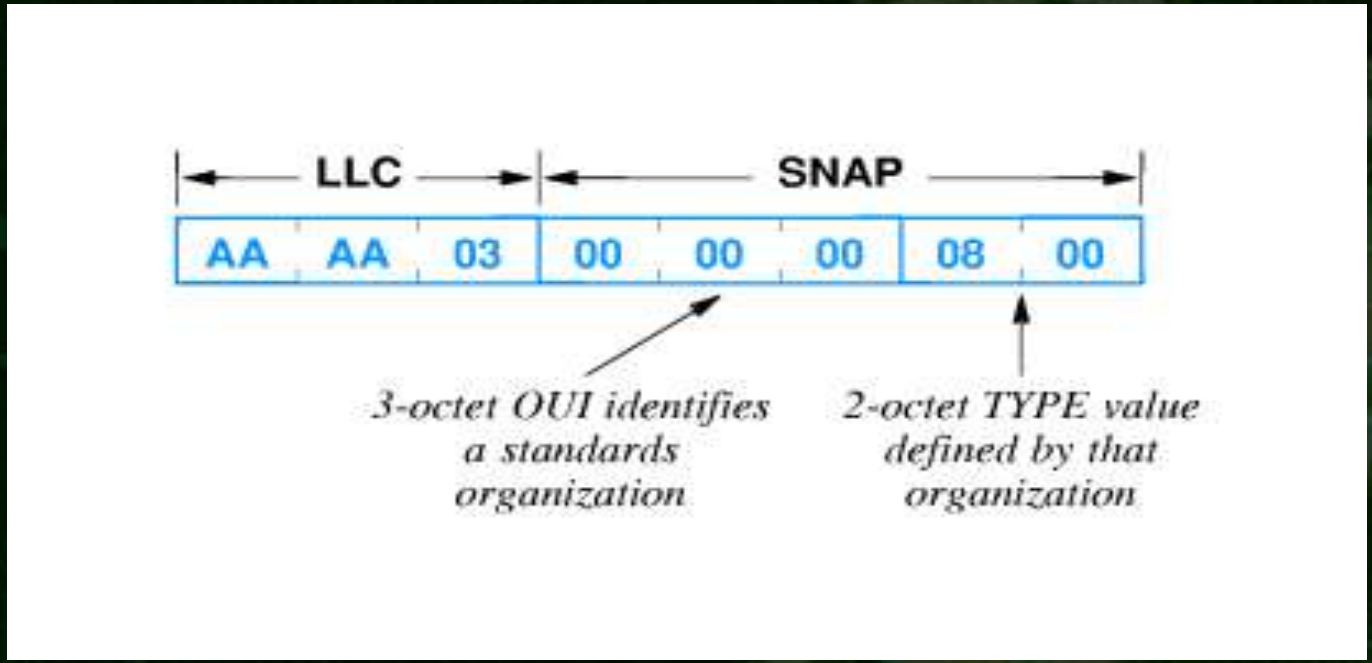
Encoding the Data Type

- To ensure interoperability, format of encoding area must be universally agreed upon.
- Format typically set by standards body.

IEEE 802.2 LLC

- To ensure IEEE 802.2 standard includes Logical Link Control (LLC) Sub Network Attachment Point (SNAP) header.
- SNAP/LLC format widely used; e.g., by Ethernet.

IEEE 802.2 LLC



IEEE 802.2 LLC

- LLC portion indicates SNAP field to follow.
- OUI (Organizationally Unique Identifier) identifies Ethernet specification organization.
-
- TYPE field interpreted as in Ethernet (in this case, IP).

Unknown Types

- For either encoding format, some computers may not be prepared to accept frames of some types like:
  - Protocol type not installed
  - Newly defined type
- Receiving computer examines type field and discards any frames with unknown type.

Network Analyzers

- A network analyzer or network monitor or Network Sniffer is used to examine the performance of or debug a network.
- Can report statistics such as capacity utilization, distribution of frame size, collision rate or token circulation time.

Operation of Network Analyzer

- Basic idea is a computer with a network interface that receives all frames.
- Sometimes called Promiscuous Mode.
- Many desktop computers have interface that can be configured for promiscuous mode.
-
  - Combined with software, computer can examine any frame on LAN.
  - Communication across a LAN is not guaranteed to be private!

Operation of Network Analyzer

- Computer receives and displays (but does not respond to) frames on the LAN.

Filtering Incoming Frames

- Analyzer can be configured to filter and process frames.
  - Count frames of a specific type or size.
  - Display only frames from or to specific computers.
  - In general, can be configured to match value of any field and capture only those frames meeting the filter specification.

Summary

- Multicasting
- Frame Type Identification
- Ethernet Frame Format
- Frame Without Type Fields
- Network Analyzer

COMPUTER NETWORKS

CS610

Lecture-11

Hammad Khalid Khan

Review Lecture 10

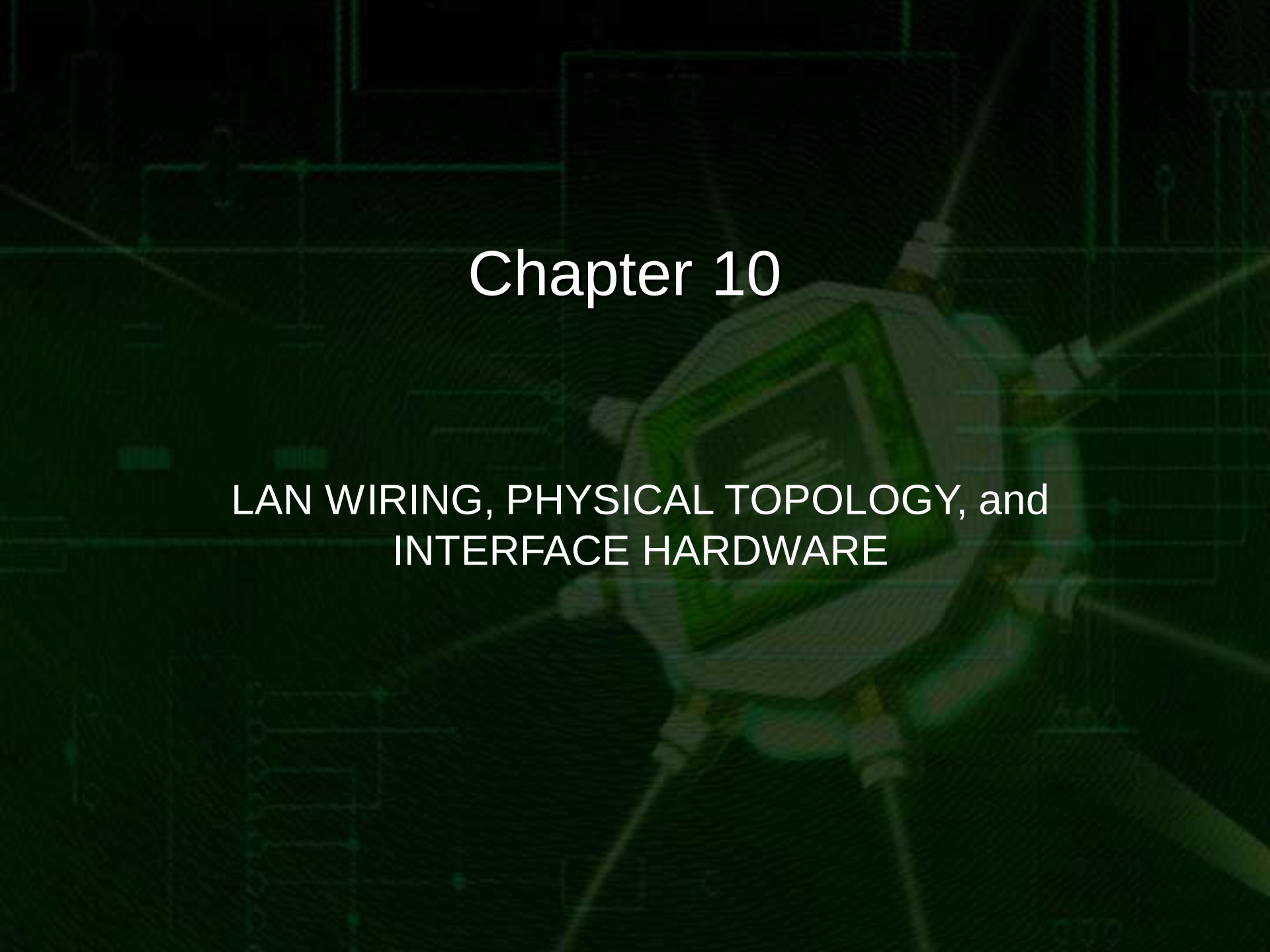
Multicasting

Frame Type Identification

Ethernet Frame Format

Frame Without Type Fields

Network Analyzer



Chapter 10

LAN WIRING, PHYSICAL TOPOLOGY, and INTERFACE HARDWARE

Introduction

- Interface cards:
  - Why a separate card?
  - How to connect the interface to the computer?
  - What is a Transceiver?
- LAN Wiring Schemes
- Logical and Physical Topology

Speeds of LANs and Computers

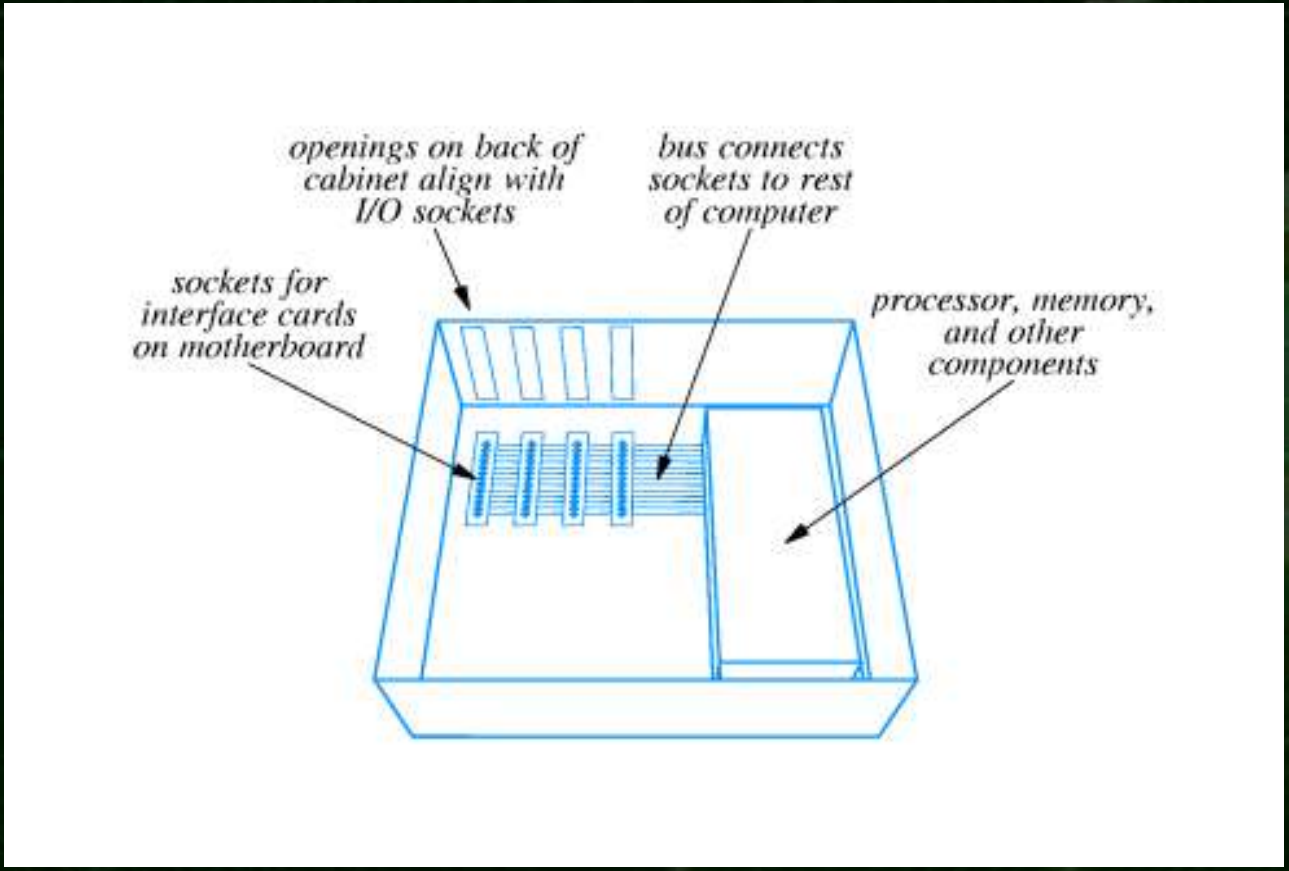
- LAN data transmission speeds are typically Fast relative to CPU speeds.

- LAN speeds are defined independent of any specific processor speeds.
  - Allows for mix of attached systems.
  - New computers can be attached without affecting LAN speeds.

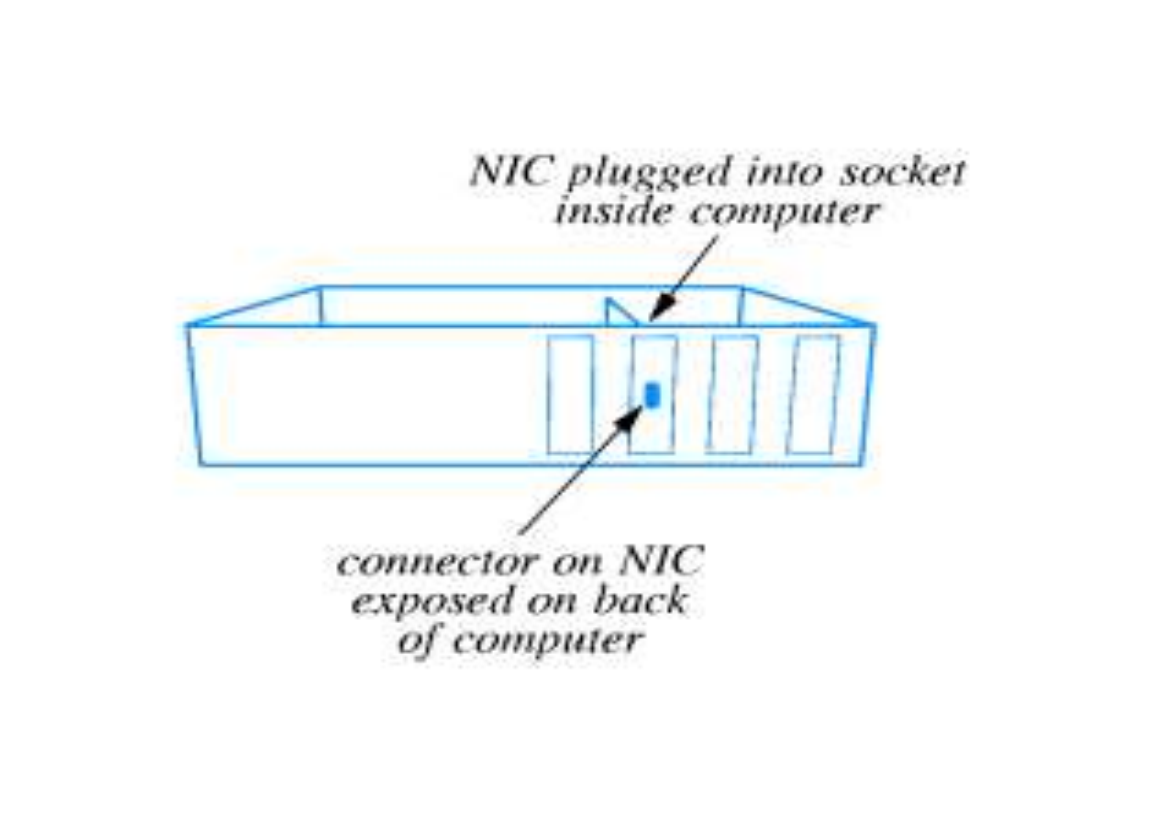
Network Interface Hardware

- CPU can't process data at network speeds.
- Computer systems use special purpose hardware for network connection.
  - Typically a separate card in the backplane.
  - Network adapter card or network interface card (NIC).
- Connector at back of computer then accepts cable to physical network.

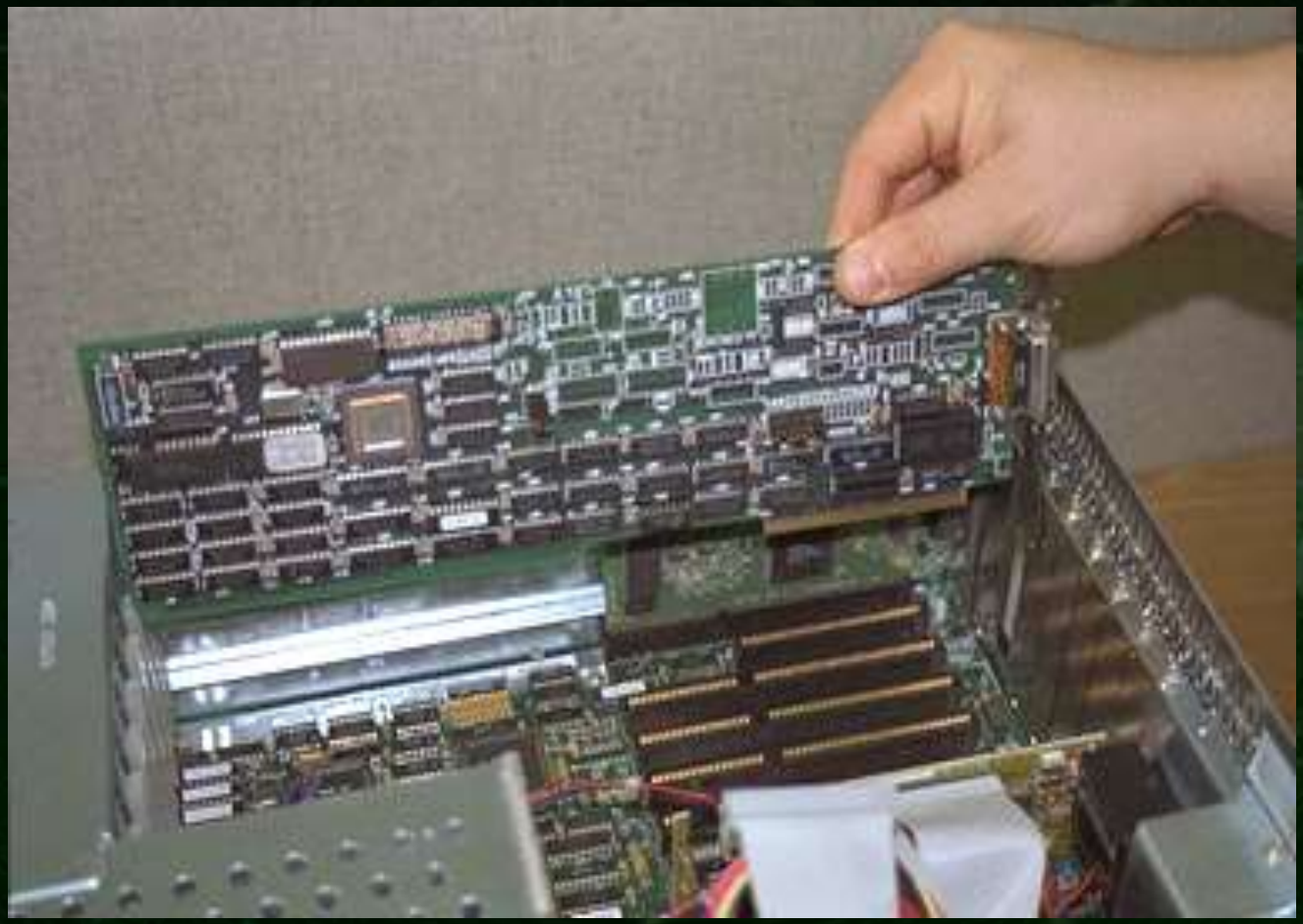
I/O Interfaces



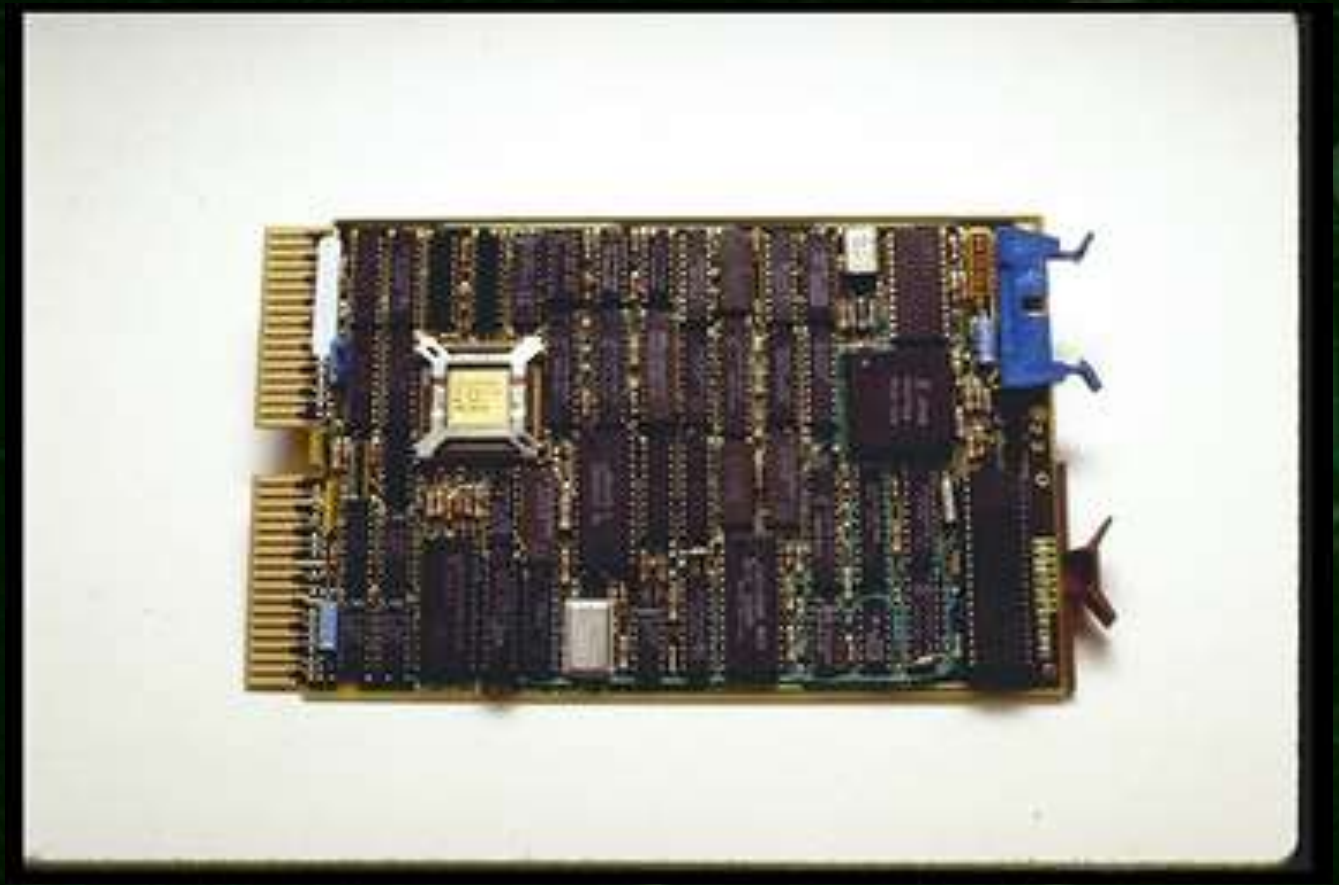
Network Connector



Network Interface Card (NIC)



Network Interface Card (NIC)



NICs and Network Hardware

- NIC is built for one kind of physical network.
  - Ethernet interface can't be used with token ring.
  - ATM interface can't be used with FDDI.

- Some NICs can be used with different, similar hardware.
  - Thick, thin and 10Base-T Ethernet.
  - 10Mbps and 100Mbps Ethernet.

NIC and CPU Processing

- NIC contains sufficient hardware to process data independent of system CPU.
  - Some NICs contain separate microprocessor.
  - Includes analog circuitry, interface to system bus, buffering and processing.

- Looks like any other I/O device to system CPU.
  - System CPU forms message request.
  - Sends instructions to NIC to transmit data.
  - Receives interrupt on arrival of incoming data.

Connection between NIC and Physical Network

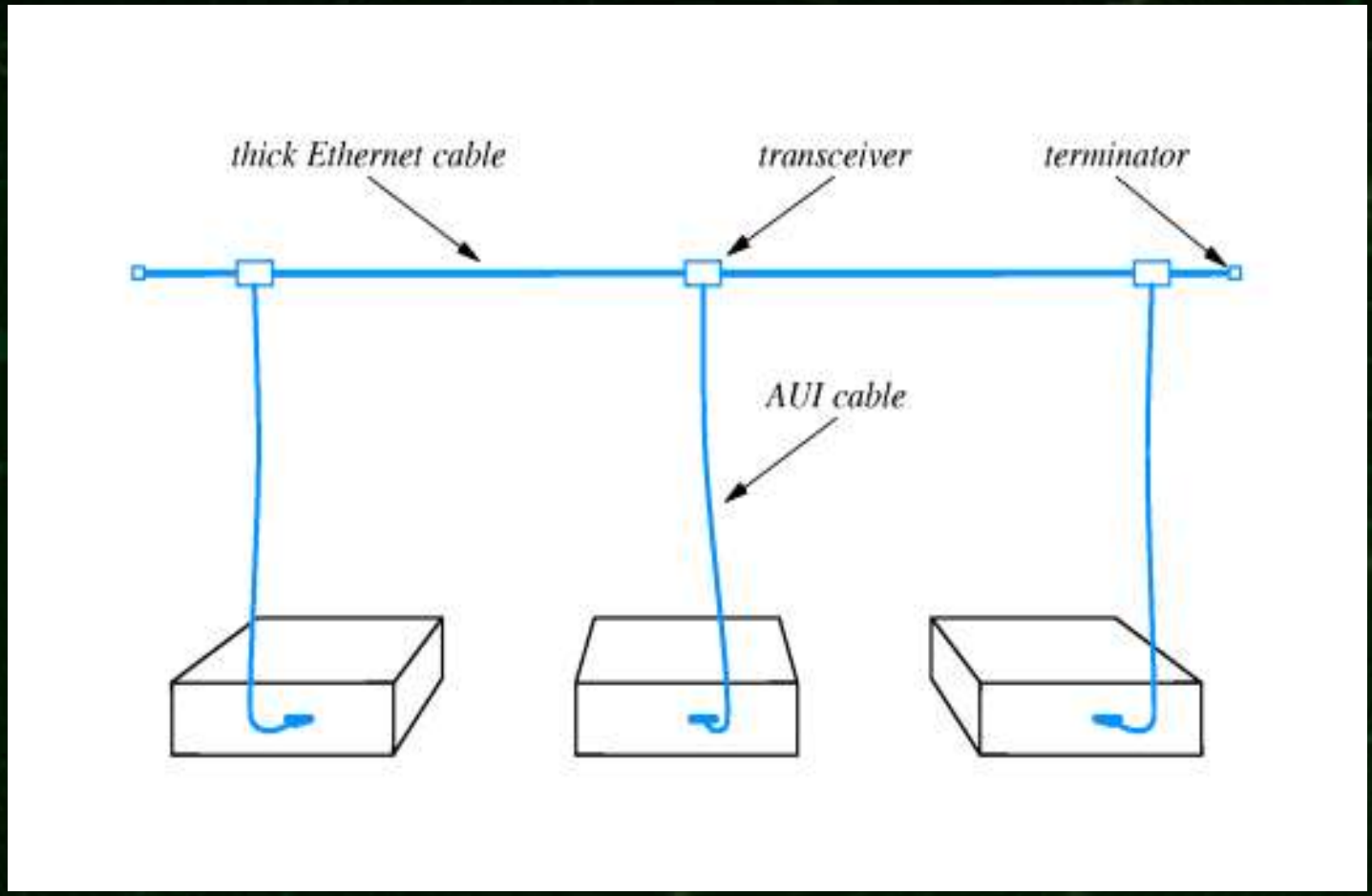
- Two alternatives:
  - NIC contains all circuitry and connects directly to network medium.
  - Cable from NIC connects to additional circuitry that then attaches to the network medium.

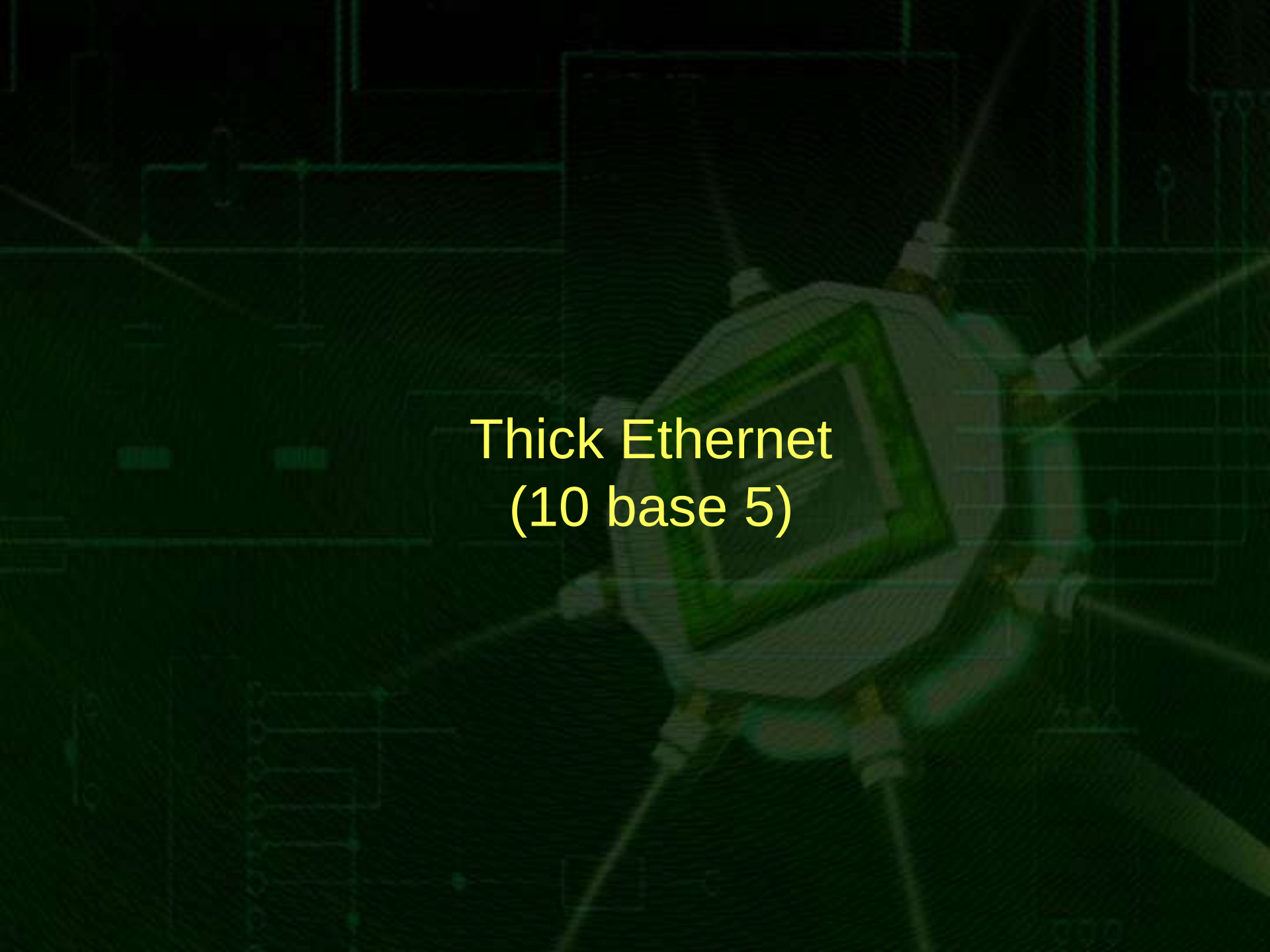
- Thin Ethernet vs. 10Base-T:
  - Both are Ethernet; network technology not limited to one style of connection.

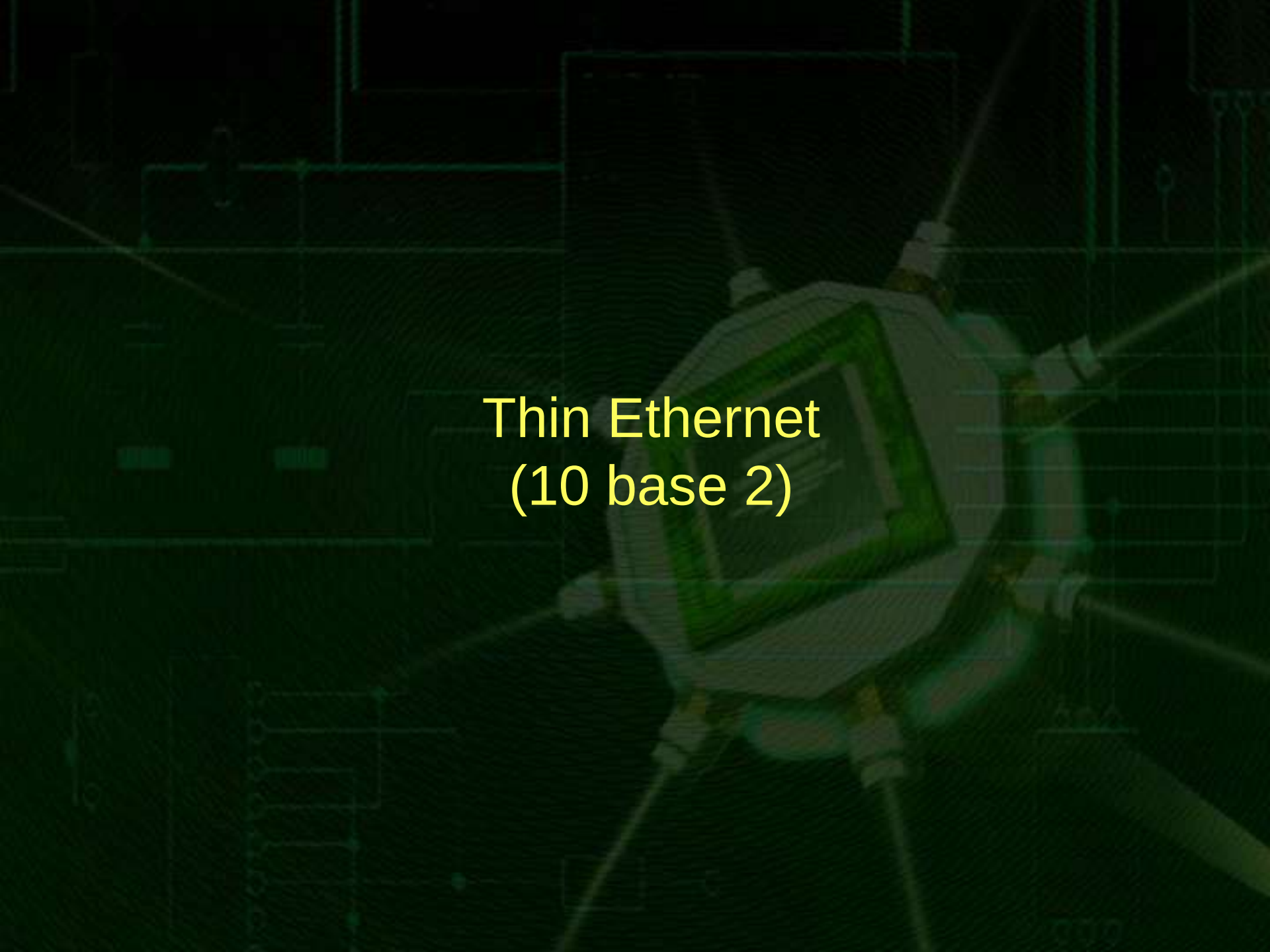
Thick Ethernet Wiring

- Uses thick coax cable.
- AUI cable (or transceiver or drop cable connects from NIC to transceiver.
- AUI cable carries digital signal from NIC to transceiver.
- Transceiver generates analog signal on coax.
- Wires in AUI cable carry digital signals, power and other control signals.
- Thick Ethernet also requires Termination to avoid signal reflectance.

Thick Ethernet Example




Thick Ethernet
(10 base 5)

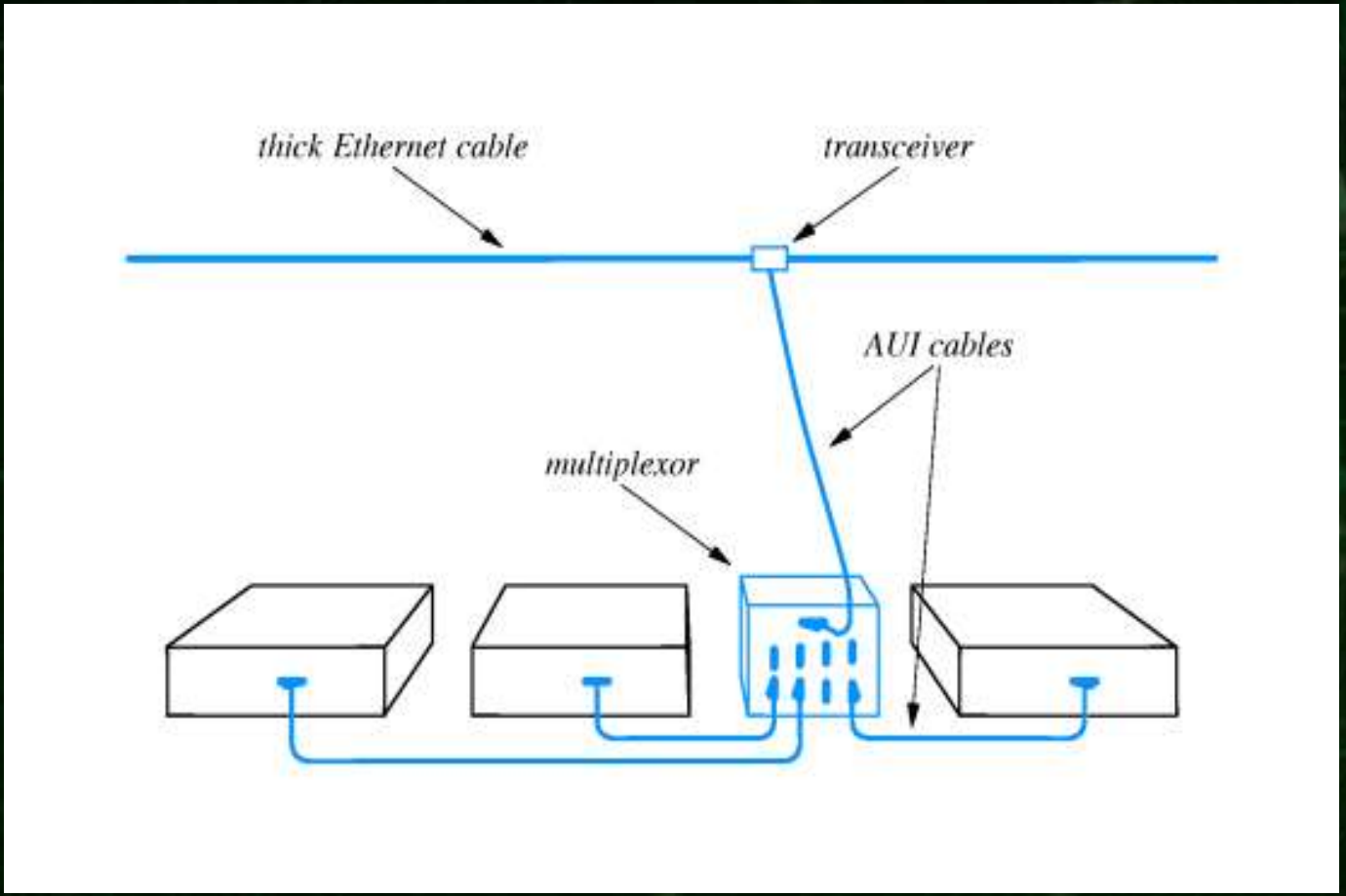

Thin Ethernet
(10 base 2)

Connection Multiplexing

- In some circumstances, transceivers may be inconvenient; e.g., workstations in a lab.

- Connection Multiplexer connects multiple computers to a single transceiver.
  - Each computer's AUI cable connects to connection multiplexor.
  - One AUI from multiplexor to Ethernet coax.

Connection Multiplexing



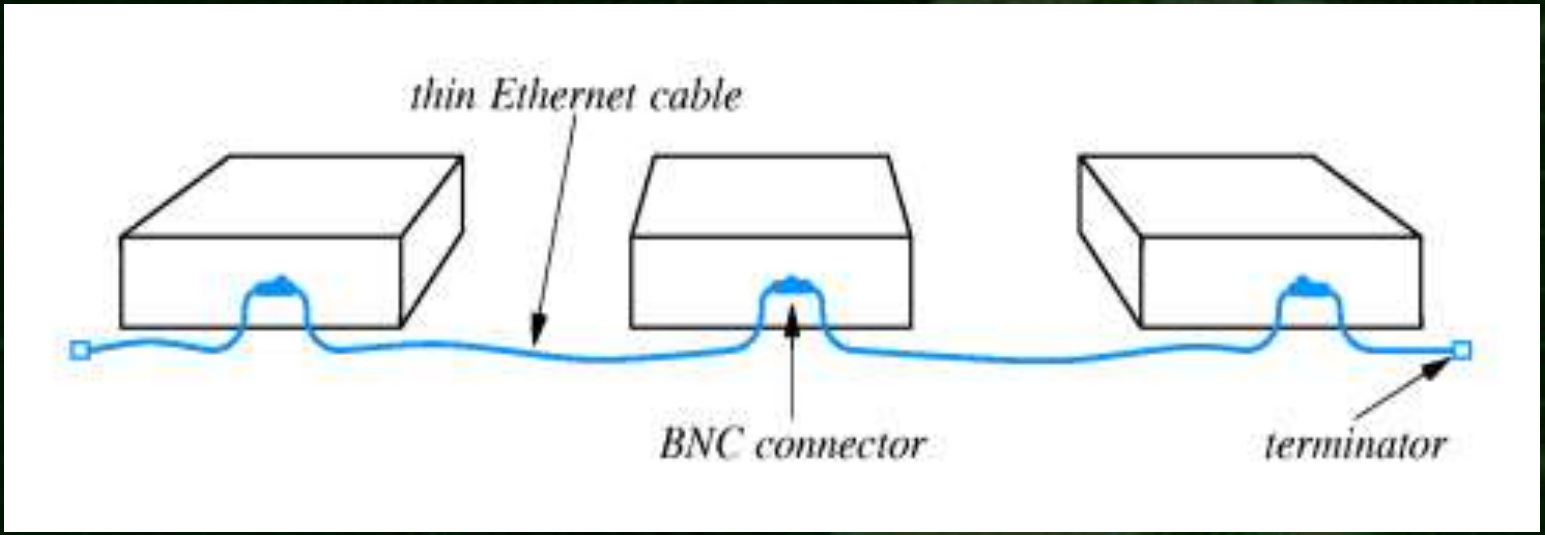
Thin Ethernet Wiring

- Uses thin coax that is cheaper and easier to install than thick Ethernet coax.
- Transceiver electronics built into NIC; NIC connects directly to network medium.
- Coax cable uses BNC connector.

Thin Ethernet Wiring

- Coax runs directly to back of each connected computer.
- T-Connector attaches directly to NIC.

Thin Ethernet Wiring



Thin Ethernet



Thin Ethernet



Thin Ethernet

- Useful when many computers are located close to each other.
- May be unreliable - any disconnection disrupts entire net.

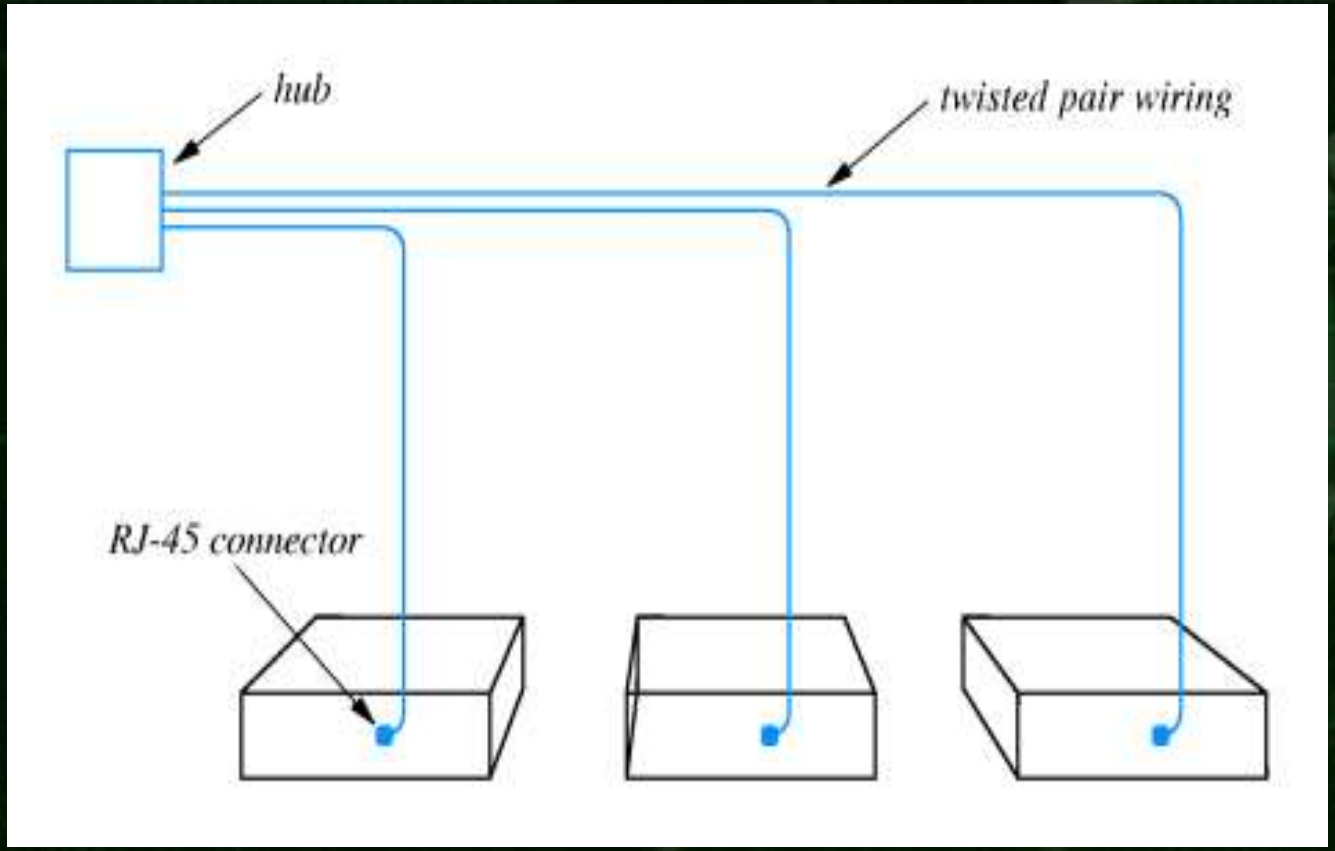
Thin Ethernet



10Base-T

- Commonly called 10Base-T, Twisted Pair or TP Ethernet.
- Replaces AUI cable with twisted pair cable.
- Replaces thick coax with Hub.

10Base-T



10Base-T



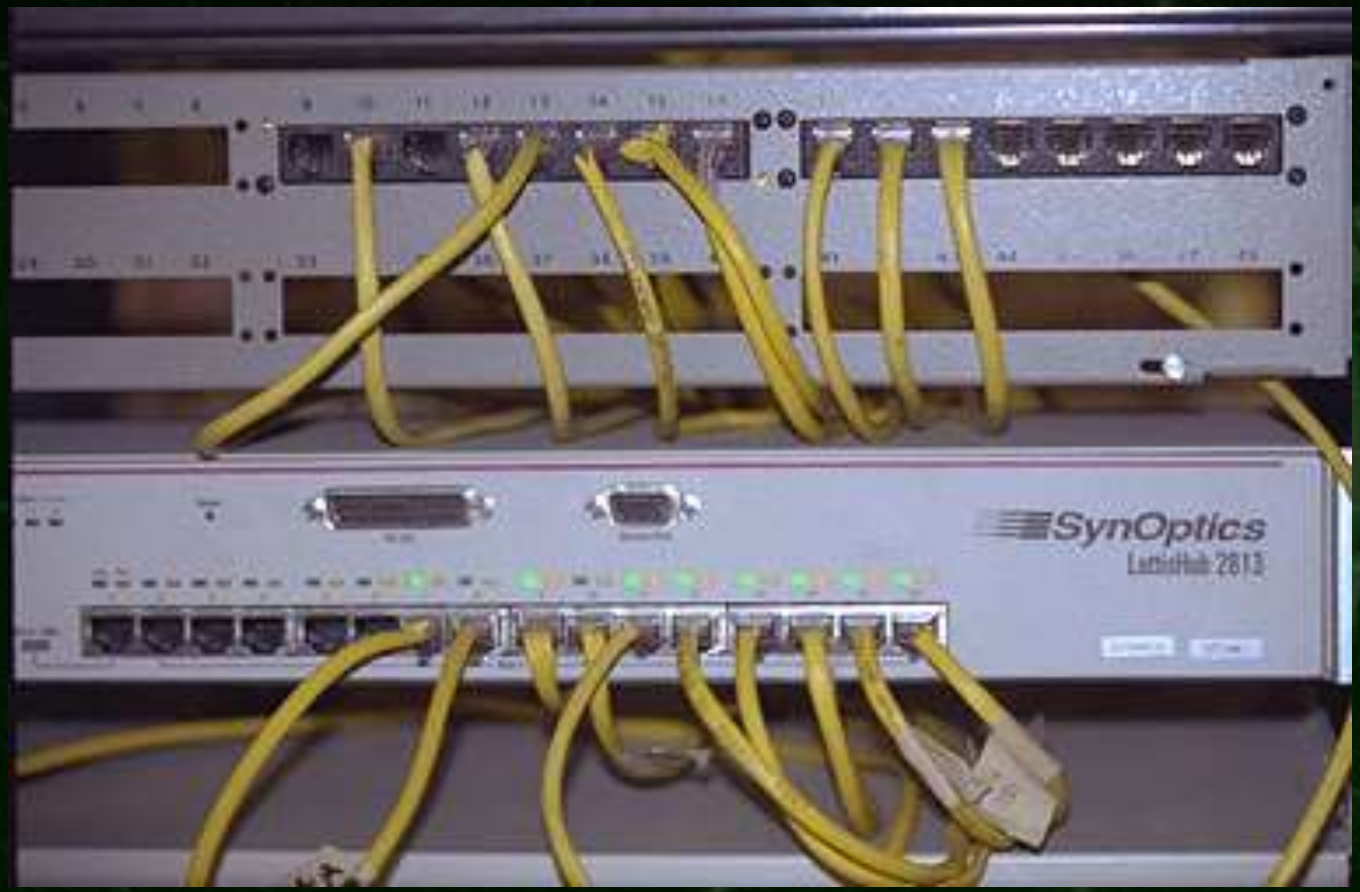
10Base-T



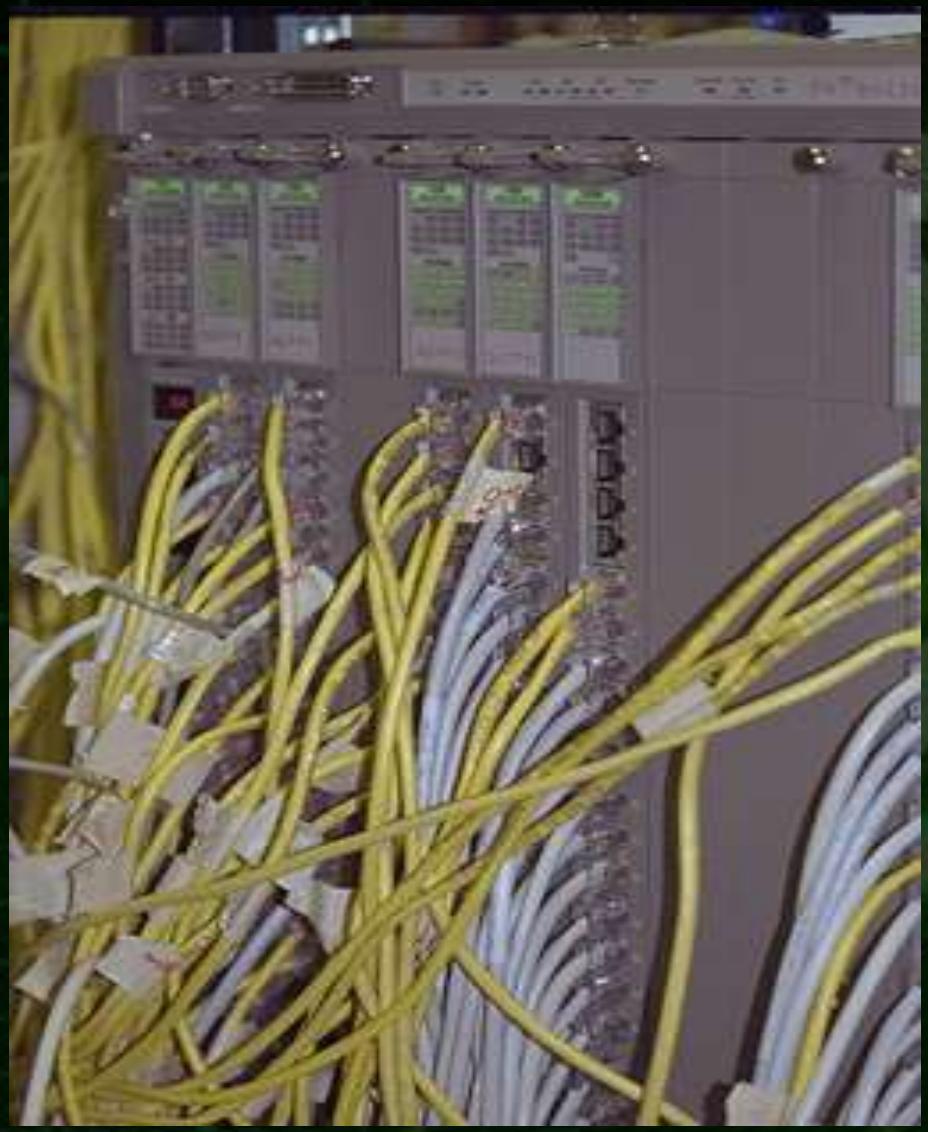
Hubs

- Extension of connection multiplexing concept.
- Sometimes called “Ethernet-in-a-Box”.
- Effectively a very short Ethernet with very long AUI cables.
- Can be connected into larger Ethernets.

Hubs



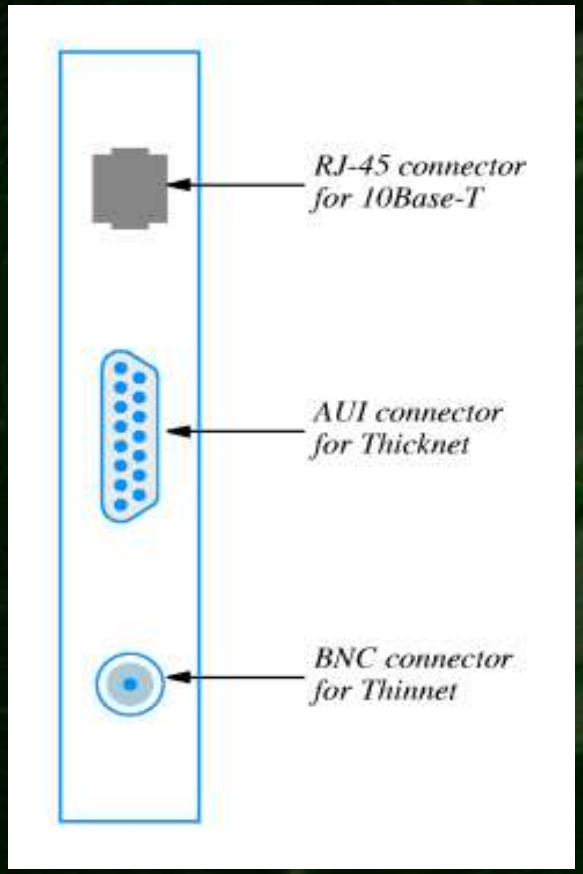
Hubs



Protocol Software and Ethernet Wiring

- All wiring technologies use identical Ethernet specification.
  - Same frame format.
  - Same CSMA/CD algorithms.
- Can mix different technologies in one Ethernet.
- NICs can provide all three connection technologies.
- Protocol software can't differentiate among wiring technologies.

Protocol Software and Ethernet Wiring



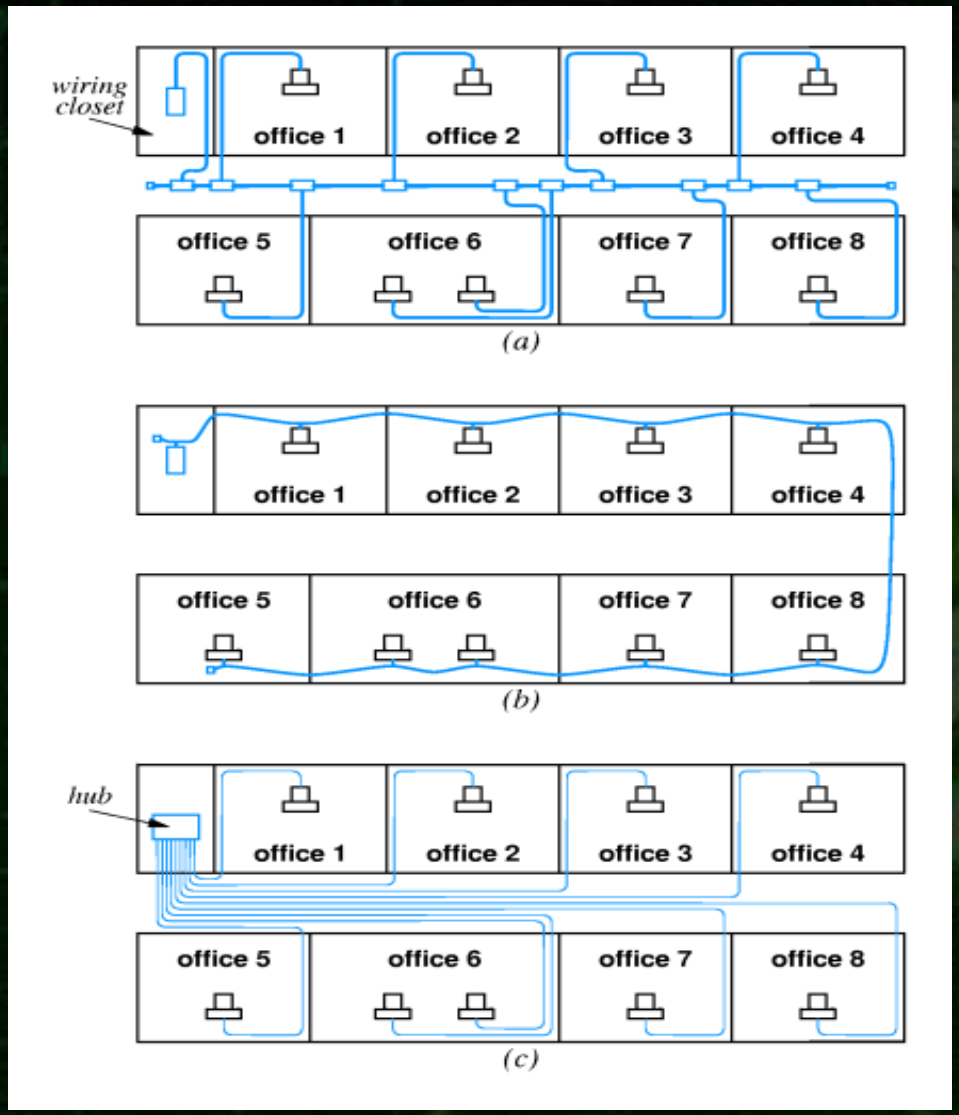
Comparison of Wiring Schemes

- Separate transceiver allows computer to be powered off or disconnected from network without disrupting other communication.
- Transceiver may be located in an inconvenient place.
- Finding malfunctioning transceiver can be hard.

Comparison of Wiring Schemes

- **Thin coax takes minimum of cable.**
- **Disconnecting one computer (or one loose connection) can disrupt entire network.**
- **Hub wiring centralizes electronics and connections, making management easier.**
- **Bottom line - 10Base-T most popular because of cost.**

Comparison of Wiring Schemes



Topologies and Network Technologies

- **10Base-T network topology is a bus; wiring topology is a star.**
- **Token ring network topology is a ring; wiring topology is a star.**
- **Remember to distinguish between Logical and Physical Topologies.**

Filtering Incoming Frames

- Analyzer can be configured to filter and process frames.

Summary

- Network Interface Hardware
- Connection Between NIC and Physical Network
- Thick Ethernet (10 BASE 5)
- Thin Ethernet (10 BASE 2)
- 10BASE-T
- Comparison Of Wiring Schemes

COMPUTER NETWORKS

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Lecture-12

Hammad Khalid Khan

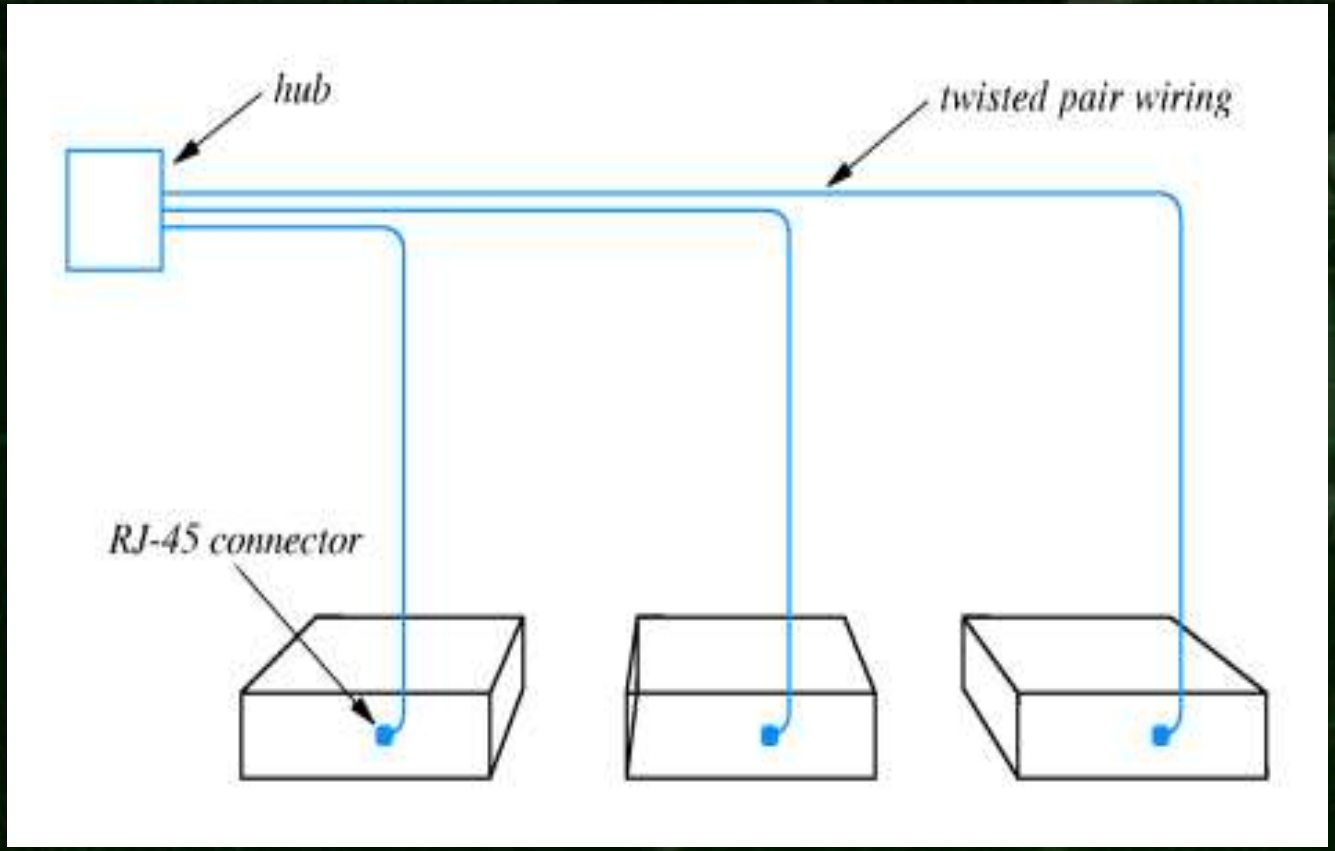
Review Lecture 11

- Network Interface Hardware
- Connection Between NIC and Physical Network
- Thick Ethernet (10 BASE 5)
- Thin Ethernet (10 BASE 2)
- 10BASE-T
- Comparison Of Wiring Schemes

10Base-T

- Formally known as 10BaseT
- Does not use coaxial cable, uses twisted pair wiring with RJ-45 connectors
- Each computer has a dedicated connection to a central electronic device called an Ethernet Hub

10Base-T



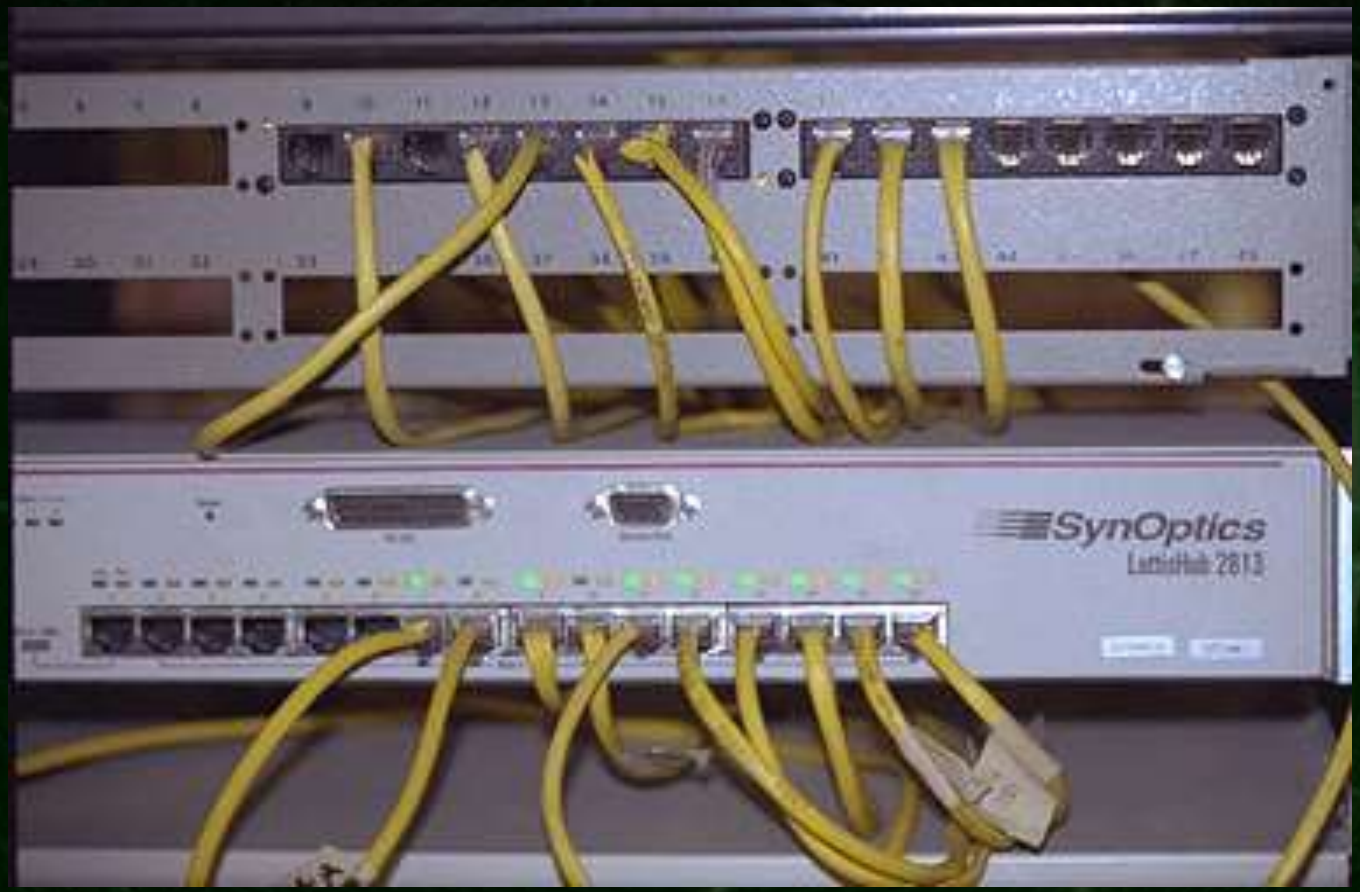
10Base-T



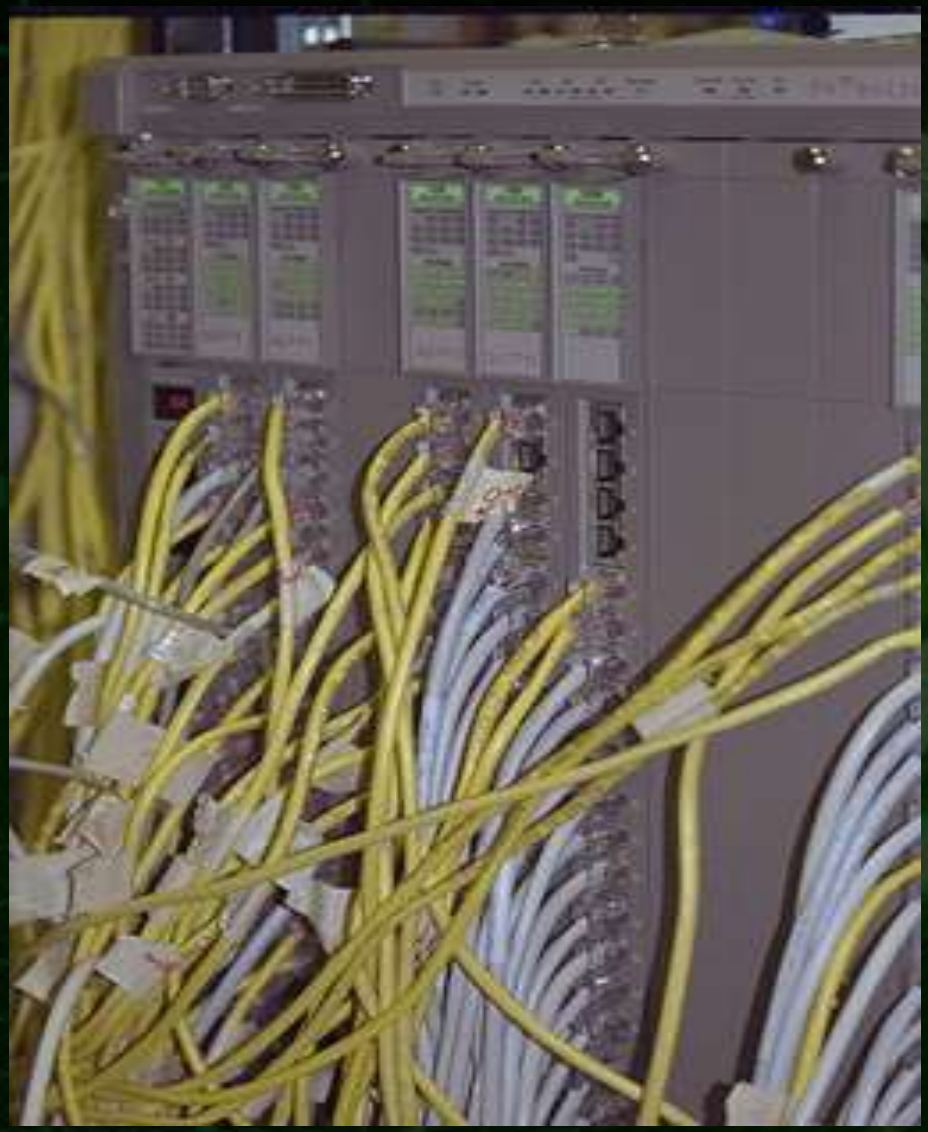
Hubs

- Extension of connection multiplexing concept
-
- Sometimes called “Ethernet-in-a-Box”
- Electronic components in a Hub simulate a shared physical cable, making the entire system operate like a conventional Ethernet

Hubs



Hubs



Advantages and Disadvantages of Wiring Schemes

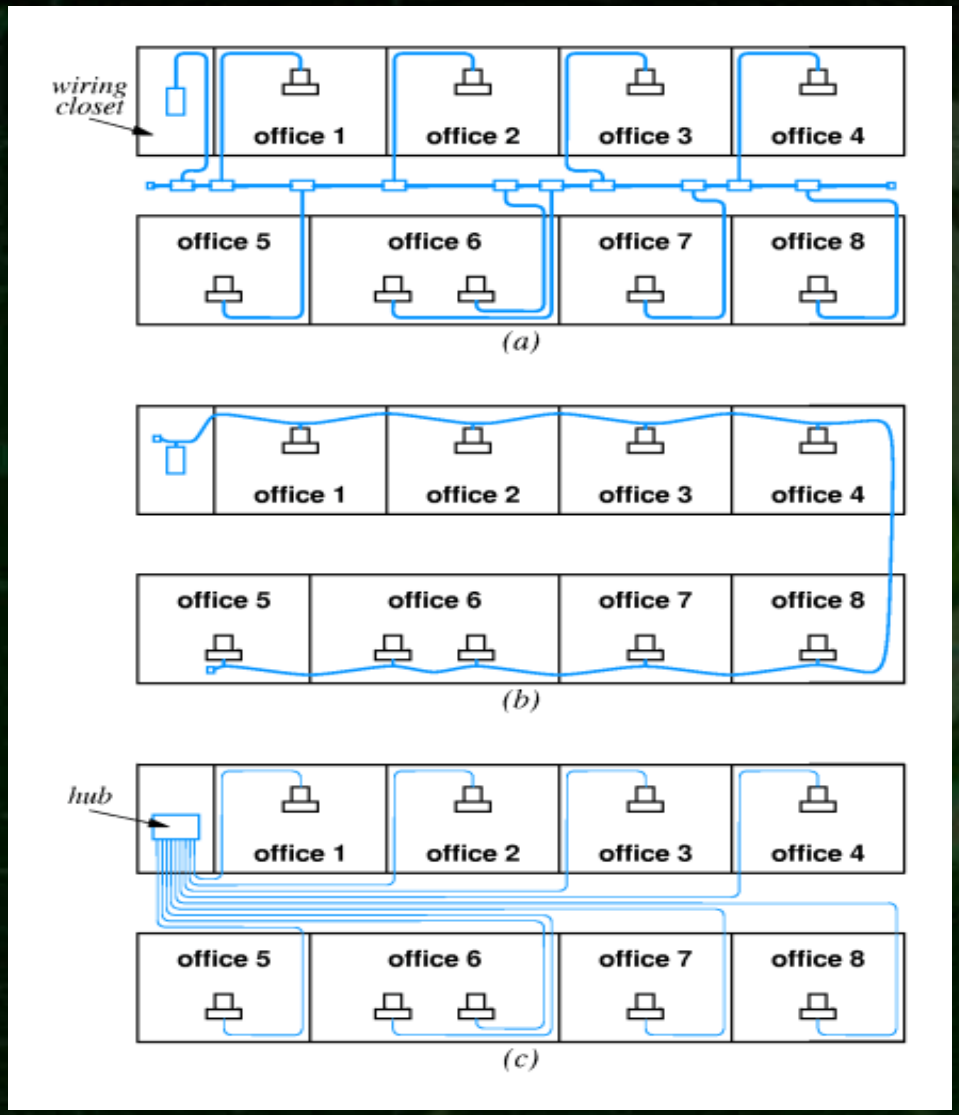
- Each of the three wiring schemes has advantages and disadvantages:
  - **Reliability issues:**
    - Wiring that uses a transceiver for each connection does not affect the entire network if a transceiver cable is disconnected
    - A cable cut occuring in a Hub wiring only affects one computer.

Advantages and Disadvantages of Wiring Schemes

- **Cost issues:**
  - Twisted pair Ethernet is the cheapest wiring that makes it so popular.
  - Thicknet is the most costly wiring which is no longer used

- **Example**
  - Eight offices wired with a) Thick Ethernet, b)Thin Ethernet, c) Twisted Pair Ethernet

Comparison of Wiring Schemes



The Topology Paradox

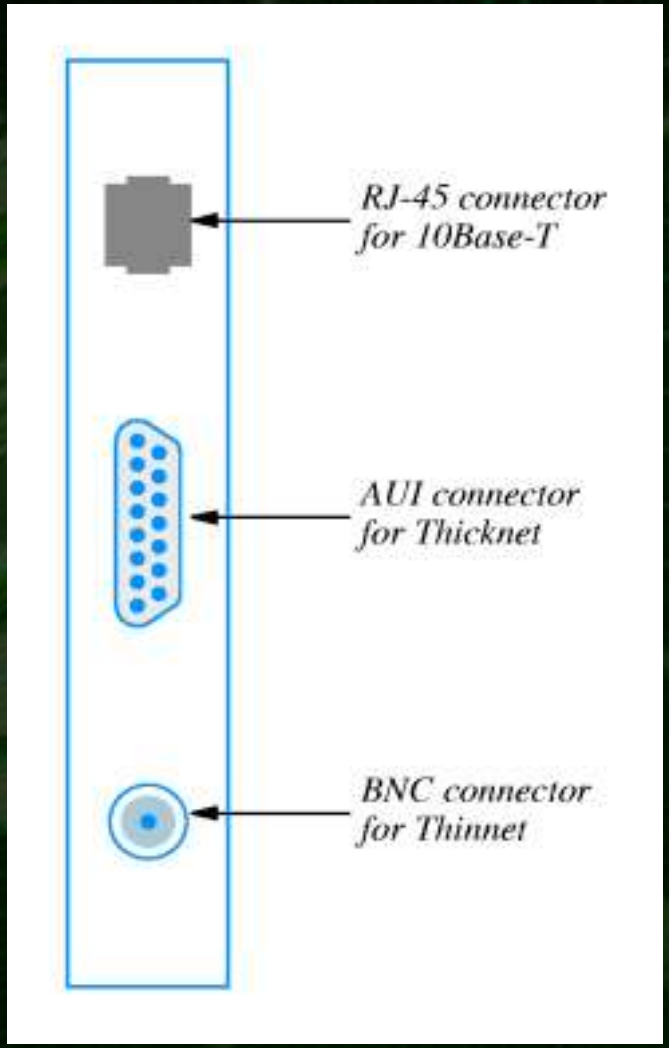
- Question: Is Ethernet a bus topology, or does the topology depend on the wiring?
- Answer: Both. Twisted pair Ethernet forms a classic star topology, however functions like a bus
- 10BaseT Ethernet is often called a star-shaped bus

The Topology Paradox

- Two different types of topologies:
  - Logical Topology: defined by the specific network technology
  - Physical Topology: depends on the wiring scheme

Network Interface Cards and Wiring Schemes

- To allow changing the wiring without changing the interface hardware, NICs support multiple wiring schemes



Network Interface Cards and Wiring Schemes



10/100 Network Interfaces and Autonegotiation

- 10 Base-T version of Twisted pair Ethernet operated at 10 Mbps
- 100 Base-T Twisted pair Ethernet operates at 100 Mbps
- 100 Base-T technology is backward compatible and allows the participants to negotiate a speed when connection is established
- Process is known as Autonegotiation

10/100 Ethernet NIC



Categories of Wires

- Cable used for wiring should match:
  - The intended data rate
  - The distance between the devices
  - The amount of electromagnetic noise
  - Anticipated future needs
  - COST

Categories of Wires

Category	Bandwidth	Typical Uses
3	16 MHz	older, low-speed networks; analog telephones
4	20 MHz	short distance 10Base-T
5	100 MHz	10Base-T Ethernet; some 100Base-T
5E	100 MHz	100Base-T (Fast Ethernet); some 1000Base-T
6	250 MHz	1000Base-T Gigabit Ethernet) or ATM
7	600 MHz	future (possibly 10 Gigabit Ethernet)

Summary

- Wiring Schemes
  - 10 Base-T
- Advantages and Disadvantages of Wiring Schemes
- The Topology Paradox
- NICs and Wiring Scheme
- Categories of Wires

COMPUTER NETWORKS

CS610

Lecture-13

Hammad Khalid Khan

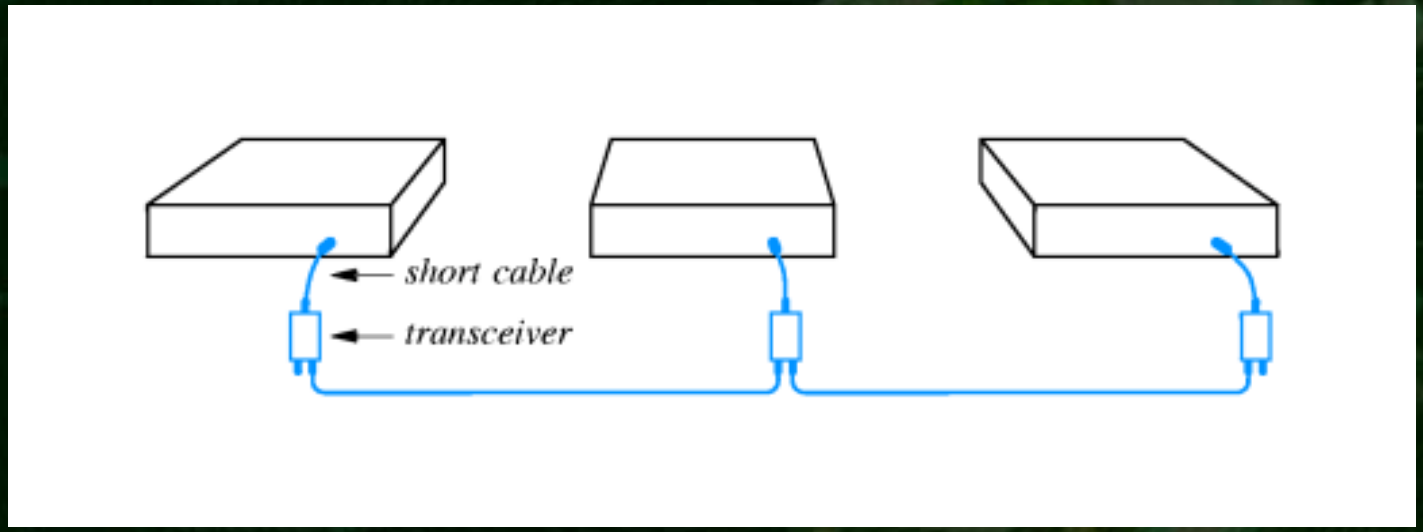
Review Lecture 12

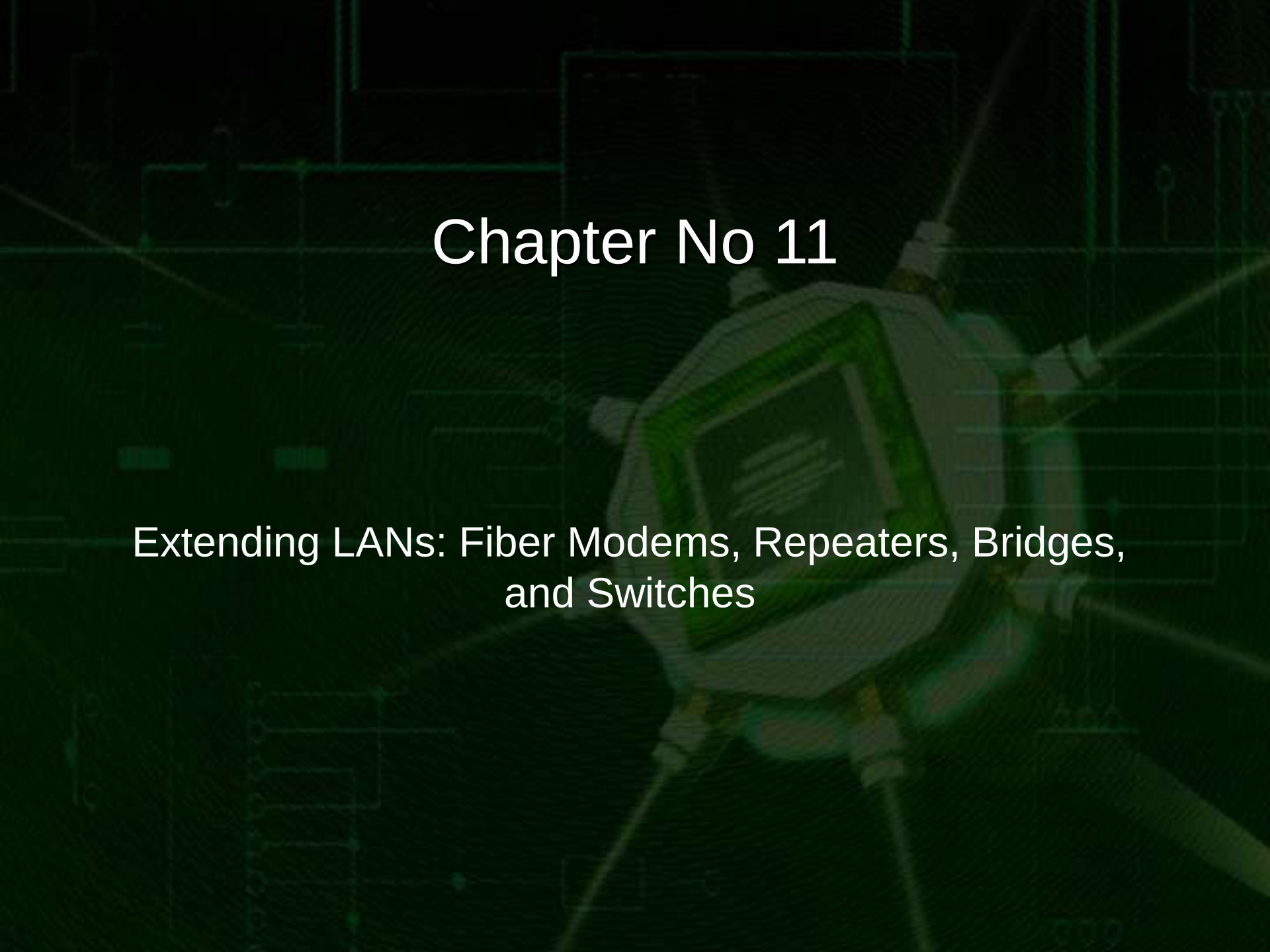
- Wiring Schemes
  - 10 Base-T
- Advantages and Disadvantages of Wiring Schemes
- The Topology Paradox
- NICs and Wiring Scheme
- Categories of Wires

Wiring Schemes and Other Network Technologies

- Multiple wiring schemes are not limited to Ethernet technology. Almost all other network technologies use different wiring schemes
- Localtalk uses Hubs (Physical Star) to simulate a bus topology
- IBMs Token Ring also uses Hubs (Physically a Star Topology) to simulate a logical ring network

Wiring Schemes and Other Network Technologies





Chapter No 11

Extending LANs: Fiber Modems, Repeaters, Bridges, and Switches

Introduction

- LAN technologies are designed to operate within the same building
- However most companies or institutions have offices located far apart from each other

Distance Limitation And LAN Design

- The maximum cable length of a LAN is fixed, because:
  - The electrical signal level gets weaker as it travels
  - The delays must be short to allow access mechanisms (CSMA/CD, token passing), work properly

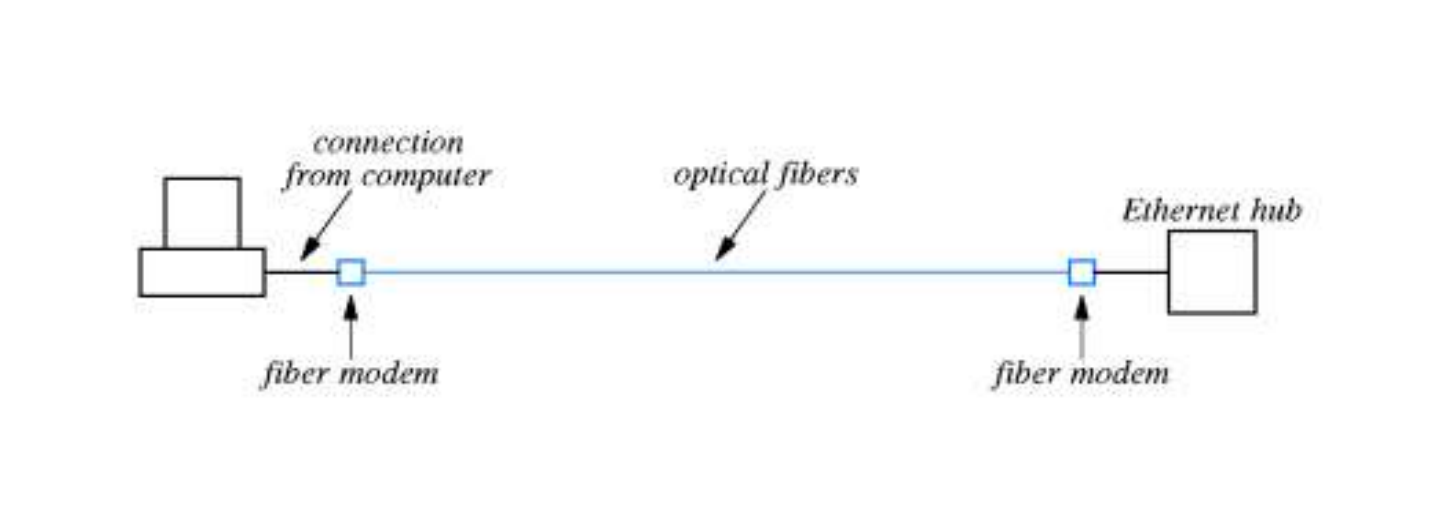
Distance Limitation And LAN Design



LAN Extensions

- Several techniques extend diameter of LAN medium
- Most techniques use additional hardware
- LAN signals relayed between LAN segments
- Resulting mixed technology stays within original engineering constraints while spanning greater distance

Fiber Optic Extensions



Fiber Optic Extensions

- The Fiber Modems:
  - Convert digital data into pulses of light
  - Transmit over the optical fiber
  - Receive light and convert into digital data
- Because
  - Delays on optical fiber is low and
  - Bandwidth is high
  - This mechanism will successfully extend the LAN across several kilometers

Fiber Optic Extensions



Fiber Optic Extensions

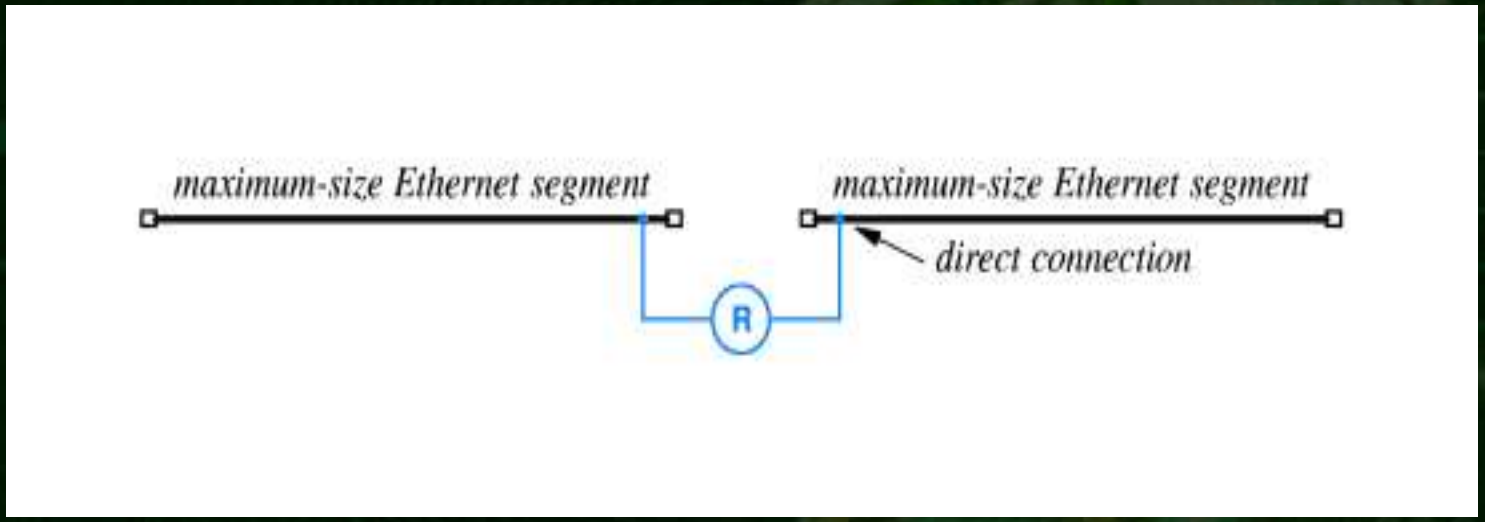


Repeaters

- Fact: Electrical signals gets weaker while travelling over copper

- A repeater (a hardware device)
  - Amplify the weakenning signals received from one segment
  - And retransmit them on to another segment

Repeaters


maximum-size Ethernet segment
maximum-size Ethernet segment
direct connection
R

Repeaters

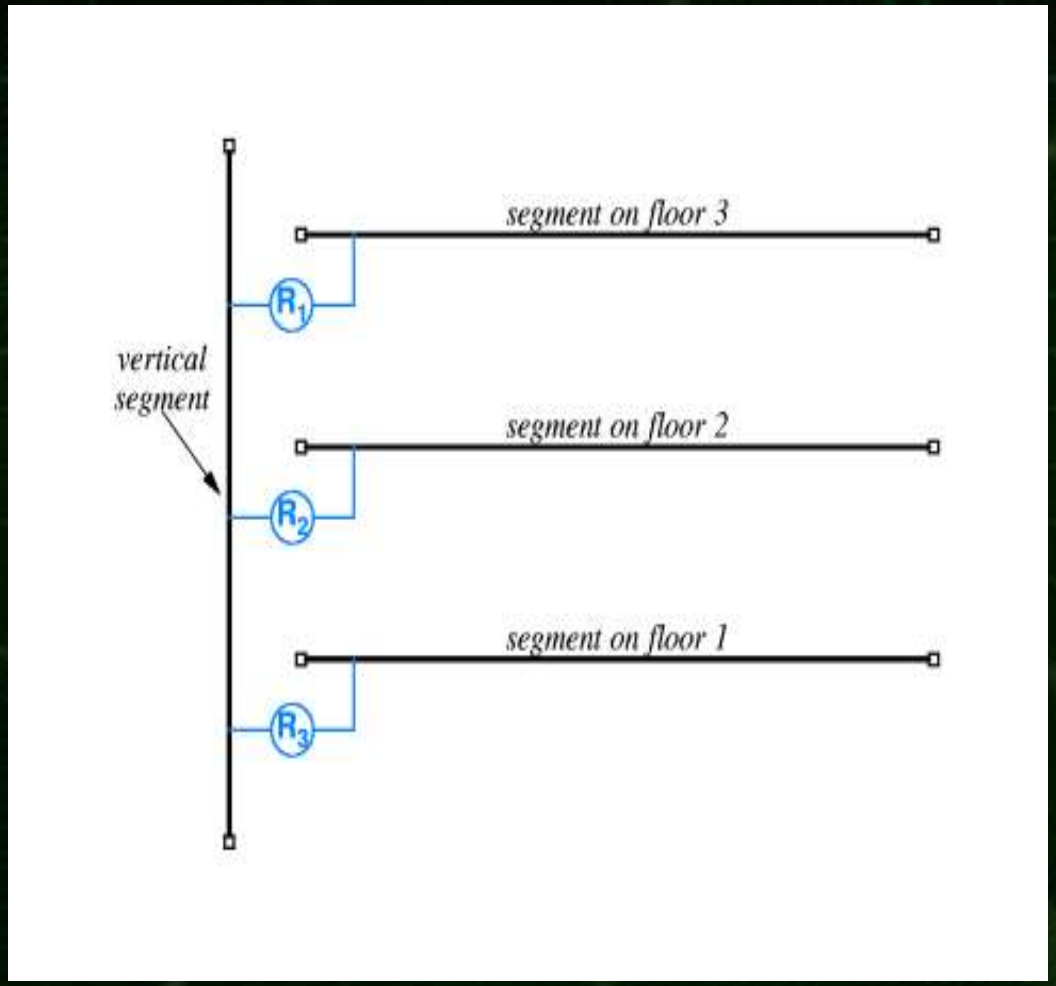
- One repeater doubles, two repeaters triples the maximum cable lenght limitation
- Computers attached to different segments communicate as if they are connected to the same cable

Repeaters

- Question: Can we increase the maximum cable lenght as many times as we wish by just adding repeaters?
- Answer: No
  - Every repeater introduces a delay
  - Access mechanisms such as CSMA/CD does not work with long delays
  - Ethernet standard specifies that any two stations cannot be seperated by more than four repeaters

Repeaters

- Using a vertical segment:
  - Only two repeaters seperate any two stations in this scheme



Repeaters

- Disadvantages:
  - Repeaters do not recognize frame formats, they just amplify and retransmit the electrical signals.
  - If a collision or error occurs in one segment,
    - Repeaters amplify and retransmits also the error onto the other segments

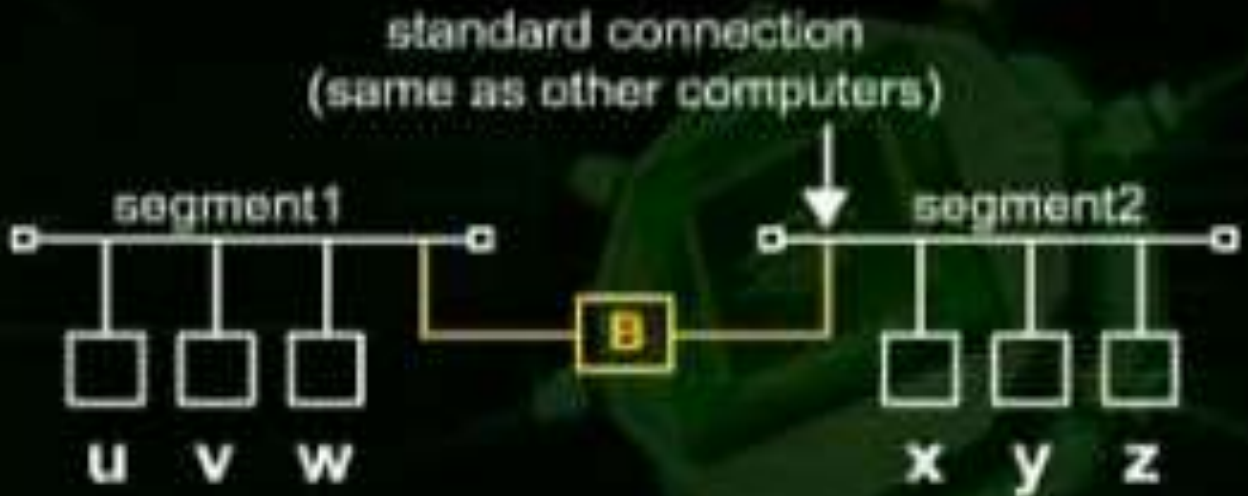
An Ethernet Repeater (Front)



An Ethernet Repeater (Back)



Bridges


standard connection
(same as other computers)
segment1
segment2
B
u v w
x y z

Bridges

- A Bridge is a hardware device also used to connect two LAN segments to extend a LAN
- Unlike a repeater, a bridge:
  - Uses two NICs, to connect two segments
  - Listens to the all traffic
  - Recognizes frame format
  - Forwards only correct, complete frames
  - Discards the collided and errored frames

An Ethernet Bridge





Frame Filtering

- A typical bridge has two NICs, a CPU, a memory and a ROM
- It only runs the code stored in its ROM
- The most important task a bridge performs is frame filtering

Frame Filtering

- If both the source and destination are on the same segment, it does not forward the frame to the other segment
- A frame is forwarded to the other segment, if the frame is destined to the that segment
- Broadcast and multicast frames are also forwarded

Frame Filtering

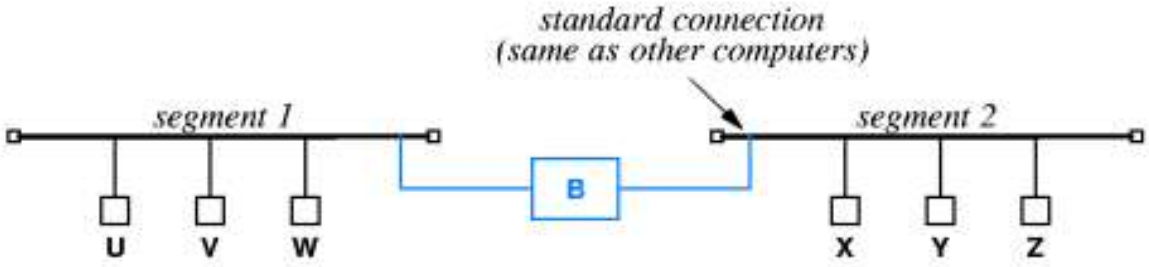
- Question: How does a bridge know on which segment a destination computer is attached?
- Answer: It keeps a list for each segment that consists of physical addresses of the computers attached to that segment

Frame Filtering

- Question: How does a bridge build up address lists?

- Answer: Most bridges are self learning bridges
  - As soon as a frame arrives to a bridge
  - It extracts the source address from its header and
  - Automatically adds it in the list for that segment

Frame Filtering



Event	Segment 1 List	Segment 2 List
Bridge boots	—	—
U sends to V	U	—
V sends to U	U, V	—
Z broadcasts	U, V	Z
Y sends to V	U, V	Z, Y
Y sends to X	U, V	Z, Y
X sends to W	U, V	Z, Y, X
W sends to Z	U, V, W	Z, Y, X

Summary

- Distance Limitation on LANs
- Fiber Optic Extensions
- Repeaters
- Bridges

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Lecture-14

Hammad Khalid Khan

Review Lecture 13

- Distance Limitation on LANs
- Fiber Optic Extensions
- Repeaters
- Bridges

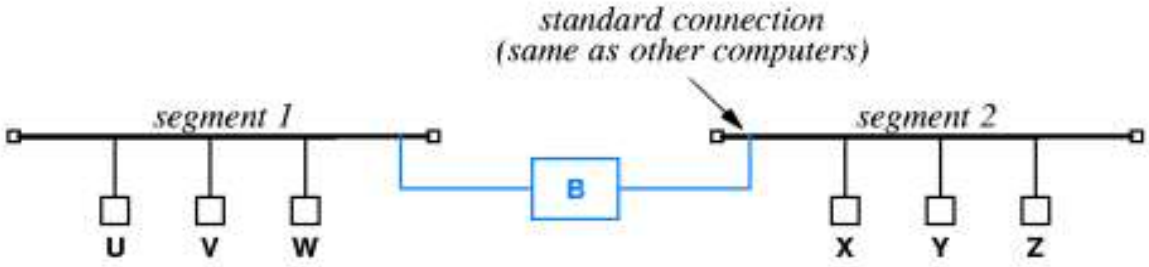

Bridges

Frame Filtering

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Frame Filtering



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U sends to V	U	—
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Y sends to V	U, V	Z, Y
Y sends to X	U, V	Z, Y
X sends to W	U, V	Z, Y, X
W sends to Z	U, V, W	Z, Y, X

Startup And Steady State

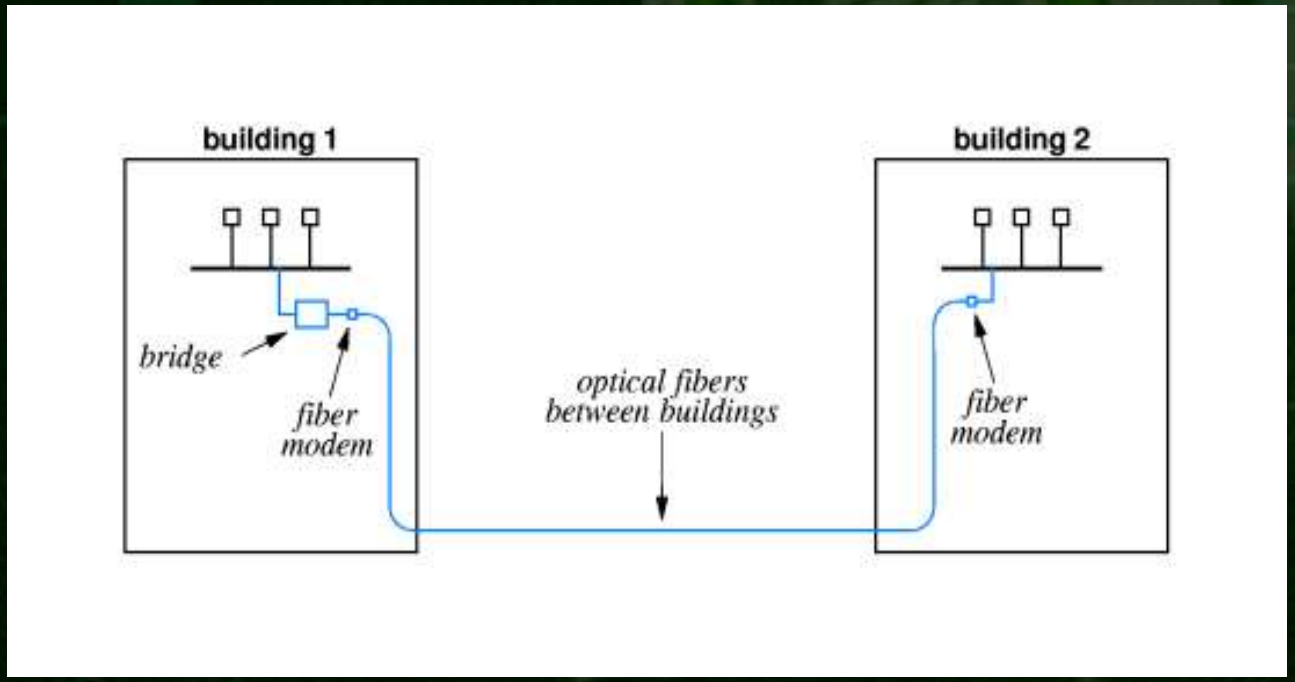
- When a bridge first boots, the address lists are empty (in startup state)
  - The bridge forwards frames to the other segment if it can not find its destination address in its lists
- After sometime, when the bridge has received at least one frame from every computer, it has the lists built (steady state), and
  - It forwards frames as far as it is necessary

Planning a Bridged Network

- In a steady state, a bridge allows simultaneous use of each segment
- When designing a LAN, bridges can be installed to divide the LAN into segments to improve performance
- For example: Frequently contacting computers can be attached to the same segment
- The frame traffic on one segment does not affect the other segment

Bridging Between Buildings

- If two buildings are located far from each other, a bridge, a pair of fiber modems and an optical fiber can be used to connect two LANs



Bridging Between Buildings

- Advantages:
  - Cost: An optical fiber and fiber modems are sufficient to connect many computers located on seperate buildings
  - Maintanence: No need to change the wiring between the buildings when installing and removing a computer
  - Performance: The traffic on each building does not affect the other

Bridging Across Longer Distances

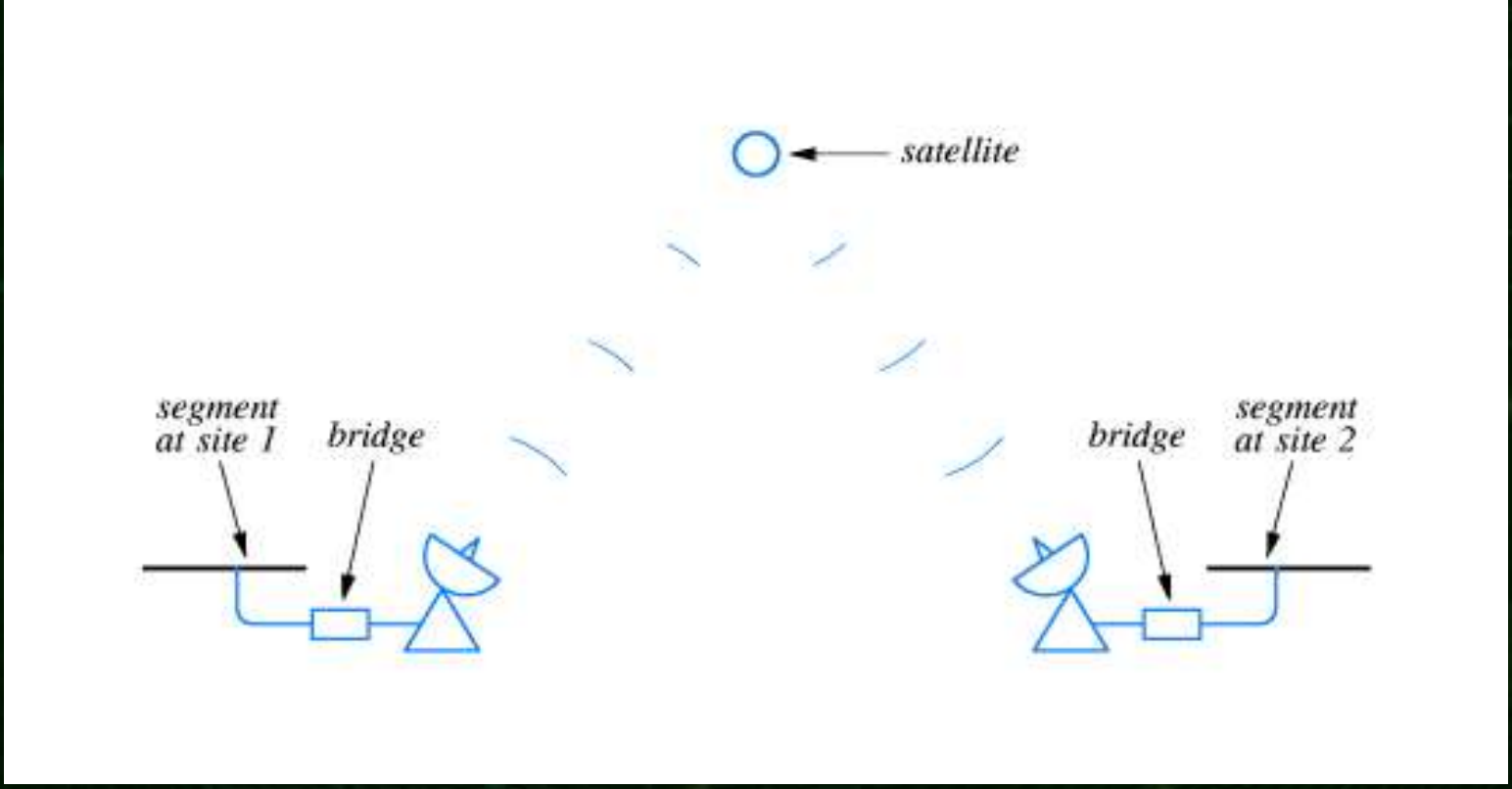
- Question: Is it always possible to connect two sites with optical fiber?
- Answer: No
  - The distance may be too long
  - It is usually not allowed to lay an optical fiber if the land does not belong to you

Bridging Across Longer Distances

- There are two common methods to connect two distant sites:
  - Leased serial line connection
    - Less expensive
  - Leased satellite channel
    - Can span arbitrarily long distance

Bridging Across Longer Distances

- Leased Satellite Connection:



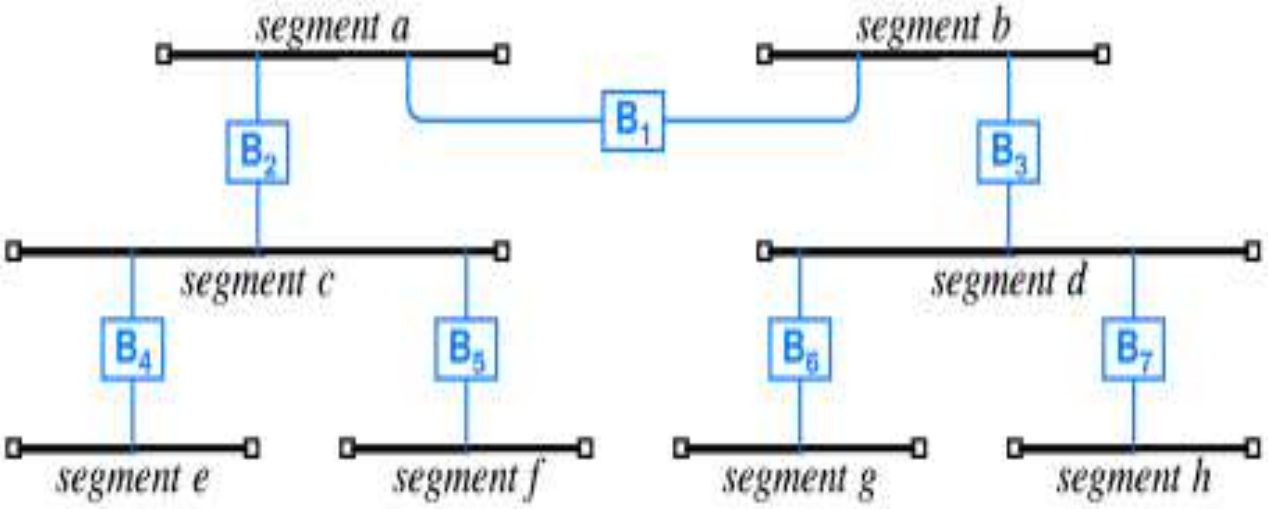
- Unlike optical fibers, satellite connections are low-bandwidth to save cost

- Because the frames arrive from local network much faster than they can be sent across a satellite link,
  - The bridge must use buffering(saving a copy of frame into memory until it can be sent)
  - The bridge may run out of memory
  - The communication software usually waits for a response after sending a few frames

A Cycle of Bridges

- A bridged network can connect many segments
- One bridge is needed to connect each segment to the rest of the bridged network

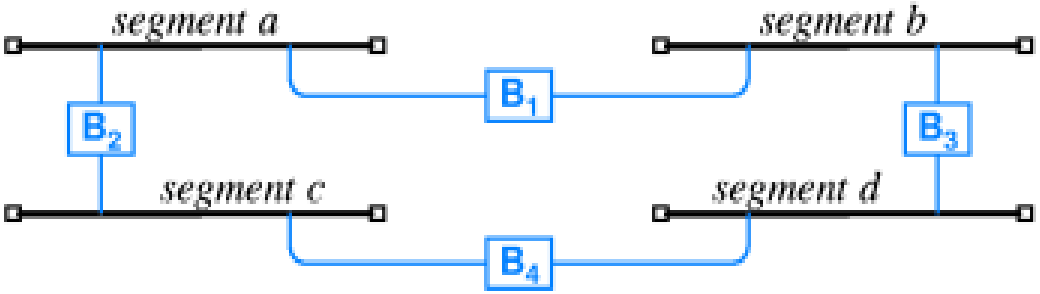
A Cycle of Bridges



A Cycle of Bridges

- If the bridges on a large network form a cycle, then broadcasting frames introduce a problem
- Copies of a broadcast frame continuosly flows around the cycle, each computer receiving an infinite number of copies

A Cycle of Bridges




Bridges

Summary

- Frame Filtering
- Planning a Bridged Network
- A Cycle of Bridges
- Distributed Spanning Tree

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Lecture-15

Hammad Khalid Khan

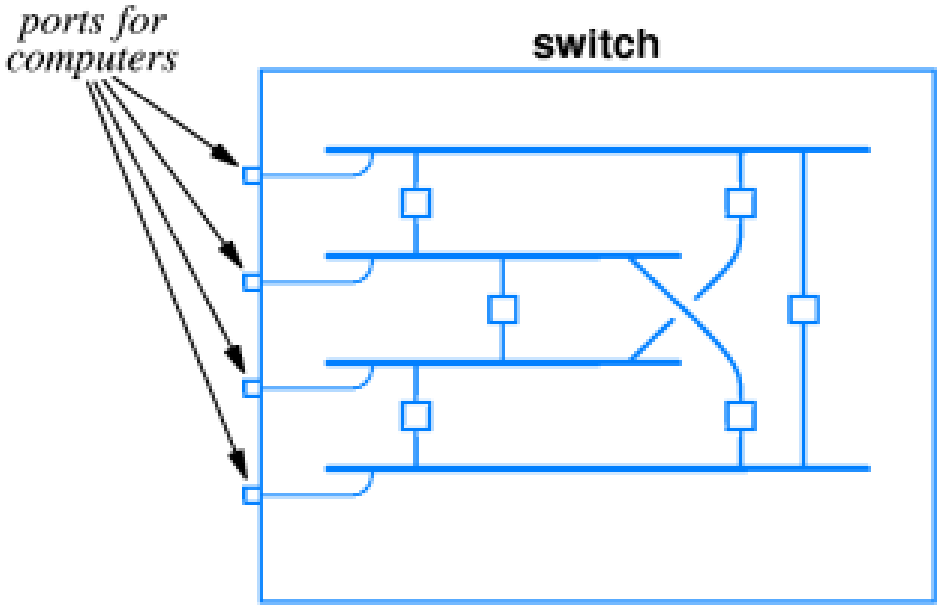
Review Lecture 14

- Start up and Stand by Modes of a Bridge
- Planning a Bridge
- Bridging between Buildings
- Bridging over Long Distance
- Cycle of Bridges and DST

Switching

- A switched LAN consists of a single electronic device that transfers frames among the connected computers
- A hub with multiple ports simulates a single shared medium
- However a switch simulates a bridged LAN with one computer per segment

Switching



Switching

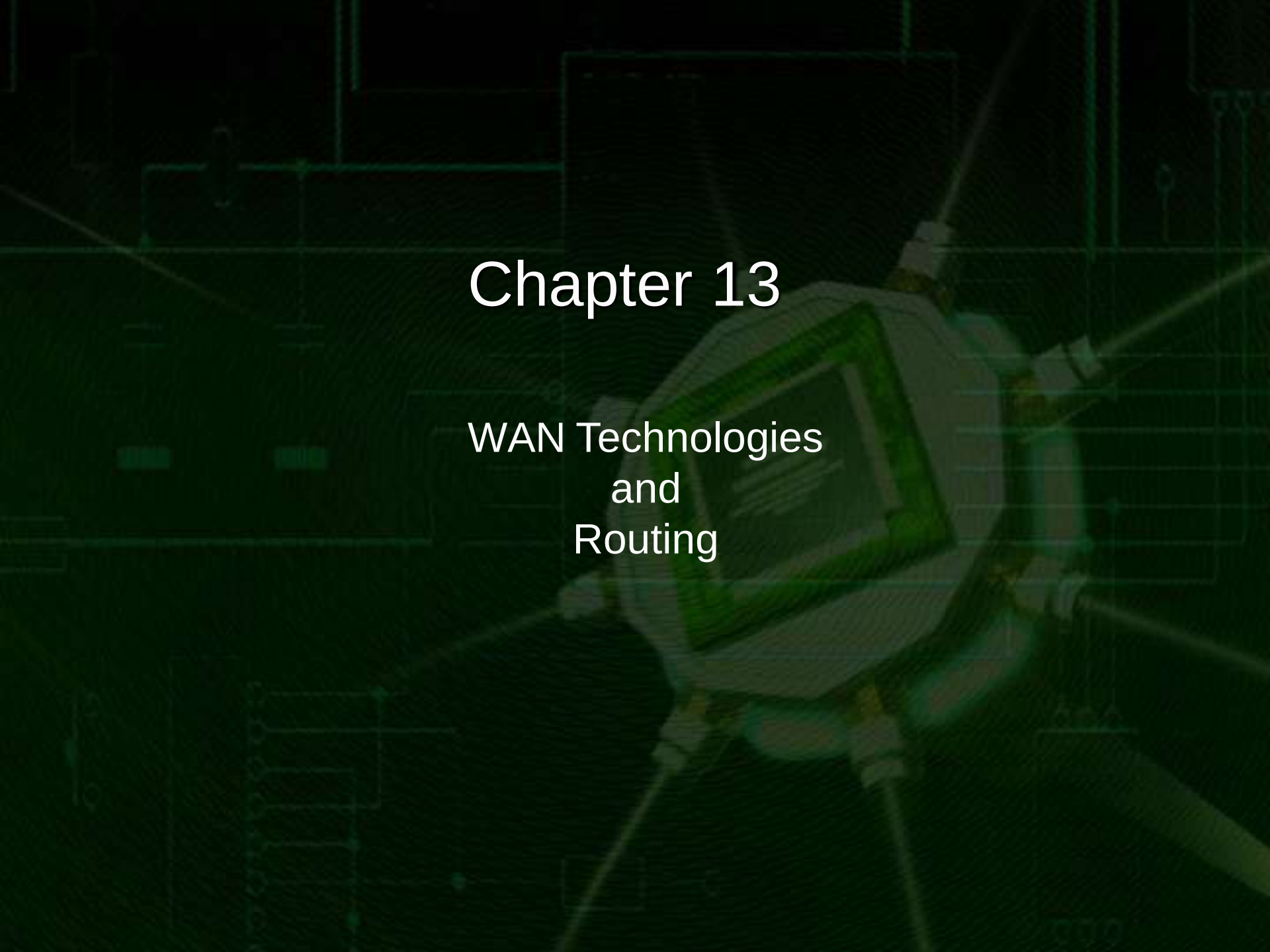
- If a hub is used to connect many computers on a LAN, then only two computers can communicate at a given time
- However if a switch is used, the communication between two computers will not affect the others, pairs of computers can communicate at the same time

Combining Switches and Hubs

- To reduce costs, computers can be connected and distributed to a number of hubs, then the hubs can be connected to a switch
  - Hubs simulate single shared LAN segments
  - Switch simulates a bridged LAN connecting segments

Bridging And Switching With Other Technologies

- Hubs, bridges and switches are not limited to Ethernet logical bus topology
- They are available also for other networking technologies such as Token ring, FDDI etc.
  - FDDI hub
  - Token ring hub



Chapter 13

WAN Technologies
and
Routing

Introduction

- LANs can be extended using techniques in previous chapter
- Can not be extended arbitrarily far or to handle arbitrarily many computers
  - Distance limitations even with extensions
  - Broadcast a problem
- Need other technologies for larger networks

Characterization of Networks

- Local Area Network (LAN)
  - Single building
-
- Metropolitan Area Network (MAN)
  - Single city

- Wide Area network (WAN)
  - Country
  - Continent

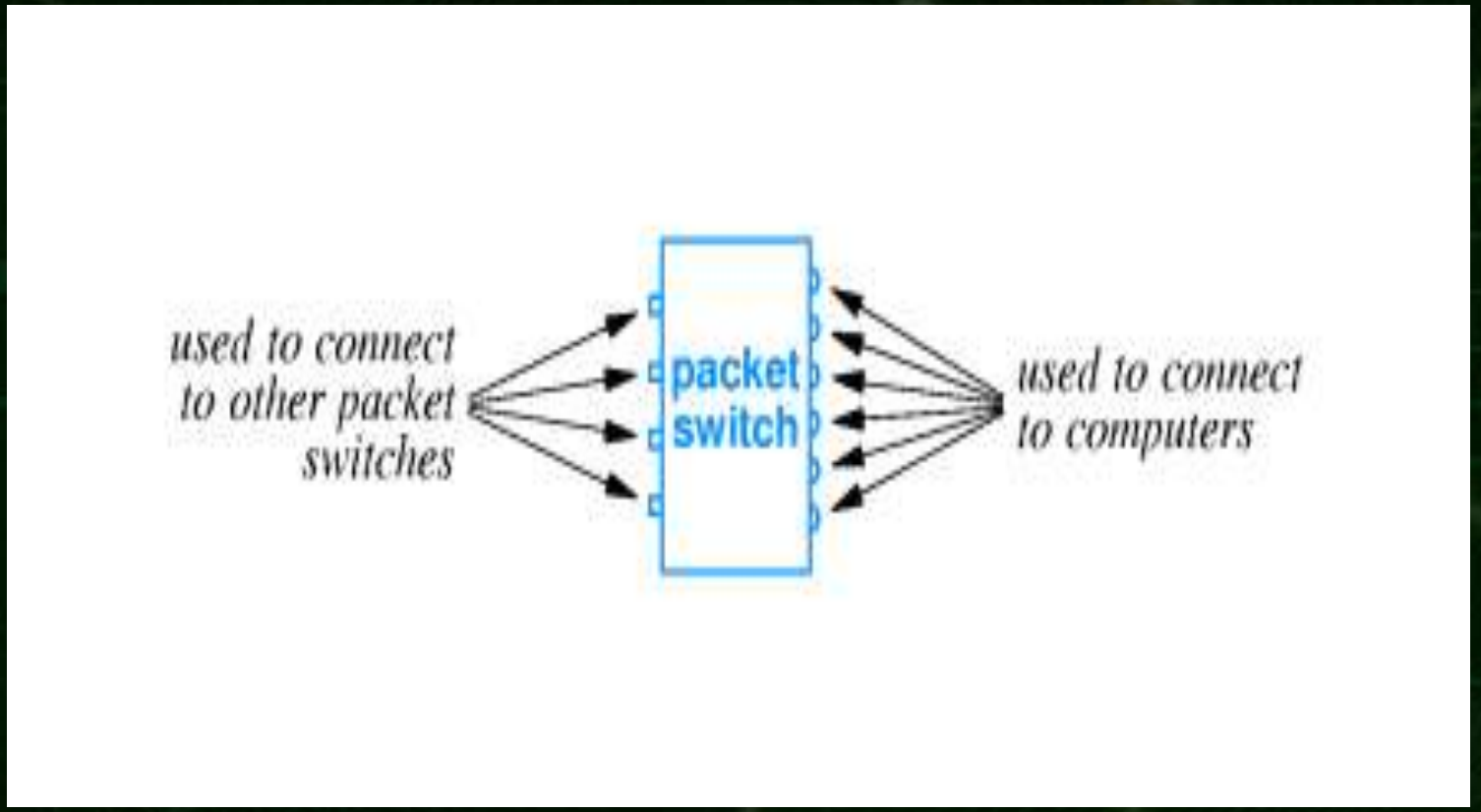
WAN Technology and Routing



Packet Switches

- To span long distances or many computers, network must replace shared medium with packet switches
-
  - Each switch moves an entire packet from one connection to another
  - A small computer with network interfaces, memory and program dedicated to packet switching function

Packet Switches



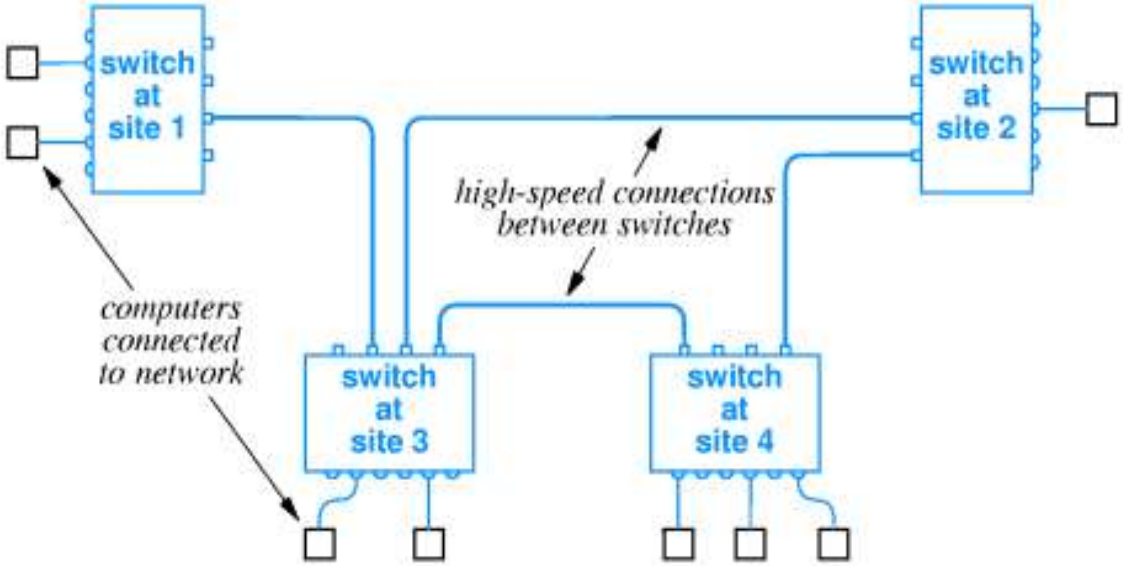
Connections to Packet Switches

- Packets switches may connect to computers and to other packet switches
- Typically high speed connections to other packets switches, lower speed to computers
- Technology details depend on desired speed

Packet Switches as Building Blocks

- Packet switches can be linked together to form WANs
- WANs need not be symmetric or have regular connections
-
- Each switch may connect to one or more other switches and one or more computers

Packet Switches as Building Blocks



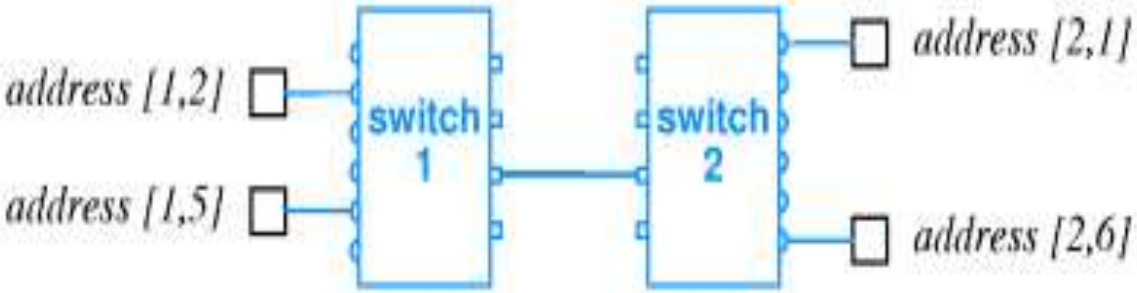
Store and Forward

- Data delivery from one computer to another is accomplished through store-and-forward technology
  - Packet switch stores incoming packet
  - ... and forwards the packet to another switch or computer
  -
- Packet switch has internal memory
  - Can hold packet if outgoing connection is busy
  - Packets for each connection held on queue

Physical Addressing in a WAN

- Similar to LAN
  - Data transmitted in packets (equivalent to frames)
  - Each packet has format with header
  - Packet header includes destination and source addresses
- Many WANs use hierarchical addressing for efficiency
  - One part of address identifies destination switch
  - Other part of address identifies port on switch

Physical Addressing in a WAN



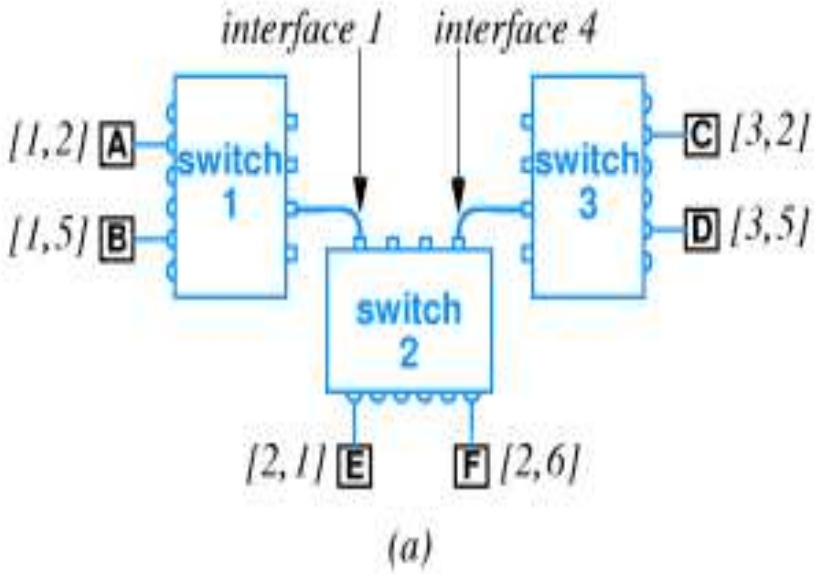
Next-Hop Forwarding

- Packet switch must choose outgoing connection for forwarding
-
  - If destination is local computer, packet switch delivers computer port
  - If destination is attached another switch, this packet switch forwards to next hop through connection to another switch
- Choice based on destination address in packet

Choosing Next Hop

- Packet switch doesn't keep complete information about all possible destination
-
- Just keeps next hop
-
- So, for each packet, packet switch looks up destination in table and forwards through connection to next hop

Choosing Next Hop



(a)

destination	next hop
[1,2]	interface 1
[1,5]	interface 1
[3,2]	interface 4
[3,5]	interface 4
[2,1]	computer E
[2,6]	computer F

(b)

Summary

- Wide Area Networks
- Packet Switches
- Physical Addressing in a WAN
- Store and Forward
- Next Hop Forwarding

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Lecture-16

Hammad Khalid Khan

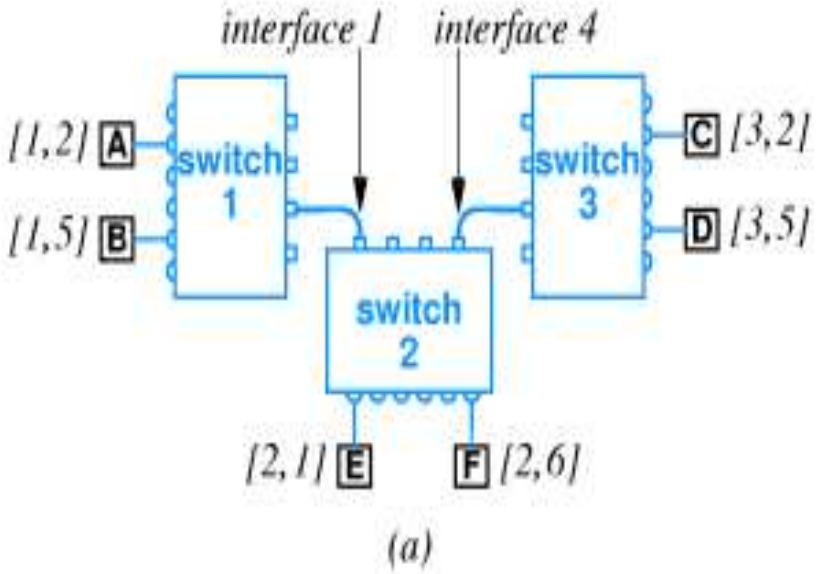
Review Lecture 15

- Wide Area Networks
- Packet Switches
- Physical Addressing in a WAN
- Store and Forward
- Next Hop Forwarding

Choosing Next Hop

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[3,2]	interface 4
[3,5]	interface 4
[2,1]	computer E
[2,6]	computer F

(b)

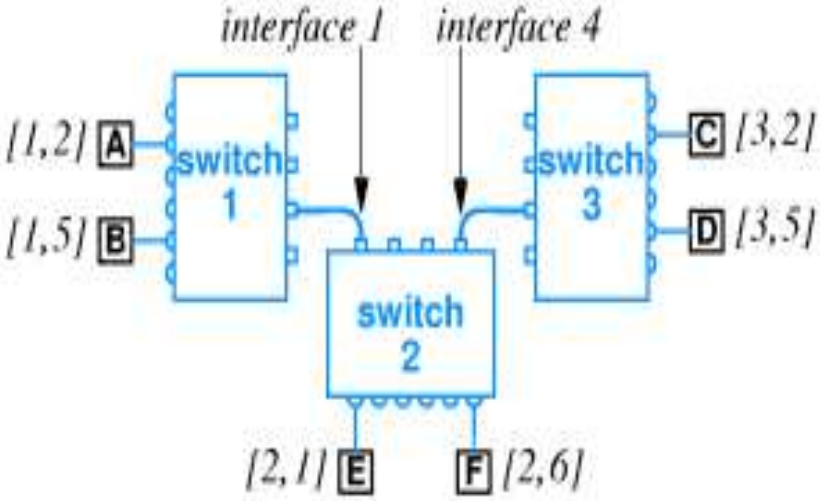
Source Independence

- Next hop to destination does not depend on source of packet, called Source Independence
- Allows fast, efficient routing
- Packet switch need not have complete information, just next hop
  - Reduces total information
  - Increases dynamic robustness - network can continue to function even if topology changes without notifying entire network

Hierarchical Addressing and Routing

The Process of forwarding the packets from one hop to the next is known as Routing

Hierarchical Addressing and Routing



(a)

destination	next hop
[1,2]	interface 1
[1,5]	interface 1
[3,2]	interface 4
[3,5]	interface 4
[2,1]	computer E
[2,6]	computer F

(b)

Hierarchical Addressing and Routing

- In particular, all destinations on same switch have same next hop
- Thus, routing table can be collapsed:

Destination	Next Hop
(1, anything)	interface 1
(3, anything)	interface 4
(2, anything)	local computer

Routing in a WAN

- More computers == more traffic
- Can add capacity to WAN by adding more links and packet switches
- Packet switches need not have computers attached
- Interior switch - No attached computers
- Exterior switch - Attached computers

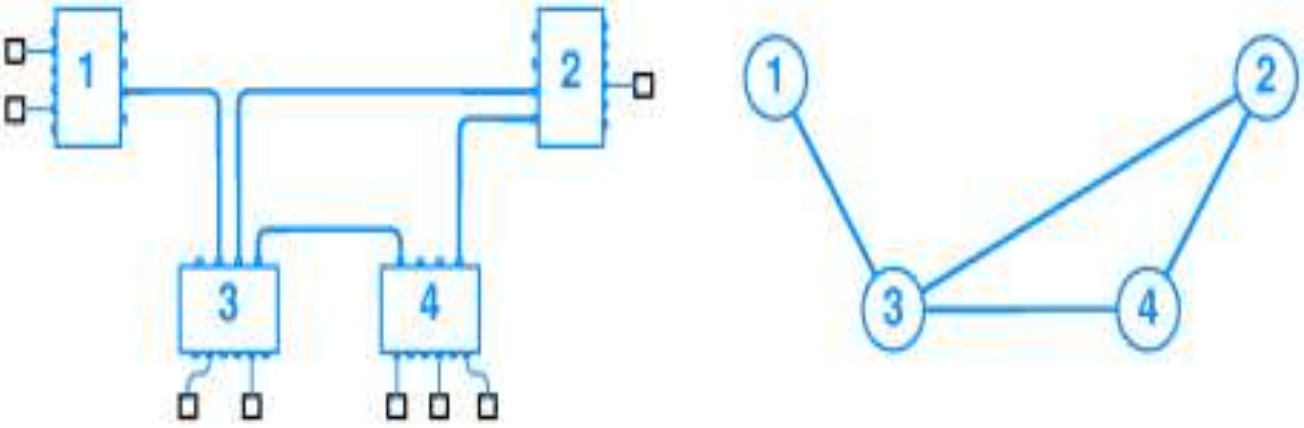
Routing in a WAN

- Both Interior and Exterior switches:
  - Forward Packets
  - Need Routing Tables
- Must have:
  - Universal Routing - Next hop for each possible destination
  - Optimal Routes - Next hop in table must be on shortest path to destination
-

Modeling a WAN

- Use a graph:
  - Nodes model switches
  - Edges model direct connections between switches
- Captures essence of network, ignoring attached computers

Modeling a WAN



Route Computation with a Graph

- Can represent routing table with edges:

destin-ation	next hop
1	-
2	(1,3)
3	(1,3)
4	(1,3)

node 1

destin-ation	next hop
1	(2,3)
2	-
3	(2,3)
4	(2,4)

node 2

destin-ation	next hop
1	(3,1)
2	(3,2)
3	-
4	(3,4)

node 3

destin-ation	next hop
1	(4,3)
2	(4,2)
3	(4,3)
4	-

node 4

- Graph algorithms can be applied to find routes

Redundant Routing Information

- Notice duplication of information in routing table for node 1:

destin-ation	next hop
1	-
2	(1,3)
3	(1,3)
4	(1,3)

node 1

destin-ation	next hop
1	(2,3)
2	-
3	(2,3)
4	(2,4)

node 2

destin-ation	next hop
1	(3,1)
2	(3,2)
3	-
4	(3,4)

node 3

destin-ation	next hop
1	(4,3)
2	(4,2)
3	(4,3)
4	-

node 4

- Switch 1 has only one outgoing connection; all traffic must traverse that connection

Default Routes

- Can collapse routing table entries with a Default Route

- If destination does not have an explicit routing table entry, use the default route:

Default Routes

destination	next hop
1	-
*	(1,3)

node 1

destination	next hop
2	-
4	(2,4)
*	(2,3)

node 2

destination	next hop
1	(3,1)
2	(3,2)
3	-
4	(3,4)

node 3

destination	next hop
2	(4,2)
4	-
*	(4,3)

node 4

Summary

- Source Independence
- Hierarchical Addressing and Routing
- Routing in a WAN
- Default Routes

COMPUTER NETWORKS

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Lecture-17

Hammad Khalid Khan

Review Lecture 16

- Source Independence
- Hierarchical Addressing and Routing
- Routing in a WAN
- Modeling a WAN
- Route Computation and Default Routes

Building Routing Tables

- How to enter information into routing tables:
  - Manual entry
  - Software

- How to compute routing table information:
  - Static routing - At boot time
  - Dynamic routing - Allow automatic updates by a program

Building Routing Tables

- Static Routing
  - Simple
  - Low Network Overhead
  - Inflexible
- Dynamic Routing
  - Can work around network failures automatically

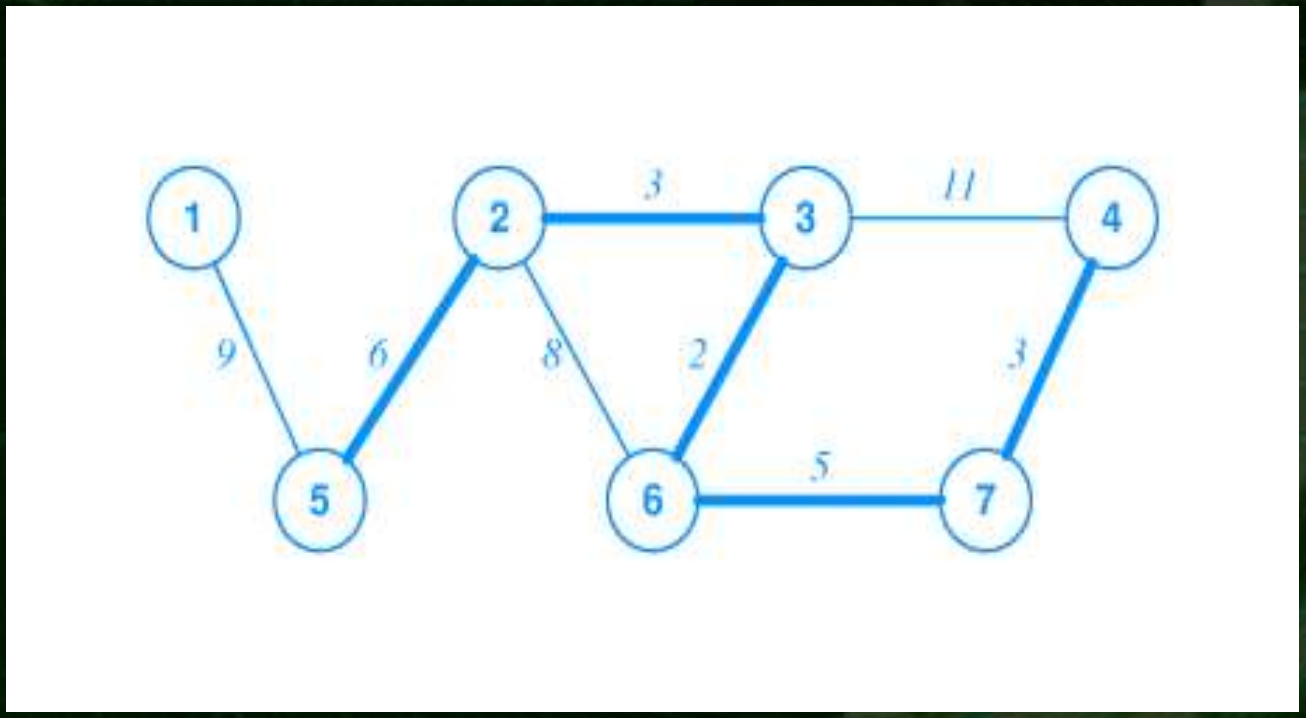
Computing Shortest Path in a Graph

- Assume graph representation of network at each node
- Use Djikstra's algorithm to compute shortest path from each node to every other node
- Extract next-hop information from resulting path information
- Insert next-hop information into routing tables

Weighted Graph

- Djikstra's algorithm can accommodate weights on edges in graph
- Shortest path is then the path with lowest total weight (sum of weights of all edges)
- Shortest path not necessarily fewest edges (or hops)

Weighted Graph



Distance Metrics

- Weights on graph edges reflect "cost" of traversing edge
  - Time
  - Dollars
  - Hop count (weight == 1)
- Resulting shortest path may not have fewest hops

Distributed Route Computation

- Each packet switch computes its routing table locally and sends messages to the neighbors
- Updates information periodically
- Network adapts if a link or a packet switch fails
- Packet switches modifies tables to avoid failed hardware

Distance Vector Routing

- Local information is next-hop routing table and distance from each switch
- Switches periodically broadcast topology information i.e. (*destination, distance*)
- Other switches update routing table based on received information

Vector-Distance Algorithm

- In more detail:

Wait for next update message

Iterate through entries in message

If entry has shorter path to destination:

Insert source as next hop to destination

Record distance as distance from next hop to destination

PLUS distance from this switch to next hop


Link-State Routing (SPF)

Comparison

- Distance Vector Routing
  - Very simple to implement
  - Packet Switch updates its own routing table first
  - Used in RIP
- Link-State Algorithm
  - Much more complex
  - Switches perform independent computations
  - Used in OSPF

Examples-WAN Technology

- ARPANET
  - Began in 1960s
  - Funded by Advanced Research Projects Agency, an organization of the US Defense Department
  - Incubator for many of current ideas, algorithms and internet technologies

Examples-WAN Technology

- X.25
  - Early standard for connection-oriented networking
  - From ITU, which was originally CCITT
  - Predates computer connections, used for terminal/timesharing connection

Examples-WAN Technology

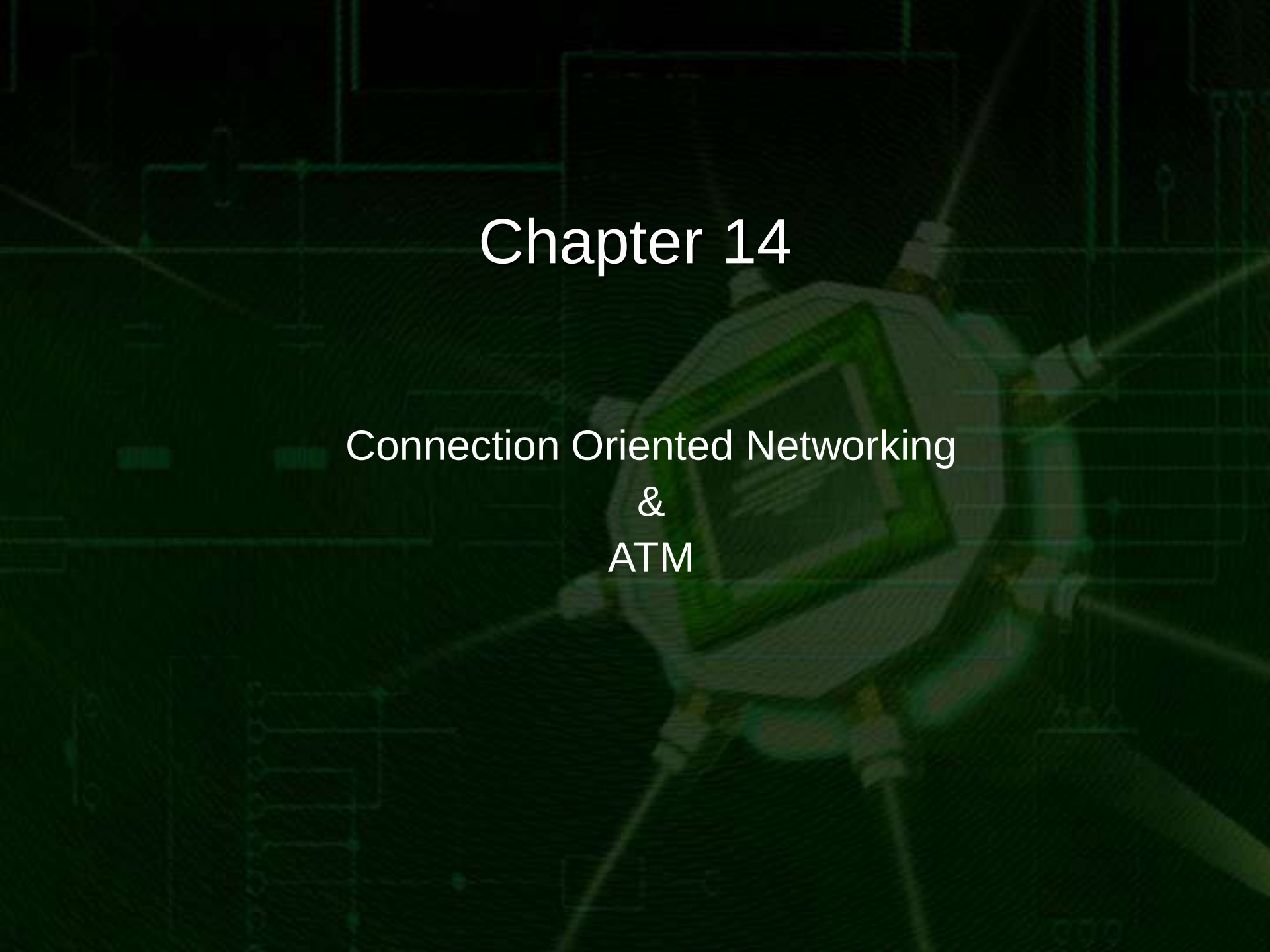
- Frame Relay
  - Telco service for delivering blocks of data
  - Connection-based service; must contract with telco for circuit between two endpoints
  - Typically 56Kbps or 1.5Mbps; can run to 100Mbps

Examples-WAN Technology

- SMDS - Switched Multi-megabit Data Service
  - Also a Telco service
  - Connectionless service; any SMDS station can send a frame to any other station on the same SMDS "cloud"
  - Typically 1.5-100Mbps

Examples-WAN Technology

- ATM - Asynchronous Transfer Mode
  - Designed as single technology for voice, video, data, ...
  - Low jitter (variance in delivery time) and high capacity
  - Uses fixed size, small cells - 48 octets data, 5 octets header
  - Can connect multiple ATM switches into a network



Chapter 14

Connection Oriented Networking
&
ATM

Summary (CH. 13)

- Packet Switch
- Next-Hope Forwarding
- Source Independence
- Hierarchical Addressing

Introduction

- LANs and WANs can both connect multiple computers, but they have different base technologies and meet different goals
- ATM is a single technology that is designed to meet the goals of both LANs and WANs
- ATM uses the concept of Connection-Oriented Networking

Summary

- Routing Table Computation
- Shortest Path Computation in a Graph
  - Dijkstra’s Algorithm
- Distributed Route Computation
  - Distance Vector Routing
  - Link State Routing
- Example WAN Technologies

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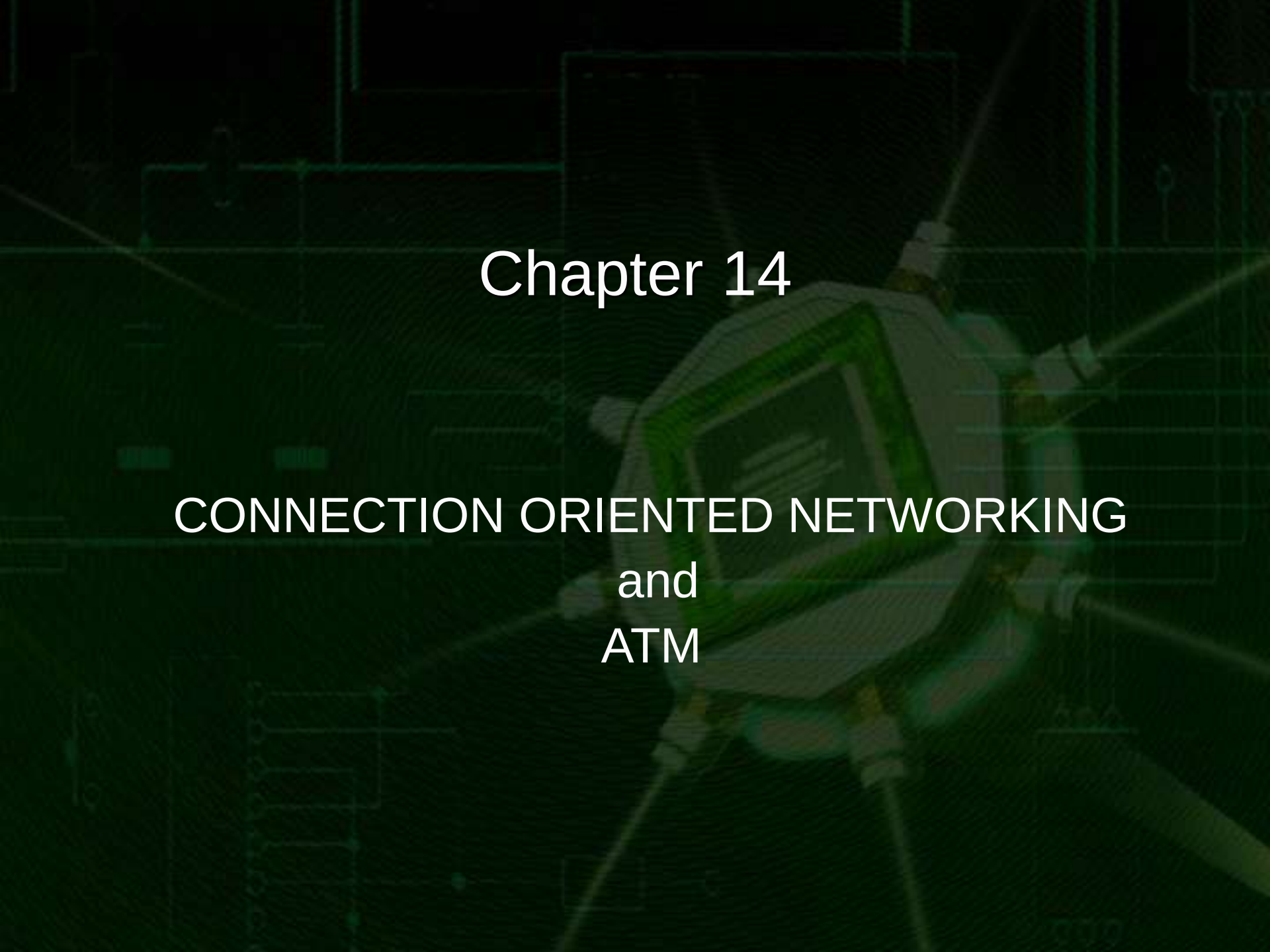
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Lecture-18

Hammad Khalid Khan

Review Lecture 17

- Routing Table Computation
- Shortest Path Computation in a Graph
  - Dijkstra's Algorithm
- Distributed Route Computation
  - Distance Vector Routing
  - Link State Routing
- Example WAN Technologies



Chapter 14

CONNECTION ORIENTED NETWORKING
and
ATM

Introduction

- LANs and WANs can both connect multiple computers, but they have different base technologies and meet different goals
- ATM is a single technology that is designed to meet the goals of both LANs and WANs
- ATM uses the concept of Connection-Oriented Networking

Asynchronous Transfer Mode

- Telephone companies (Telcos) introduced ATM to meet several goals:
  - Universal service – for all subscribers
  - Support for all uses – voice, video, and data
  - A single, unified infrastructure (no separate LANs & WANs)
  - Guaranteed service, when appropriate
  - Support for low-cost devices

Jitter

- Jitter is the term used for variance in transmission delays
- Jitter is significant for voice, video, and data
- Jitter in LANs can occur when a packet is delayed because the network is busy

Packet Sizes

- Large packets result in less overhead because a smaller fraction of the packet is used for header information
- Optimum networks use 4KB packets or larger
- Large packets can’t easily be used for voice
  - 8 bit samples (at 125 usec per sample) would require half a second to fill a 4KB packet.
  - Echo cancellation can only be used with low transmission delays

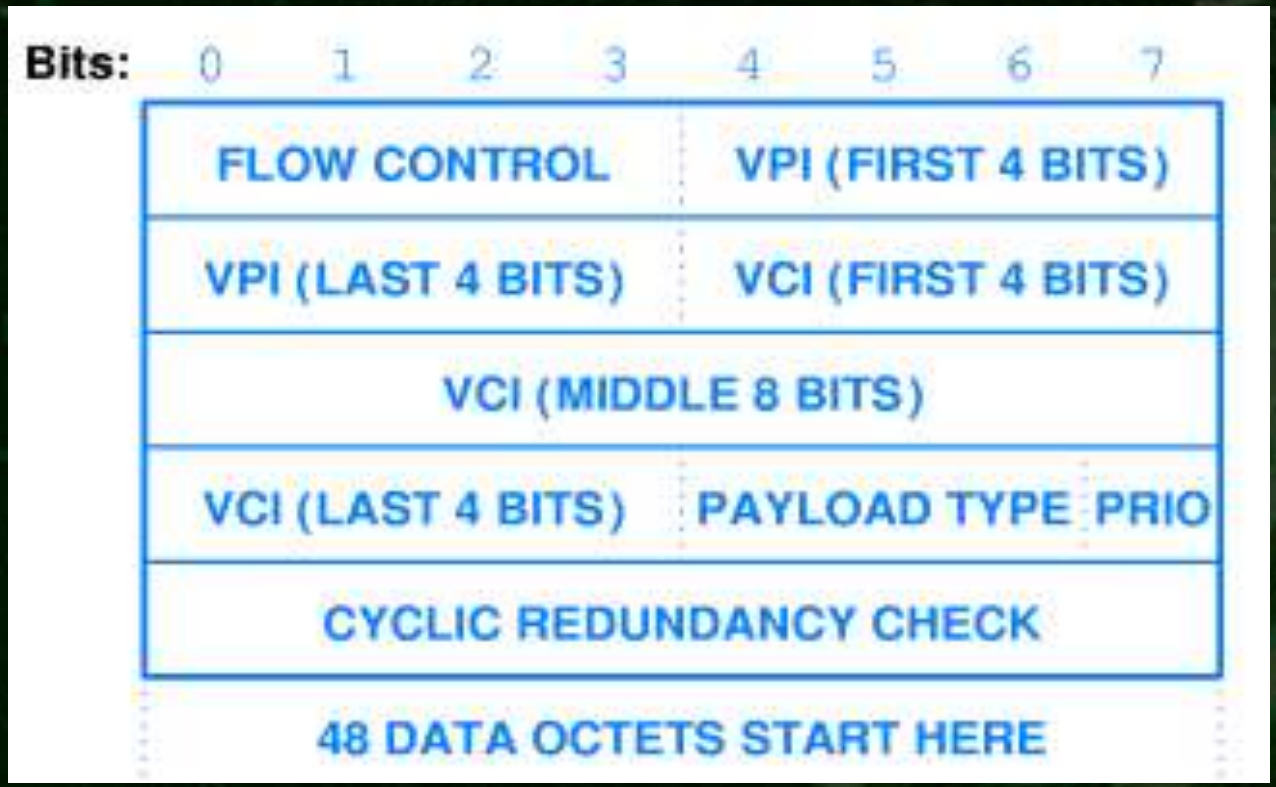
ATM Cells

- To meet its goals, ATM uses small, fixed-sized packets called cells
- Each cell has 53 octets
- VCI/VPI fields identify the cell’s destination
- PRIO tell if cell can be discarded

ATM Cells

- CRC checks the header's bits only!
- ATM header is about 10 percent of the cell
- Ethernet can have overhead of only 1 %
- Engineers sometime call ATM overhead the Cell Tax

ATM Cells


Bits:
0 1 2 3 4 5 6 7
FLOW CONTROL
VPI (FIRST 4 BITS)
VPI (LAST 4 BITS)
VCI (FIRST 4 BITS)
VCI (MIDDLE 8 BITS)
VCI (LAST 4 BITS)
PAYLOAD TYPE
PRIO
CYCLIC REDUNDANCY CHECK
48 DATA OCTETS START HERE

Connection-Oriented Service

- The connection-oriented service paradigm for networking is similar to the manner in which telephones are used:
  - A caller dials the number of the destination
  - The telephone at the destination signals the arrival of a Connection Request
  - If the called person doesn’t answer, the caller gives up after waiting for a Timeout
  - If the called person does answer, then the connection is established, and data is exchanged

Connection-Oriented Service

- In data communications, a binary connection identifier is given to each of the two parties to the connection to enable identification of the connection

Virtual Channels (or Circuits)

- Connections in ATM are called virtual channels (VC), or virtual circuits (a term preferred by some)
- These are called virtual, since connections are formed in ATM by storing values in memory locations (tables) in ATM switches, as opposed to making actual electrical connections

Virtual Channels (or Circuits)

- The VC is identified by a 24-bit value formed from:
  - The VPI, or virtual path indicator (8 bits), which identifies a particular path through the network; &
  - The VCI, or virtual channel indicator (16 bits), which identifies the channel in the virtual path is being used by the connection
- Most frequently, the 24-bit pair is treated as just a single connection identifier by computers

Summary

- Connection-Oriented Networking and ATM
- Asynchronous Transfer Mode
- ATM Design and Cells
- Connection-Oriented Services
- Virtual Channels

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Lecture-19

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Review Lecture 18

- A single Global Technology
- ATM Design and Cells
- Connection Oriented Service
- VPI/VCI

Labels and Label Switching

- An ATM network is built from interconnected ATM switches
- Attachment points on switches, or ports, can be connected to computers or other ATM switches

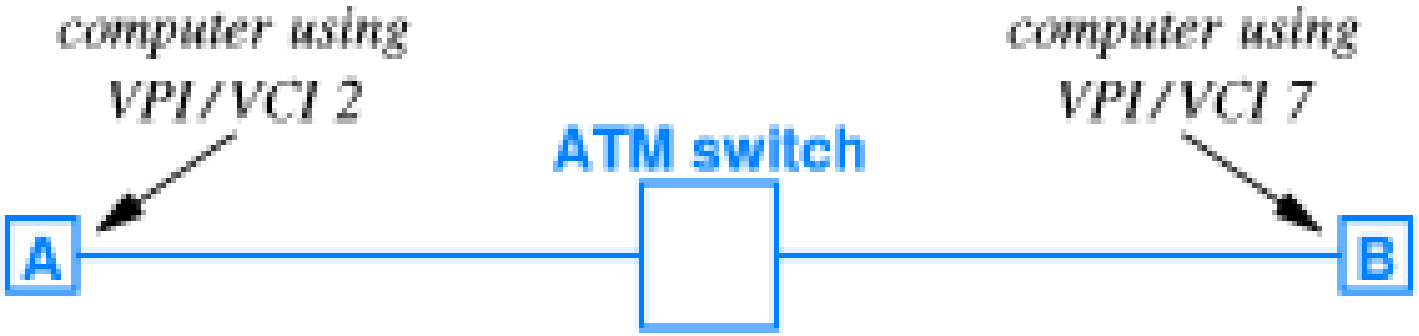
Labels and Label Switching

- As cells arrive at an ATM switch, their VPI/VCI is modified using a forwarding table which gives the new VPI/VCI for the next leg of the cell's trip
- The forwarding table is essentially indexed by the incoming cell's VPI/VCI, and the contents yield the new VPI/VCI

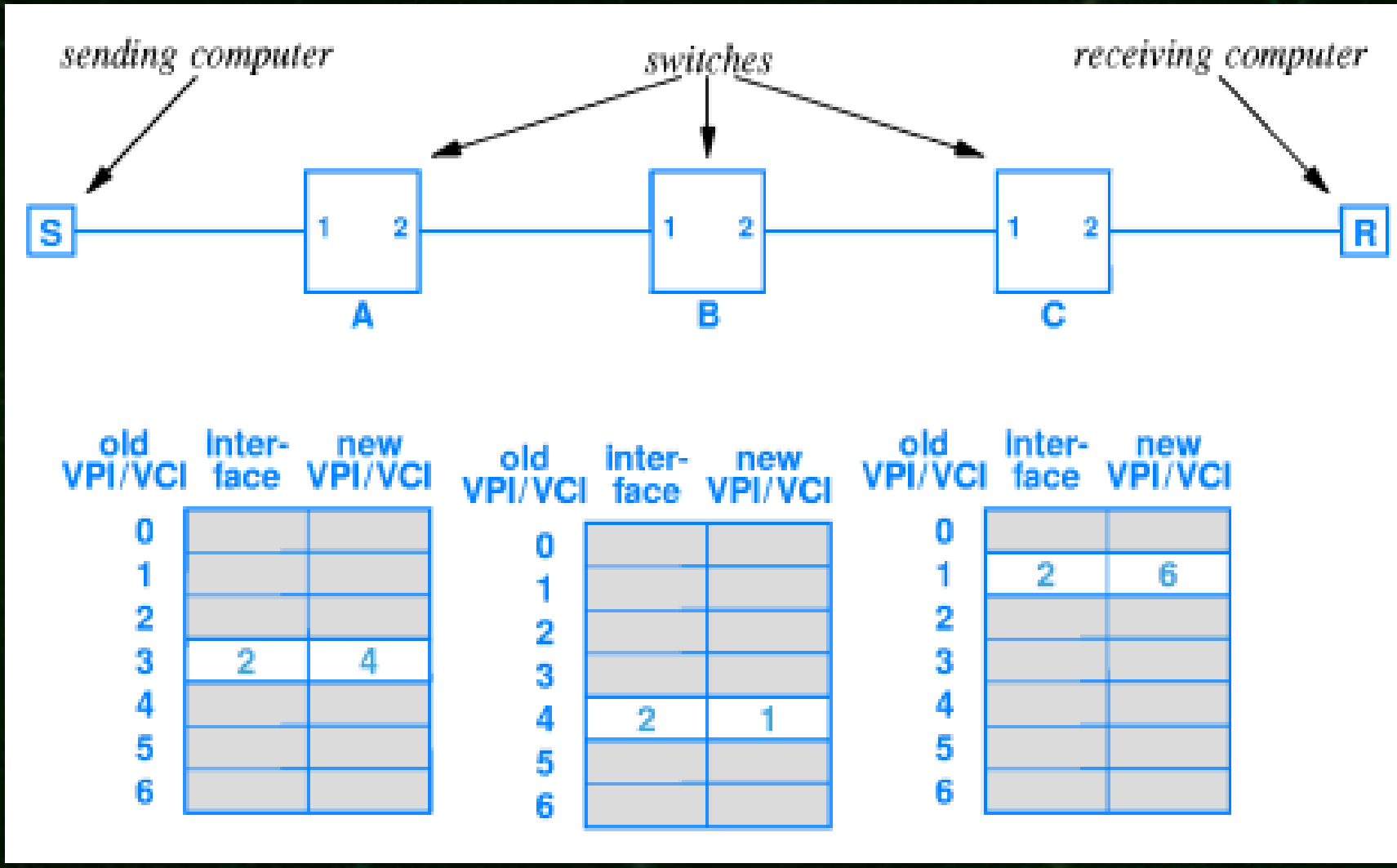
Label Rewriting

- The replacement of the incoming cell's VPI/VCI with a probably different VPI/VCI is called Rewriting
- ATM is thus called a Label Rewriting, or Label Switching system
- Thus two computers with a connection through an ATM network will likely have different VPI/VCI values for each end of the connection!

Label Switching



Example

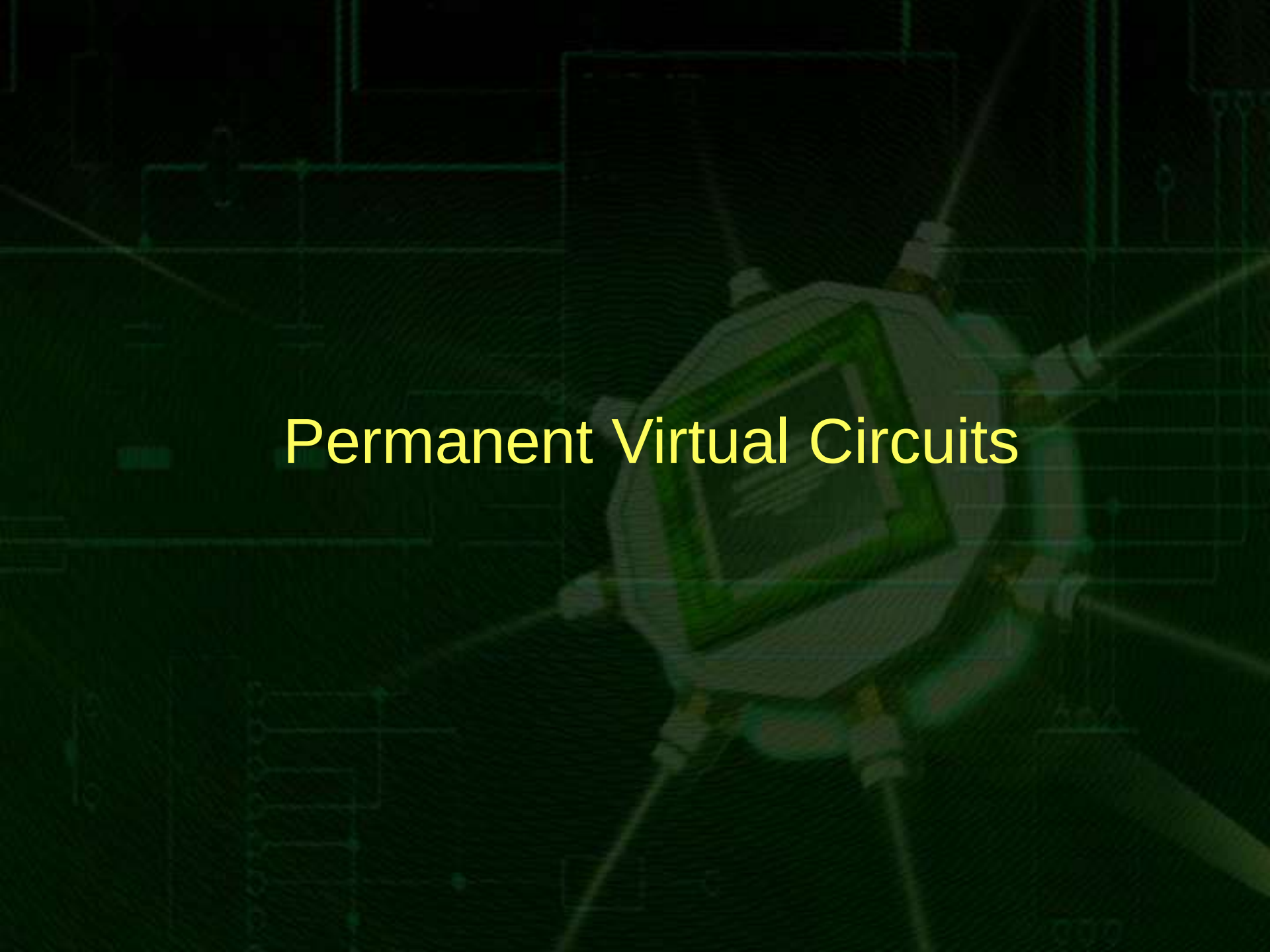

sending computer
switches
receiving computer
S
1
2
A
1
2
B
1
2
C
R
old VPI/VCI
Inter-face
new VPI/VCI
0
1
2
3
4
5
6
2
4
old VPI/VCI
Inter-face
new VPI/VCI
0
1
2
3
4
5
6
2
1
old VPI/VCI
Inter-face
new VPI/VCI
0
1
2
3
4
5
6
2
6

Example

- The sending computer, S, uses a VPI/VCI of 3 and sends a cell
- Switch A receives the cell and looks up 3, then rewrites the VPI/VCI as 4, and sends the cell out of its port 2
- Switch B receives the cell and looks up 4, then rewrites the VPI/VCI as 1, and sends the cell out of its port 2

Example

- Switch C receives the cell and looks up 1, then rewrites the VPI/VCI as 6, and sends the cell out of its port 2
- The receiving computer, R, receives the cell with a VPI/VCI of 6, which is the value it is using for the connection
- Forwarding tables in each switch must be coordinated to define meaningful “paths” through the network


Permanent Virtual Circuits

Permanent Virtual Circuits

- Provisioning requires two steps:
  - Determine a complete path (that is, identify the switches that will be used)
  - Choose appropriate VCI/VPI for each step in the path, and configure each adjacent pair of switches (easy, since each switch rewrites the VCI/VPI)

Switched Virtual Circuits

- Most networks offer dynamic connections which last for a relatively short time
- To handle this, ATM can dynamically establish a Switched Virtual Circuit (SVC), allow it last as long as necessary, and then terminate it
- The terminology comes from the telcos, where switching system normally refers to call switching

Establishing an SVC

- Computer sends a connection request to the switch to which it is attached
- Software in the switch finds a network path to the destination and sends along the connection request
- Each pair of switches in the path communicate to choose a VPI/VCI for their tables

Establishing an SVC

- Once the connection is accepted by the destination, a message is sent back to the originating computer to indicate the SVC is ready
- If any switch, or the destination computer, does not agree to setting up the VC, an error message is sent back and the SVC is not established

Signaling

- The term signaling is used to describe communication about the network, as opposed to communication that just uses the network
- A computer uses signaling, with reserved VCI/VPI values, to communicate with a switch to establish a connection or send other network control messages

Signaling

- These connection requests and network control messages are also sent in cells, and since the VPI/VCI used in these cells are reserved, there is no confusion between data and control cells

Quality of Service (QoS)

- Networks are more frequently being designed to allow specification of the quality of service required by users

- For example, a typical voice telephone call might give a QoS parameter for througput of 64Kbps and delay less than 500 msec. A user sending video might require throughput of 2Mbps

Providing Desired QoS

- QoS desires are specified at connection setup time, and are never altered for the connection duration
- Switches along the path must reserve resources to guarantee the QoS
- If a switch cannot provide the resources, it rejects the connection request and an appropriate error message is returned

ATM QoS Specifications

- Constant Bit Rate (CBR): Used for uncompressed audio and video, since these have predefined maximum data rates
- Variable Bit Rate (VBR): Used for compressed audio and video, where the data rate depends on the level of compression that can be achieved

ATM QoS Specifications

- Available Bit Rate (ABR) is used for typical data applications (where the data rate may be unknown and bursty), and allows use of whatever bandwidth is available at a given time

Summary

- Label switching and Label Rewriting
- Permanent Virtual Circuits
- Switched Virtual Circuits
- Quality Of Service

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Lecture-20

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Review Lecture 19

- Label Switching
- An Example ATM Network
- Permanent Virtual Circuits (PVC's)
- Switched Virtual Circuits (SVC's)
- Quality of Service (QoS)

Cells vs Packets

- ATM designers chose cells over packets because:
  - Cells aren't variable length, and memory management for them is simpler. Handling variable length packets leads to memory fragmentation
  - Variable length packets require hardware to accommodate the largest possible packet, and thus to detect the end of the packet. With cells, bits can just be counted as they arrive

Cells vs Packets

- The length of time required to send a variable length packet is variable, and requires complicated interrupt schemes to detect completion of transmission
- QoS can’t be guaranteed with variable length packets as easily as it can with fixed length cells

ATM Speed

- ATM designers also chose cells to meet the need for speed, since it was designed to handle arbitrarily large numbers of users, each of which could be willing to pay for high throughput

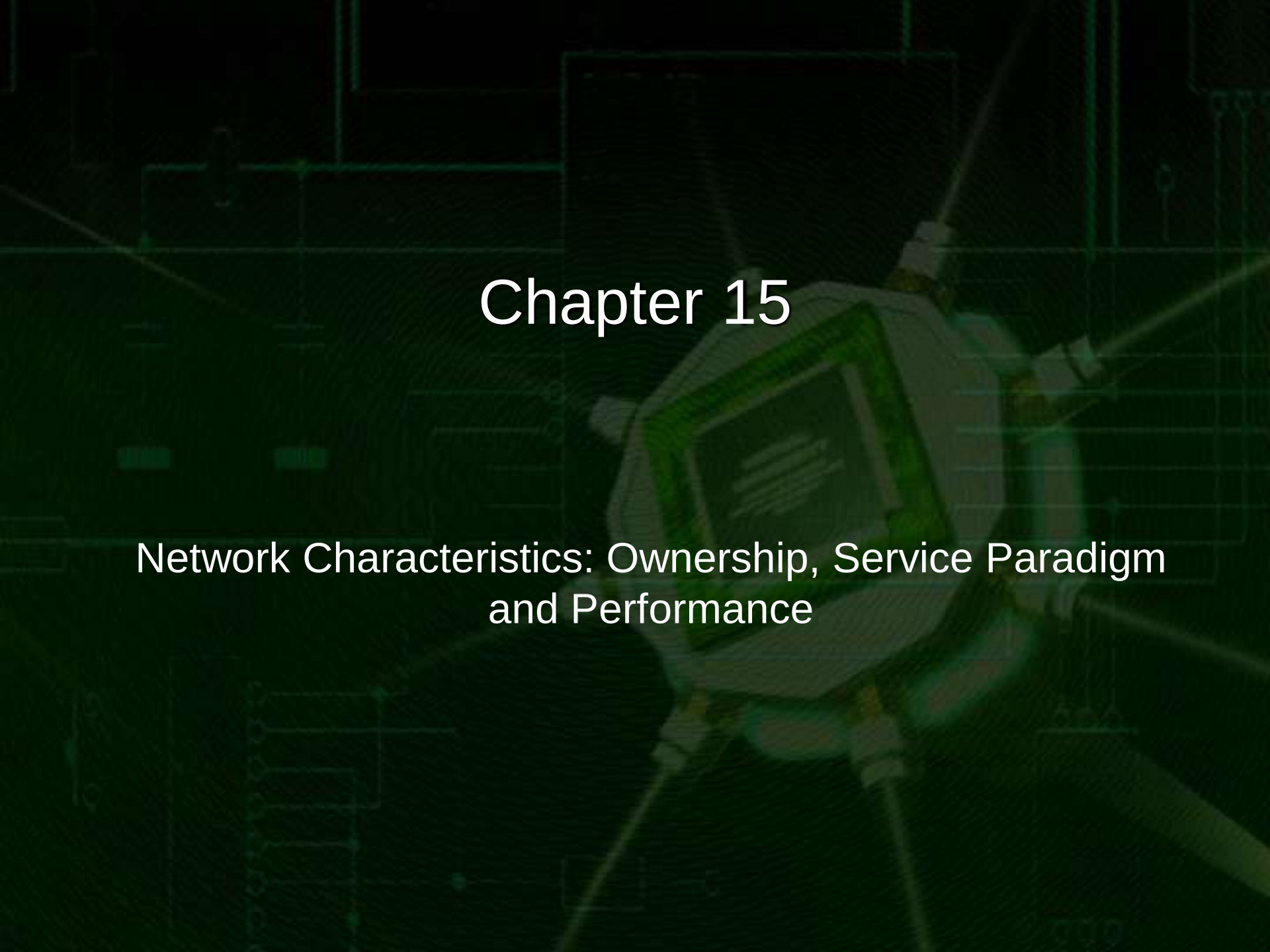
- ATM is designed to work on fiber (but can be used with twisted pair). A typical port on an ATM switch operates at OC-3 speed (155 Mbps) or higher

ATM Critique

- ATM is far more expensive than typical LAN hardware
- Connection setup time may be excessive for short communications
- Cell tax consumes 10 % of network capacity
- QoS requirements might be unknown, leading to applications picking values that are too high or too low

ATM Critique

- Broadcast is inefficient, and has to be simulated by sending the same message to each computer separately
- ATM, as a single universal networking technology, has only minimal provision for interoperation with other technologies



Chapter 15

Network Characteristics: Ownership, Service Paradigm and Performance

Network Ownership

- Private Network
  - Owned by single organization or company
- Public Network
  - Owned by common carrier e.g. Phone Company)

Private Network

- Often LAN Technology
- Multiple LANs in a building or campus linked together
- Sometimes called Intranet

Private Network Architecture

- Operates autonomously from other networks (e.g. Internet)
- Usually includes one or few closely managed external connections
- May restrict access at connections

Managing Private Networks

- Organization buys own equipment
- Hires staff to design, implement, maintain and upgrade network
- Responsible for all network management

Extending Private Networks

- Large organizations may have multiple buildings or campuses
- Can only install cables on own property
- May contract for leased lines from common carrier

Public Networks

- Operated by common carrier
- May be telephone company or other organization that builds network out of leased lines
- Multiple organizations subscribe and connect
- Data transits public network to other organizations

Advantages & Disadvantages

Private

- Advantages
  - The owner has complete control over both the technical decision and polices
- Disadvantages
  - Expensive to install and maintain

Public

- Advantages
  - Flexibility
- Disadvantages
  - No decision making power equipment or policies

Virtual Private Network

- VPN (Virtual Private Network) combines features of private and public networks
  - Limited to single organization
  - Uses public network for connectivity
- Connections, sometimes called Tunnels, connect sites
  - Each site sees tunnel as point-to-point link
  - No access for other users of public network

Service Paradigm

- Connection-Oriented
  - Similar to telephone system; endpoints establish and maintain a connection as long as they have data to exchange
- Connectionless
  - Similar to postal system; endpoint puts data to send into a packet and hands to network for delivery


Public and Private Networks

Summary

- Cells vs. Packets
- Label switching vs. Routing
- ATM Critique
- Network Characteristics (Chapter 15)
- Network Ownership
- Private and Public Networks

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Lecture-21

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Review Lecture 20

- Cells vs Packets
- Label Switching vs Routing
- ATM Critique
- Network Ownership
  - Private vs Public Networks

Virtual Private Network

- VPN (Virtual Private Network) combines features of private and public networks.
  - Limited to single organization
  - Uses public network for connectivity
- Connections, sometimes called Tunnels, connect sites
  - Each site sees tunnel as point-to-point link
  - No access for other users of public network

Guaranteeing Absolute Privacy

- In addition to restricting packets, VPN systems use encryption to guarantee absolute privacy.

- Even if an outside does manage to obtain a copy of the packet, the outside will be unable to interpret the contents.

Service Paradigm

- At the lowest level, most networks transfer individual packets of data and the network requires each packet to follow an exact format dictates by the hardware.

- Other offer additional facilities the hide may of the details of the packets.
  - Interface Paradigms Or Service Paradigms

Service Paradigm

- Connection-Oriented
  - Similar to telephone system; endpoints establish and maintain a connection as long as they have data to exchange.
- Connectionless
  - Similar to postal system; endpoint puts data to send into a packet and hands to network for delivery.

Connection-Oriented Service

- One endpoint requests connection from network.
- Other endpoint agrees to connection.
- Computers exchange data through connection.
- One endpoint request network to break connection when transmission is complete.

Continuous and Bursty Traffic

- Networks handling voice or video are engineered to accept and deliver continuous data at fixed rate.
- Others are designed to handle burst traffic typical of computer networks.
- Connection does not disappear when no data is sent.

Simplex & Full Duplex Connections

- Some connection-oriented technologies provide full duplex while other allow on simplex connection.

  - To communicate using a simplex design a pair of computers must establish two connection; one from Computer A to Computer B and another from Computer B to A.

Connection Duration & Persistence

- Connections can be made on-demand or set up permanently.
  - Switched connection or switched virtual circuit.
  - Permanent connection or provisioned virtual circuit.

- Permanent connections
  - Originally hard-wired
  - Now configured at system unit time.

Connection Duration & Persistence

- Switched Connections
  - Computer maintains permanent connection to network
  - Network makes connection on demand

- Internal components are switches; network is a switched data network.

Service Guarantees

- Some connection-oriented networks provide guarantees about the service that computer will receive.
- They may guarantee a throughput rate or maximum packet loss rate.
- For example, ATM provides statistical guarantee about performance.

Stream or Message Interface

- Some connection-oriented networks provide stream interfaces
  - No boundaries are recorded; that receive may receive a single block of 60 characters even though the send sends three blocks of 20 characters.

- Others provide a message interface
  - Delivers data in the same size chunks that the sender transmitted.

Connectionless Service

- No connection necessary
- Source of data adds destination information to data and delivers to the network.
- Network delivers each data item individually.

Interior and Exterior Service Paradigms

- A network providing one service paradigm to the attached computers can use an entirely different service paradigm internally.

- For Example: ARPANET
  - Connection-Oriented Internally
  - Connection-less externally

Comparison

- Connection-Oriented
  - Accounting easier.
  - Application can learn of network problems immediately.
- Connectionless
  - Less overhead.
  - Easier to implement network.

Examples of Service Paradigms

Technology	Connection-Oriented	Connectionless	used for LAN	used for WAN
Ethernet		•	•	
Token Ring		•	•	
FDDI		•	•	
Frame Relay	•			•
SMDS		•		•
ATM	•		•	•
LocalTalk		•	•	

Addresses & Connection Identifiers

- Address is a complete, unique identifier.
- Connectionless delivery requires address on each packet.
- Connection-Oriented delivery can use a shorthand that identifies the connection rather than the destination.

Addresses & Connection Identifiers

- ATM - 160-bit address, 24-bit connection identifier
  - Connection identifier includes:
    - 8-bit virtual path identifier (VPI)
    - 16-bit virtual circuit identifier (VCI)
  - Connection identifier local to each computer.
  - May be different in different parts of the ATM switch.

Summary

- Virtual Private Networks
- Service Paradigm
- Connection-Oriented Service
- Connection-less Service

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Lecture-21

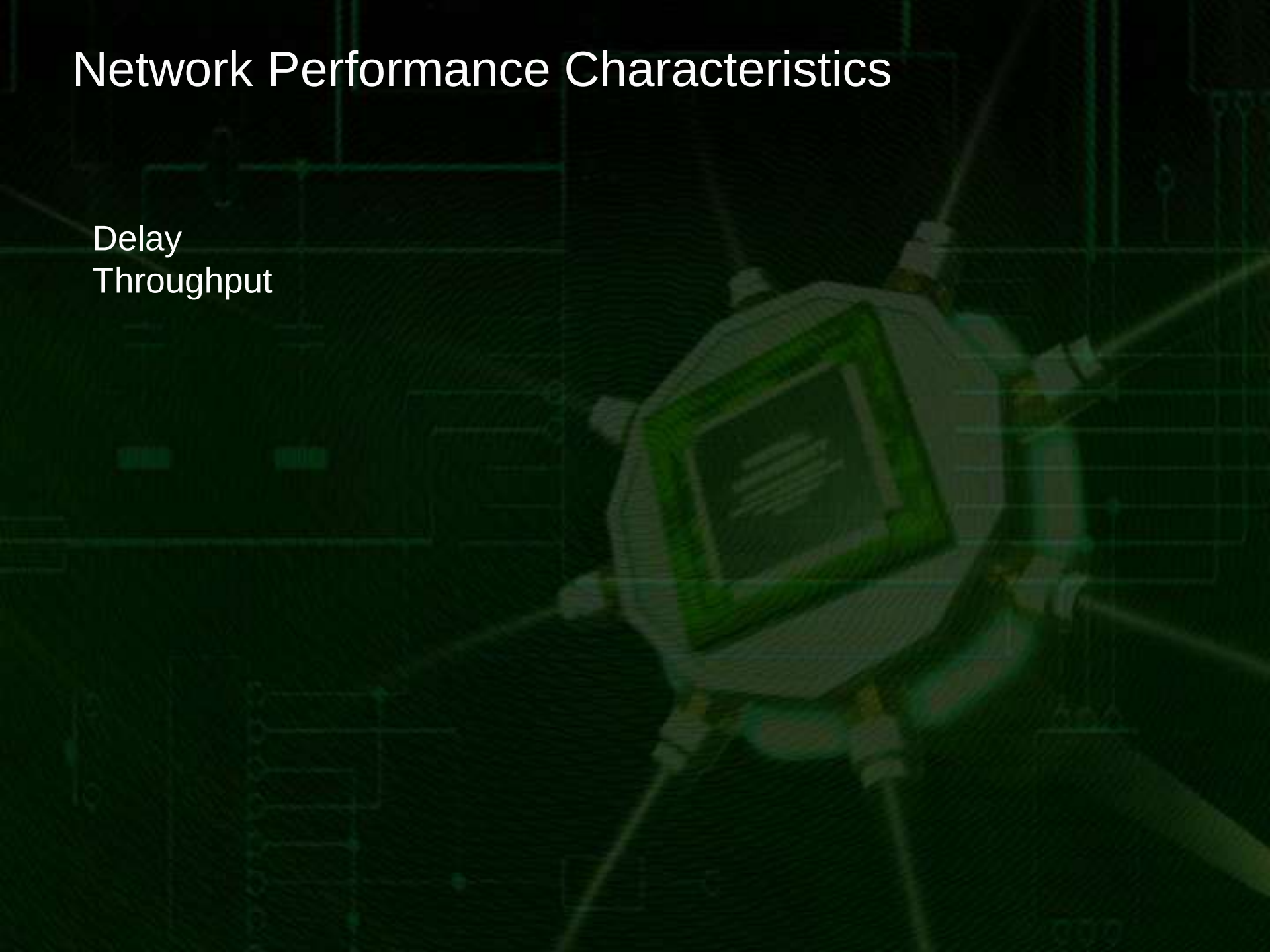
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Review Lecture 21

- Virtual Private Networks (VPNs)
- Guaranteeing Absolute Privacy
- Service Paradigm
  - Connected-Oriented Service
  - Connectionless Service
- Address and Connection Identifiers

Network Performance Characteristics

Delay
Throughput



Delay

- Important quantitative property of networks.
- Delay is a measure of how long it takes for a bit of data to travel across the network from one computer to the other.
- Measured in seconds or fractions of seconds (milliseconds).
- Maximum and Average Delay.

Types of Delay

- Propagation delay - time to travel across medium.
- Switching delay - time for network. component (hub, bridge, packet switch) to forward data.
- Access delay - time to get control of medium (CSMA/CD, token).
- Queuing delay - time enqueued in packet switches.

Throughput

- Throughput is a measure of the rate at which data can be sent through the network.

- The throughput capability of the underlying hardware is called bandwidth.

Throughput

- Because each frame contain headers the effective throughput is less than the hardware bandwidth.

- Networking professional often use the term speed as a synonym for throughput.

Relationship between Delay and Throughput

- If a packet switch had a queue of packets waiting when a new packet arrives the new packet will be placed on the ail of the queue and will need to wait while the switch forward the previous packets.

$$D = \frac{D_0}{(1-U)}$$

D = Effective Delay

D_0 = Delay when the network is idle

U = Value between 1 and 0 denoting the current utilization

Relationship between Delay and Throughput

- Throughput and delay are not completely independent.
- As traffic in a computer network increase, delays increase; a network that operates at close to 100% of its throughput capacity experiences severe delay.

Delay-Throughput Product

- Computed as delay time multiplied by effective throughput.
- Measures amount of data that can be present in the network.
- In fast networks with long delay times, sending computer can generate large amounts of data before destination receives first bit.

Jitter

- The amount of delay that a network introduces.
- A network with zero jitter takes exactly the same amount of time to transfer each packet.
- A network with high jitter takes much longer to deliver some packets than others.

Summary

- Delay
- Throughput
- Relationship between Delay and Throughput
- Delay-Throughput Product
- Jitter

COMPUTER NETWORKS

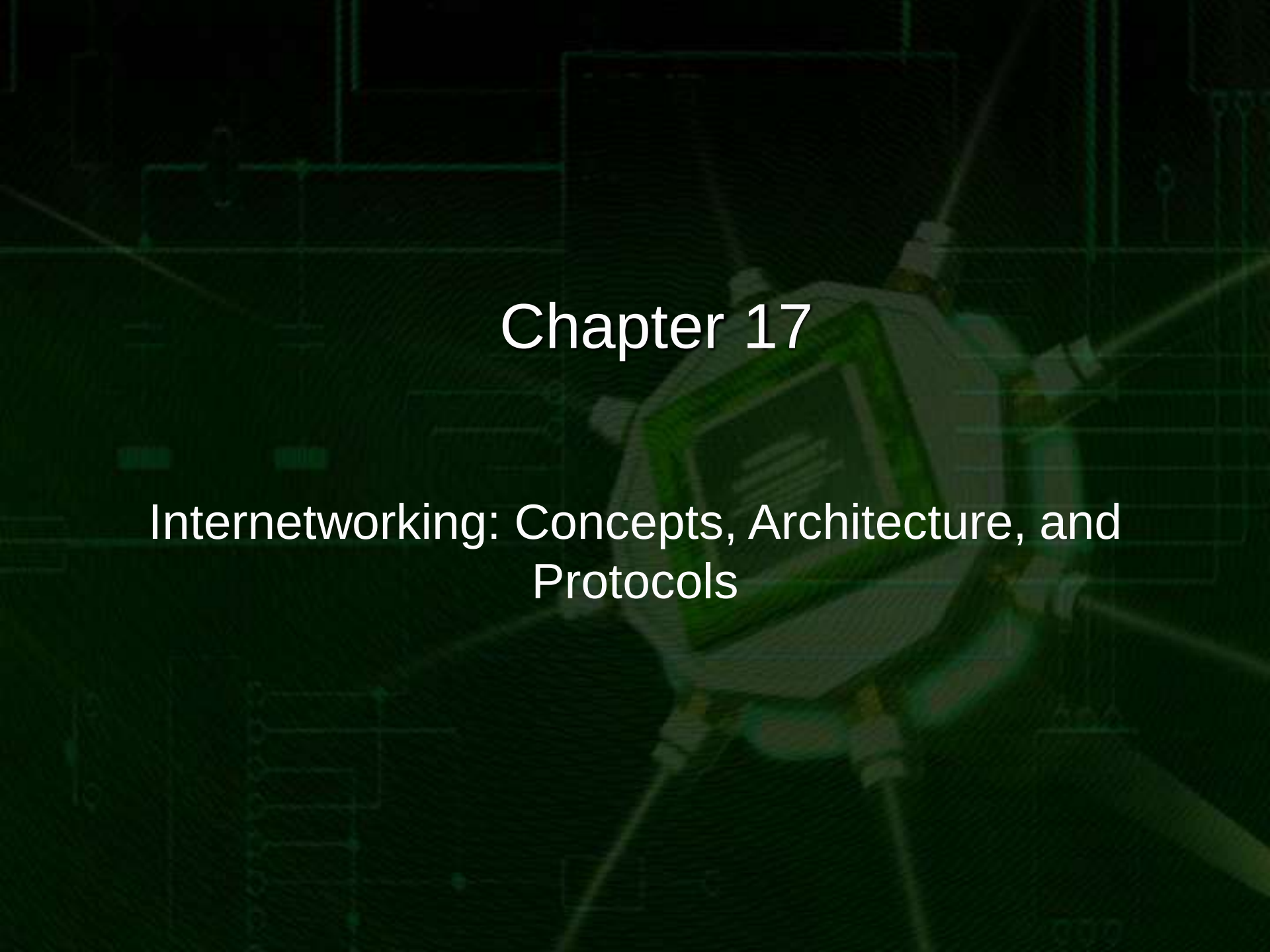
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Lecture-22

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Review Lecture 22

- Delay
- Throughput
- Relationship between Delay and Throughput
- Jitter



Chapter 17

Internetworking: Concepts, Architecture, and Protocols

The Motivation For Internetworking

- No single networking technology is best for all needs.
- A large organization with diverse networking requirements needs multiple physical networks.
- If the organization chooses the type of network that is best for each task, the organization will need several types of networks.

The Motivation For Internetworking

- The interconnection of two or more networks, usually local area networks so that data can pass between hosts on the different networks as though they were one network.
- This requires some kind of Router or Gateway.

The Concept Of Universal Services

- The chief problem with multiple networks:
  - A computer attached to a given network can only communicate with other computers attached to the same network.
  - In the 1970s large organizations began to acquire multiple networks.
    - Each network in the organization formed an island.
    - Employees had to choose a computer appropriate for each task.
    - Multiple screens, keyboards, and computers.

The Concept Of Universal Services

- Universal Services
  - A communication system that supplies universal services allows arbitrary pairs of computers to communicate.
  - Universal service is desirable because it increases individual productivity.

The Concept of Universal Service

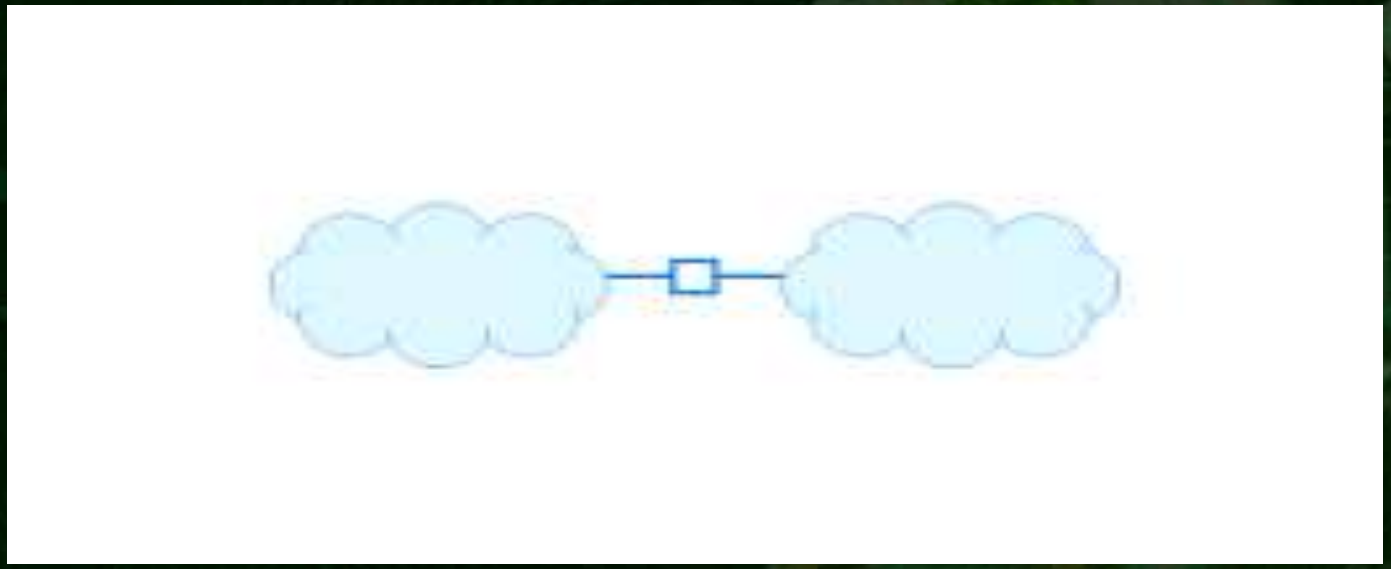
Internetworking

- Despite the incompatibilities among networks, researchers have devised a scheme that provides universal services among heterogeneous networks.
- Called INTERNETWORKING
  - Uses both hardware and software.

Physical Network Connection With Routers

- A router is a special-purpose system dedicated to the task of interconnecting networks.
- A router can interconnect networks that use different technologies, including different media, physical addressing schemes, or frame formats.

Physical Network Connection With Routers



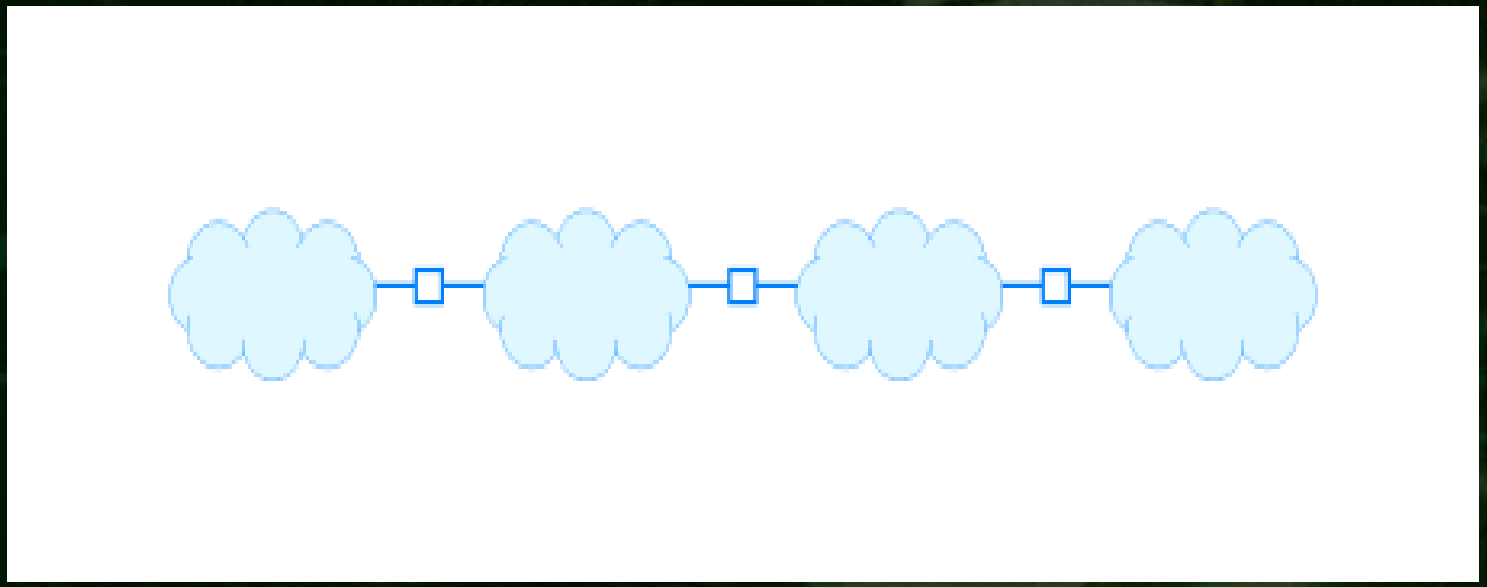
Internet Architecture

- Organization seldom use a single router to connect all its network for two reasons.
  - Because the router must forward each packet, the processor in a given router is insufficient to handle the traffic.
  - Redundancy improved internet reliability.

Internet Architecture

- An Internet consists of a set of networks interconnected by routers.
- The internet scheme allows each organization to choose the number and type of network, the number of routers to use to interconnect them, and the exact interconnection topology.

Internet Architecture



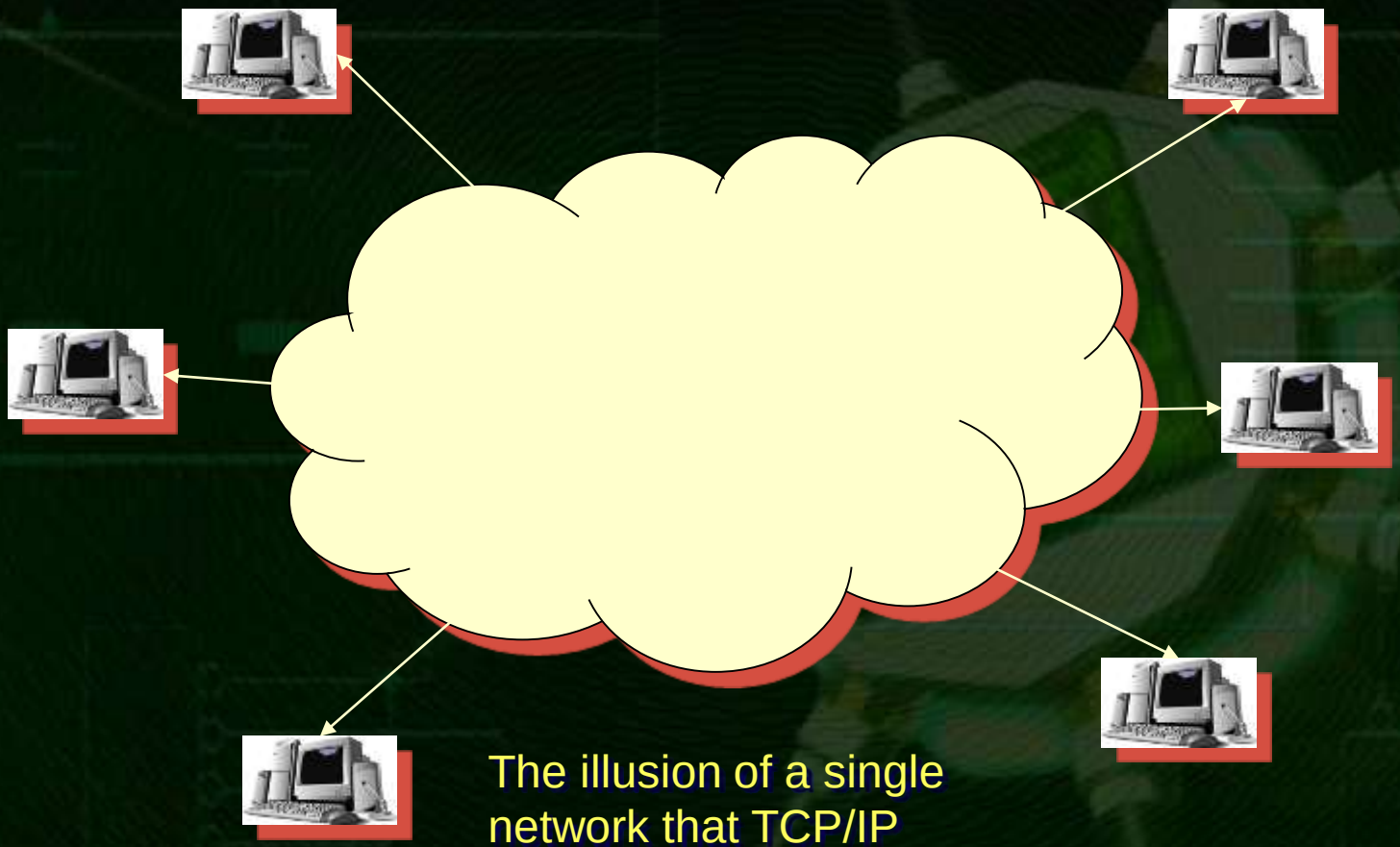
Achieving Universal Services

- **The goal of internetworking is universal service across heterogeneous networks.**
- **To provide this service all computers, and routers must agree to forward information from a source on one network to a specified destination.**

Achieving Universal Services

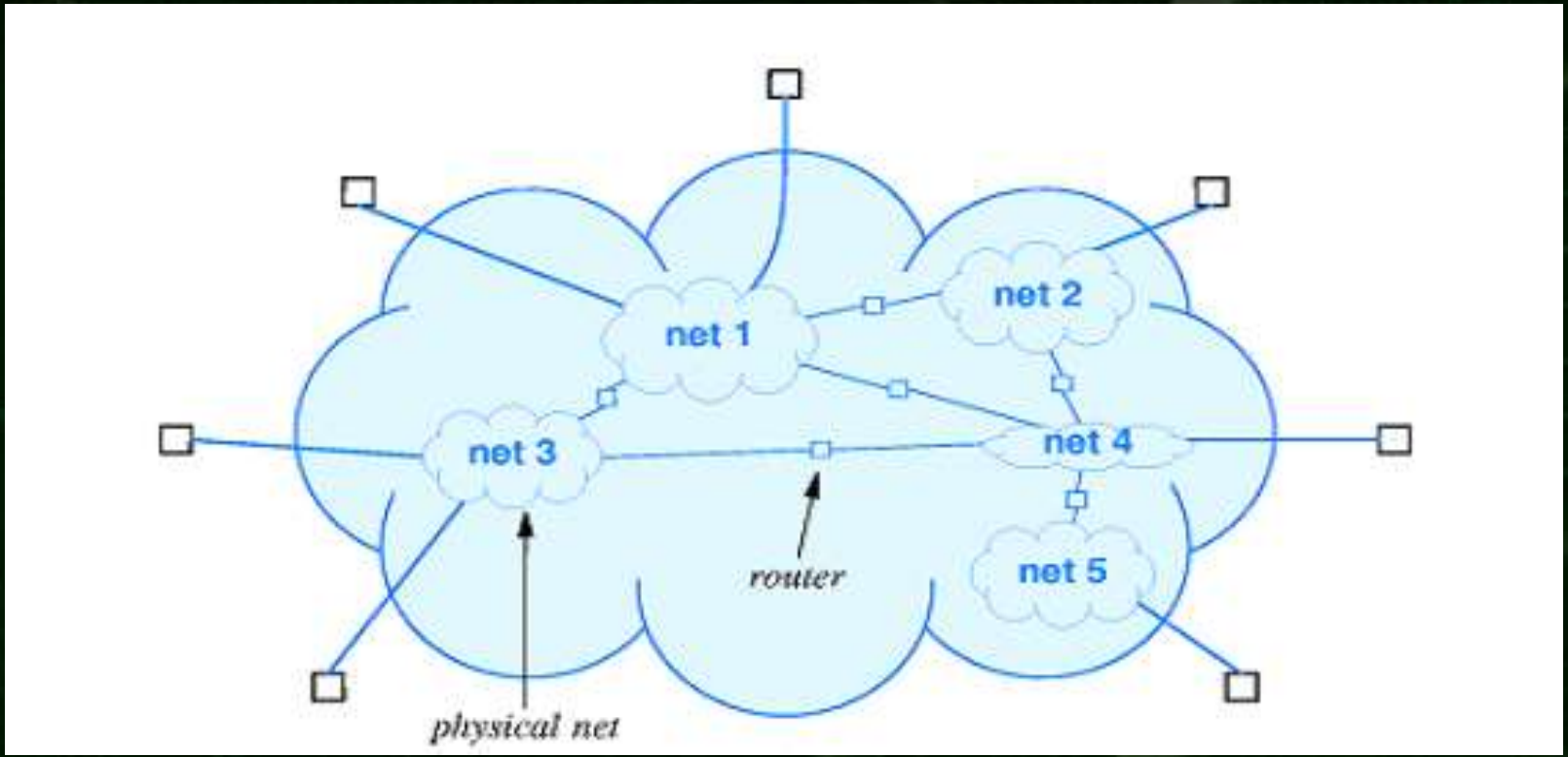
- **The task is complicated as frame formats and addressing schemes may differ.**
- **The key is universal protocol software (TCIP/IP).**

A Virtual Network



The illusion of a single network that TCP/IP software provides to users and applications

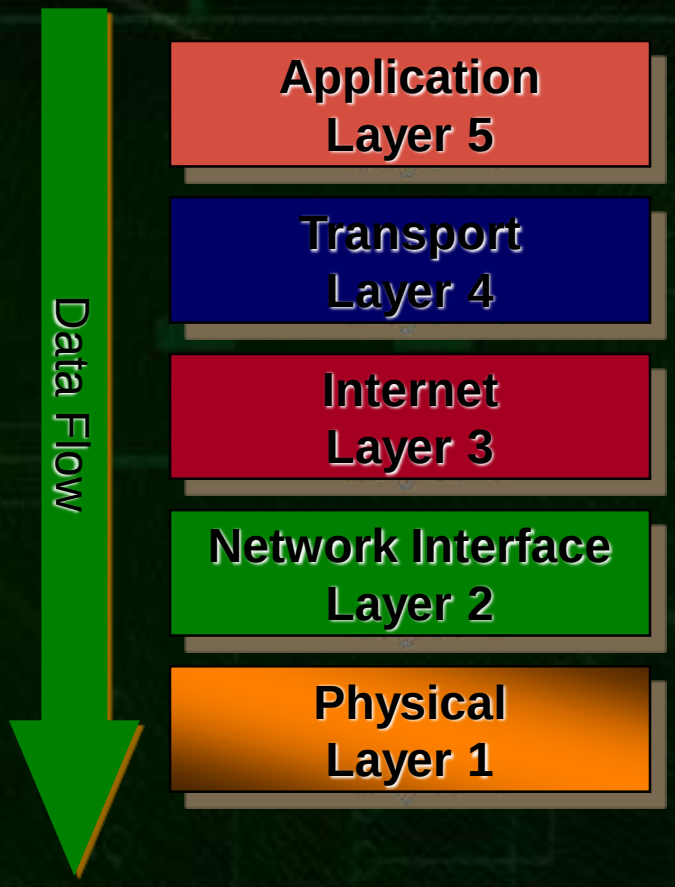
A Virtual Network



Layering And TCP/IP Protocols

- TCP/IP protocols are organized into five conceptual layers.
- Although some layers of the TCP/IP reference model correspond to layers of the ISO reference model, the ISO layers scheme does not have a layer that corresponds to TCP/IP Internet Layer.

Layering And TCP/IP Protocols



- Layer 1 Corresponds To Basic Network Hardware Layer In OSI..
- Layer 2 Specifies How To Organize Data Into Frames.
- Layer 3 Specifies The Format Of Packets Sent Across An Internet And Forward Packets.
- Layer 4 Specifies How To Ensure Reliable Transfer.
- Layer 5 Like 6 And 7 In OSI Species How One Application Uses An Internet.

Host Computers, Routers, and Protocol Layers

- TCP/IP defines the term host computer to refer to any computer system that connect to an internet and runs applications.

- TCP/IP protocols makes it possible for any pair of host to communicate despite hardware differences.

- Both host and routers need TCP/IP protocol software.
  - Routers do not need Layer 5 protocols for applications as they do not run applications.

Summary

- Motivation for internetworking
- Concept of Universal Services
- Internet Architecture
- A Virtual Network
- Layering and TCP/IP Protocols
- Host Computers, Routers and Protocol Layers